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AFFECTIVE BEHAVIOR AND SOCIAL INTERACTION ANALYSIS

PHD THESIS

2025

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DOI: 10.15476/ELTE.2025.390

April 2025

ABSTRACT

Human interaction has been deeply investigated in psychology and social sciences, focusing on explaining the complex underlying mechanisms of communication. From a computational point of view, research in affective behavior and the analysis of dyadic and small-group interactions are enabling the development of human-centered applications. These include healthcare systems for diagnosing mental health conditions and meeting analysis tools that can improve team collaboration by capturing and analyzing various communication cues, such as gestures and blinking patterns. In the field of Personality Computing, challenges include accurately estimating personality traits, especially from first impressions. Research in group interactions explores the complexities and subtleties of the communicative space. The detection and interpretation of non-verbal cues, such as blinking, is key and involves a number of challenges, including noisy data, temporary occlusions, or extreme head poses. Together, these research areas contribute significantly to advancing human-centered technologies, and we are getting one step closer to characterizing human behaviors comprehensively.

In this thesis, I propose three approaches to address the aforementioned tasks and challenges. First, a linear complexity multimodal transformer, called LinMulT, is proposed to model automatic sentiment analysis and personality perception. It uses action units, among others, to mitigate biases while satisfying near-real-time and invariant appearance conditions. Next, I study the perceived personality state of individuals during dyadic and group interactions. I show that LinMulT performs reliably on out-of-domain sessions, and then phenomena related to group affect are investigated. Lastly, a novel transformer-based model, BlinkLinMulT, is proposed for blink presence detection and eye state recognition tasks. It performs efficient fusion and addresses challenges such as noisy data and a wide range of head poses. The network demonstrates state-of-the-art performance and strong generalization capabilities. Additionally, feature significance and the impact of head pose on the blink presence detection performance of the model are also studied.

A prerequisite for developing human-centered, high-quality applications is to analyze and understand the complexity of affective behavior and social interactions, which I contribute to by the methods proposed in this thesis.

KIVONAT

Az emberi interakciókat mélyrehatóan vizsgálják mind a pszichológia, mind a társadalomtudományok területén: a kommunikáció mögöttes, összetett mechanizmusainak magyarázatára összpontosítva. Az affektív viselkedés, a diadikus, valamint kis csoportos interakciók elemzése emberközpontú alkalmazások fejlesztését teszik lehetővé. Többek között olyan egészségügyi rendszerek tartoznak ide, amelyek mentális egészségi állapotok diagnosztizálására szolgálnak, továbbá értekezlet-elemző eszközök, melyek különböző kommunikációs jelzések, például gesztusok, pislogási minták, rögzítésével és elemzésével javíthatják a csapatmunkát. A személyiség-számítás területén kihívást jelent a személyiségjegyek pontos becslése, különösen az első benyomások alapján. Csoportos interakciókra irányuló kutatások a társadalmi megnyilvánulást, csoportdinamikát és a kommunikációs tér összetettségét vizsgálják. Ehhez pedig a nem verbális jelek, mint például a pislogás, detektálása és értelmezése kulcsfontosságú; ez számos kihívással jár, beleértve a zajos adatokat, az átmeneti okklúziókat, vagy az extrém fejtartásokat.

Disszertációmban három módszert javaslok a fent említett feladatok és kihívások kezelésére. Első módszerként egy lineáris komplexitású multimodális transformert mutatok be LinMulT elnevezéssel, amely az automatikus érzelem- és személyiségészlelés modellezésére szolgál. Többek között akcióegységeket használ bemenetként, s emellett eleget tesz közel valós idejű futtatási és invariáns megjelenési feltételeknek. Ezt követően a kísérletekben résztvevők észlelt személyiségi állapotait tanulmányozom diadikus és csoportos interakciók során. Először megmutatom, hogy a LinMulT megbízhatóan teljesít, majd a csoport-hatás jelenségét vizsgálom, valamint észlelt személyiségjegyekből a csoport-teljesítmény előrejelzését is megkísérlem. Végül egy transformer alapú rendszert javaslok BlinkLinMulT elnevezéssel, a pislogás detektálására. A háló hatékony fúziót hajt végre, és olyan kihívásokkal is megbirkózik, mint a zajos adatok és a fejpózok széles skálája. A hálózat korszerű teljesítményt és erős általánosítási képességeket mutat.

Emberközpontú, minőségi alkalmazások fejlesztésének egyik alapfeltétele az affektív viselkedés és a szociális interakciók komplexitásának elemzése és megértése, aminek megvalósításához hozzájárulok a disszertációban javasolt módszerekkel.

PUBLICATIONS

Some ideas, figures, and tables have appeared previously in the following publications:

[P1] Ádám Fodor, Rachid R. Saboundji, Julio C. S. Jacques Junior, Sergio Escalera, David Gallardo-Pujol, and András Lőrincz. "Multimodal Sentiment and Personality Perception Under Speech: A Comparison of Transformer-based Architectures". In: *Proceedings of the ICCV 2021 Workshop on Understanding Social Behavior in Dyadic and Small Group Interactions*. Vol. 173. Proceedings of Machine Learning Research. PMLR, 2022, pp. 218–241. URL: https://proceedings.mlr.press/v173/fodor22a.html.

[P2] Kristian Fenech, Ádám Fodor, Sean P. Bergeron, Rachid R. Saboundji, Catharine Oertel, and András Lőrincz. "Perceived personality state estimation in dyadic and small group interaction with deep learning methods". In: *arXiv 2211.04979* (2022). DOI: 10.48550/arXiv.2211.04979.

[P3] Ádám Fodor, Kristian Fenech, and András Lőrincz. "BlinkLinMulT: Transformer-Based Eye Blink Detection". In: *Journal of Imaging* 9.10 (2023). ISSN: 2313-433X. DOI: 10.3390/jimaging9100196.

FURTHER RELATED PUBLICATIONS

The following publications are loosely connected to this Thesis:

[P4] Ádám Fodor, Rachid R. Saboundji, and András Lőrincz. "Enhancing Apparent Personality Trait Analysis with Cross-Modal Embeddings". In: *Annales Universitatis Scientiarium Budapestinensis de Rolando Eötvös Nominatae. Sectio Computatorica (MaCS Special Issue)* 57 (2024), pp. 167–185. DOI: 10.48550/arXiv.2405.03846.

[P5] Ádám Fodor, Áron Fóthi, László Kopácsi, Ellák Somfai, and András Lőrincz. "Skeletonization Combined with Deep Neural Networks for Superpixel Temporal Propagation". In: *2019 International Joint Conference on Neural Networks (IJCNN)*. 2019, pp. 1–7. DOI: 10.1109/IJCNN.2019.8852391.

[P6] László Kopácsi, Áron Fóthi, Ádám Fodor, Ellák Somfai, and András Lőrincz. "Common Fate Based Episodic Segmentation by Combining Supervoxels with Deep Neural Networks". In: *2019 International Joint Conference on Neural Networks (IJCNN)*. 2019, pp. 1–7. DOI: 10.1109/IJCNN.2019.8851697.

[P7] Ádám Fodor, László Kopácsi, Zoltán Ádám Milacski, and András Lőrincz. "Speech De-identification with Deep Neural Networks". In: *Acta Cybernetica* 25.2 (2021), pp. 257–269. DOI: 10.14232/actacyb.288282.

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LIST OF ACRONYMS

AC Affective Computing

AI Artificial Intelligence

ANN Artificial Neural Network

AMT Amazon Mechanical Turk

APP Automatic Personality Perception

APR Automatic Personality Recognition

APS Automatic Personality Synthesis

AU Action Unit

BERT Bidirectional Encoder Representations from Transformers

BFI Big Five Inventory

BFI-2 Big Five Inventory-2

BFI-10 Big Five Inventory-10

BPE Byte-Pair Encoding

CE Cross-Entropy

CLIP Contrastive Language-Image Pre-Training

CMU-MOSEI Multimodal Opinion Sentiment and Emotion Intensity

CMU-MOSI Multimodal Corpus of Sentiment Intensity

CNN Convolutional Neural Network

CPC Contrastive Predictive Coding

CV Computer Vision

DenseNet Dense Convolutional Network

DL Deep Learning

DNN Deep Neural Network

EAR Eye Aspect Ratio

eGeMAPS extended Geneva Minimalistic Acoustic Parameter Set

ELEA Emergent Leader

EMFACS Emotional FACS

FAb-Net Facial Attributes-Net

FACSAID FACS Affect Interpretation Database

FACS Facial Action Coding System

FCN Fully-Connected Network

FFM Five Factor Model

FFN Feed-Forward Network

FI ChaLearn First Impressions V2

FLOP Floating Point Operation

GAN Generative Adversarial Network

GAP Global Average Pooling

GA Group Affect

GBT Gradient Boosted Tree

GP Group Personality

HCAI Human-Centered Artificial Intelligence

HCC Human-Centered Computing

HCD Human-Centered Design

HCI Human-Computer Interaction

ICC(2,k) Intraclass Correlation Coefficient

LinMulT Multimodal Transformer with Linear Complexity Attention Mechanism

LLD Low-Level Descriptor

LSTM Long Short-Term Memory

MAE Mean Absolute Error

ML Machine Learning

MHA Multi-Head Attention

MSA Multimodal Sentiment Analysis

MSE Mean Squared Error

MULTISIMO MULTImodal and MULTIparty Social Interactions MOdelling

MulT Multimodal Transformer

NLP Natural Language Processing

NMT Neural Machine Translation

PC Personality Computing

PERMANOVA Permutational Multivariate Analysis of Variance

PP Personality Psychology

RNN Recurrent Neural Network

ResNet Residual Network

RoBERTa Robustly optimized BERT approach

SA Sentiment Analysis

SGDR Stochastic Gradient Descent with Restarts

SSE-FT Self-Supervised Embedding Fusion Transformer

SSE Self-Supervised Embeddings

SSL Self-Supervised Learning

TOST Two-One-Sided T-test

UDIVA Understanding Dyadic Interactions from Video and Audio Signals

Part I

INTRODUCTION

1 INTRODUCTION

Human-Centered Artificial Intelligence (HCAI) aims to integrate technology into our daily lives by ensuring it aligns with human needs and behaviors. This thesis presents approaches to understanding and predicting human characteristics like personality and sentiment. It also examines the dynamics of task-oriented social interactions and non-verbal cues like eye blinks, which are essential for creating a comprehensive framework for analyzing human behavior and developing human-centered applications through high-quality human-machine interactions. These include affective behavior recognition systems used in healthcare for diagnosing mental health conditions or improving customer service in call centers. Mental health chatbots use conversational AI to adjust their responses based on the user's mood, behavior, and personality traits to provide empathetic and effective support. Meeting analysis software captures and interprets individual and interpersonal social signals such as speech-, gaze-, and blinking patterns, facial gestures, and body language, which helps to understand team dynamics and improve engagement and collaboration during business meetings or therapy sessions.

In social communication, emotion and personality are two fundamental cues that shape how individuals feel and behave during interactions, as well as how they respond to others. For intelligent systems to engage and communicate in a *human-like* manner, it is required to equip them with mechanisms that mirror these human traits. This may explain why automatic sentiment and personality perception—and recognition—are receiving increasing interest from the Machine Learning (ML) and Computer Vision (CV) communities, but also Personality Psychology (PP), which is supported by easy access to large and public databases and benchmarks on related topics, in addition to the outstanding advancements of Deep Learning (DL) methods.

Personality Computing (PC) focuses on the computational quantification and modeling of human personality traits. Understanding these traits is important as they profoundly affect how individuals interact socially and perceive their environment. Systems capable of human behavior analysis could be used for self-driving cars, medical research, and surveillance, among many others. A major challenge within this field is the accurate estimation of personality traits, particularly from first impressions, which are subconscious judgments based on minimal information, that significantly shape future interactions and decisions. Furthermore, personality significantly influences how individuals interact with each other and perceive the world around them. For instance, an extroverted person might engage more openly in social settings, influencing group dynamics positively, whereas an introvert might contribute more in smaller, intimate gatherings, affecting group cohesion differently.

Researchers aim to simulate this human cognitive process by utilizing multimodal information sources, such as textual, acoustic, and visual data, which complement each other and enhance the accuracy of personality trait predictions. Although multimodal systems offer advantages compared to monomodal systems, they raise several challenges as well. For example, one faces the problems of selecting from the modalities which to be included in multimodal systems, deriving the architecture to fuse them, and attenuating errors from noisy, missing, or underrepresented data. However, resolving such cases is crucial in several applications, such as accurately diagnosing and understanding rare medical conditions like autism spectrum disorder or assessing personality traits under stress—like in interviews, where nuanced behavior can reveal vital insights.

To develop more specialized systems capable of analyzing social communication, it is essential to capture small changes in sentiment and personality states while meeting several key requirements. These systems must be responsive, requiring near-real-time processing, should be robust to noise, and invariant to appearance variations.

Previous works attempted to model temporal dynamics through frame-by-frame recognition followed by aggregation. Nonetheless, such an approach may not generalize well under speech due to the dynamics of the task. Moreover, human face-to-face communication is a complex multimodal signal, where words (language modality), gestures/pose/gaze/blink (visual modality), and changes in tone (acoustic modality) are used to convey our intentions [177]. To address this problem, different deep learning-based solutions have been proposed, from Recurrent Neural Networks [143] to the recently emerging Transformers [165], as proposed by [119, 164, 146]. Past works have already exploited the use of Transformer architecture for personality recognition [99], but in the context of Natural Language Processing (NLP).

Most existing approaches to personality computing and sentiment analysis rely on specific datasets that often pose challenges due to significant variations in appearance, background, and human-centered attributes like head pose, ethnicity, age, gender, and fashion. These factors hinder the development of robust and fair ML models that can generalize across different populations and environments. Additionally, the narrow focus of some databases may overemphasize particular problems or lack sufficient diversity to capture the full spectrum of human behavior. This lack of comprehensive data complicates building models that fairly predict and analyze human traits across diverse demographics and conditions.

Moving from the individual to dyadic and small-group sessions [131, 118], these interactions are critical for understanding fundamental social processes, including communication patterns, influence, and decision-making. The human characterization in such a social domain helps in designing more effective collaborative environments and technologies that facilitate better interpersonal connections and group functionalities. However, research in these fields faces significant challenges, primarily due to the complexity of human behaviors and the subtleties of social interactions. Variability in individual personalities, cultural backgrounds, and situational contexts makes it difficult to generalize findings.

Certain characteristics of people within a fixed group can demonstrate convergent behavior such as in emotion [122, 66] and group affect [65, 17, 24]. Progress has been made toward developing a conceptual model that merges individual and group personality traits in a framework [137]. However, there are open research questions about group affect and dynamics, such as how group members' perceived personalities align, and whether analyzing collective behavior can uncover features that predict overall group performance, helping to understand team cohesion and effectiveness.

While blinking is mostly an involuntary response, it can shape first impressions and alter the dynamics of social interactions, potentially indicating disagreement, but can also reveal a person's emotional state, or signal cognitive load during complex tasks, influencing how individuals perceive and communicate with each other. In face-to-face communication, non-verbal behaviors play a significant role for interpreting personality cues. Variations in blink duration convey different signals. For instance, long blinks can be perceived as a cue to conclude a speaking turn, while short blinks may not convey the same level of feedback, thus influencing speakers' communicative behavior [76]. Moreover, blinking is a specific muscle movement, categorized as an Action Unit (AU) (namely AU45) in the Facial Action Coding System (FACS). Interestingly, studies have found correlations between blinking and personality traits, such as conscientiousness and emotional stability. Specifically, conscientiousness is negatively correlated with blinking, as well as with other facial expressions like inner and outer brow raising, which are associated with surprise and disgust [93]. Similarly, emotional stability is negatively correlated with blinking and brow raising, suggesting that these facial cues may reflect underlying personality characteristics. Understanding these subtle relationships can provide valuable insights into how non-verbal behaviors like blinking contribute to personality perception in social interactions. Detecting eye blinks is also important for multiple facial analysis tasks, such as spotting deep fakes [100], monitoring driver alertness [169], and fatigue [4], and identifying health issues.

Traditional methods rely on face recognition and frame-by-frame eye state analysis, struggling where blinks occur partially for only a few frames or where input data are noisy or incomplete. Challenges in this area include noise from varying lighting conditions and reflections, limitations from low-resolution cameras, restricted camera-subject distances, and extreme head poses [28, 182, 134].

I propose three approaches to address the above-mentioned tasks and challenges. Each thesis point is extended in the corresponding chapter in the dissertation.

Thesis 1. I propose LinMulT, a Multimodal Transformer with a Linear Complexity Attention Mechanism optimized for Multimodal Sentiment Analysis (MSA) and Automatic Personality Perception (APP) tasks [P1]. With the goal of minimizing identity-related biases and enhancing generalization across different datasets and scenarios, I designed LinMulT to use privacy-preserving, appearance-invariant visual features like AUs and FAb-Net deep hidden representations while also achieving reduced computational complexity.

In terms of performance, LinMulT demonstrates almost identical results to the baseline model MulT on the MSA task. On CMU-MOSI, LinMulT achieves competitive F1 (0.824) and correlation (0.745) scores, while on CMU-MOSEI, it performs comparable to state-of-the-art methods with F1 (0.746), MAE (0.530), and correlation values (0.784). For the APP task, LinMulT slightly outperforms MulT, achieving an average R_{acc} of 0.909 and an R^2 of 0.372 on the First Impressions V2 dataset. T-tests confirm that this performance difference is statistically significant.

The most significant advantages of LinMulT lie in its computational efficiency. It achieves near-real-time processing with an inference time of 134 ms on GPU and 234 ms on CPU for single-person interactions. Its excellent scaling properties are demonstrated by testing across batch sizes and analysis window durations. When increasing the batch size from 1 to 8, LinMulT achieves a speedup of up to 16.2x on CPU and reduces reserved GPU memory by up to 85.1%. As the analysis window duration increases from 7.5 to 30 seconds, LinMulT reduces inference FLOPs by up to 80.9%, training FLOPs by up to 81.1%, the reserved memory reduction increases from 55.8% to 85.7%, compared to MulT. These efficiency gains make LinMulT highly suitable for resource-constrained environments and near-real-time applications.

Thesis 2. I utilized LinMulT to estimate the perceived personality state of individuals participating in dyadic and small group interactions [P2]. I aim to address several key research questions: How reliably does the perceived personality estimator perform in out-of-domain sessions compared to human annotators? Does the perceived personality of members within a group cluster? Is the group-level perceived personality a stronger predictor of group performance than self-reported personality?

To investigate these questions, I conducted experiments using the MULTISIMO, ELEA, AMI, and UDIVA datasets. The reliability of the personality estimator was evaluated through inter-rater correlation and TOST equivalence tests, showing that LinMulT's prediction error is statistically equivalent to that of the majority of human annotators, matching the level of agreement observed among the human raters themselves.

A Permutational Multivariate Analysis of Variance (PERMANOVA) revealed significant clustering of perceived personality traits in collaborative tasks, in which verbal agreement is key to solve the given task, like the winter survival task in ELEA and all four remote control design tasks in AMI, and the Animals task in UDIVA. However, significant group affect cannot be observed for the other tasks in UDIVA, which are either competitive (Ghost), do not require much verbal communication (Lego), or an open conversation with no specific goal to be achieved as a group (Talk).

Furthermore, a GBT model demonstrated that group-averaged perceived personality traits are superior predictors of group performance compared to self-reported traits, achieving a MSE of 16.17 ± 0.45 for perceived metatraits versus 43.88 ± 1.17 for self-reported metatraits, and a MSE of 22 ± 1.26 for perceived Big Five versus 40.68 ± 1.10 for self-reported Big Five traits.

While the used datasets showed promising clues of the existence of group affect for multiple tasks, despite using all publicly available data for this purpose, there is no guarantee that the results are universally applicable.

Thesis 3. I propose BlinkLinMulT, a novel transformer-based model with a linear complexity attention mechanism, designed for eye blink presence detection and frame-wise eye state recognition [P3]. My goal is to improve eye blink detection performance by integrating low-level RGB texture with high-level features including iris and eye landmarks, Eye Aspect Ratio (EAR), and head pose angles, while incorporating motion information using a sequence model that operates effectively under diverse head poses and challenging conditions like lighting changes.

To evaluate BlinkLinMulT's robustness, I conducted extensive within-dataset and cross-dataset experiments on multiple public benchmark datasets, including CEW, ZJU Eye, MRL Eye, RT-BENE, EyeBlink8, Researchers' Night (RN15 and RN30), and TalkingFace. The results show that BlinkLinMulT achieves competitive or superior performance compared to state-of-the-art methods, with notable improvements such as a 3.8% increase in F1 score on EyeBlink8 and a 4.0% increase on the Researchers' Night datasets for blink presence detection. Additionally, BlinkLinMulT demonstrated substantial improvements compared to the literature in eye state recognition on RT-BENE^{seq}, increasing the F1 score by 12.3%.

To highlight the importance of combining different feature sets for enhanced performance, I performed a comprehensive feature fusion ablation study: the combination of texture with landmark-based features and head pose angles improved performance over single-modality evaluations by 4.3% on RT-BENE^{seq}, 1.8% on EyeBlink8, 2.7% on RN15, 2.0% on RN30, and 0.8% on TalkingFace.

Finally, I demonstrated that BlinkLinMulT remains robust even under extreme head poses, such as when participants are looking down or in profile, achieving 0.744 and 0.917 F1 score, respectively.

The three theses are closely interrelated and will be further strengthened through integration, including the fusion of information sources similarly to the technological approach I employed in Thesis 1. This integration is intended not only to advance scientific understanding but also to promote the development of practical, human-centered applications aiming at rendering digital systems more intuitive and responsive to affective and social signals. The present work is an intermediate step toward adaptive technologies that learn from and adapt to individual users, enabling highly personalized human–machine interactions.

In addition to the main points of my thesis, I have also published several research papers that are only loosely connected to the central topic. I designed a multimodal system using deep metric learning to improve personality perception accuracy [P4]. This experience motivated me to explore the area further and make more focused contributions. I also proposed a video object segmentation approach [P5] that combines optical flow, depth estimation, and medial axis transformation to achieve superpixel temporal propagation. With my colleagues at ELTE NIPG, we investigated an episodic segmentation method that leverages optical flow, depth, and a hierarchical supervoxel algorithm [P6]. Finally, we developed a Deep Neural Network (DNN) solution [P7] that removes personal characteristics from human speech by converting it to the voice of a Text-to-Speech system. All of these methods can be integrated into human-centered applications; however, they contribute only indirectly to the thesis points.

Furthermore, I have engaged in other publishing efforts that, while unrelated to my thesis topics, have been instrumental in my personal development. These publications have broadened my experimental approaches and perspectives, contributing to the achievements of my actual thesis points. A complete list of these publications is provided as an attached document.

The thesis is structured as follows: Part II introduces the knowledge required to understand the foundations. The following chapters in Part III detail the proposed methods, one chapter corresponding to one thesis point. To enhance clarity, each method's introduction is immediately followed by its analysis and evaluation, rather than compiling these assessments into a separate chapter. This structure allows readers to more effectively track the progression of ideas and discussions. Finally, Chapter 8 summarizes the work and indicates options for further extensions.

To help reproducibility and further research, the methods proposed in this thesis are all publicly available, with source code and trained models[1].

1 https://github.com/fodorad

Part II

BACKGROUND

2 PSYCHOLOGICAL AND SOCIAL FOUNDATIONS

Automatic detection, understanding, and utilization of personality traits, affect, and group dynamics as contextual information are essential for developing high-quality and adaptive human-centered applications.

2.1 HUMAN-CENTERED ARTIFICIAL INTELLIGENCE

Artificial Intelligence (AI) is an interdisciplinary field aiming to create systems capable of performing tasks that would typically require human intelligence. AI encompasses various technologies and approaches, including Machine Learning (ML), Natural Language Processing (NLP), and Computer Vision (CV), all aimed at enabling machines to perceive, learn, reason, and interact in ways that support and empower users [139].

Human-Centered Artificial Intelligence (HCAI) is an approach for developing and deploying AI that prioritizes user needs, values, and capabilities [176]. This methodology focuses on creating AI systems that enhance human abilities, align with ethical principles, and serve as tools to augment rather than replace human intelligence. HCAI aims to bridge the gap between technological advancements and user-centered design, ensuring that AI systems are not only efficient but also intuitive, transparent, and beneficial to society [144].

The development of HCAI is rooted in the evolution of AI philosophies and design paradigms. Two main philosophical perspectives have shaped the early development phase and its interaction with users [174]:

- **Rationalistic:** Regarding this perspective, AI covers the theory and development of computer systems that mimic cognitive abilities and handle tasks requiring human intelligence.
- **Design:** This perspective views AI as a problem-solving tool and emphasizes the importance of Human-Computer Interaction (HCI). It focuses on the interfaces of how users interact with computers.

These philosophical foundations have significantly influenced the development of Human-Centered Design (HCD) methodologies. While the rationalistic perspective has driven advancements in AI capabilities, the design perspective has paved the way for more user-focused approaches.

HCD research is the design approach that centers people and their needs, motivations, emotions, behavior, and perspectives throughout the design development process. I encourage interested readers to explore all the designs presented in [11]. Hereby, I emphasize three methodologies that are related to this thesis.

INTERACTION DESIGN: This approach focuses on understanding and developing machine-user interactions by examining people's behavior, actions, and cognitive processes while they're using technology [11]. When we explore research on AI interactions—such as how self-driving cars collaborate with users or how we engage with robots—we're examining the human impact of these technologies, including our behaviors, emotions, and responses to them.

PERSUASIVE TECHNOLOGY: This design attempts to intentionally change human attitudes, behavior, or both through technology [59]. These technologies leverage psychological principles such as social comparison, commitment, and consistency to influence user behavior. They can function as tools, media, or social actors, employing various strategies like personalization, goal-setting, and feedback mechanisms. While persuasive technologies have shown potential in areas like public health and education, they also raise ethical concerns regarding privacy and manipulation.

HUMAN-CENTERED COMPUTING: An interdisciplinary field that focuses on designing, developing, and deploying computing systems that prioritize human needs, abilities, and experiences. [85]. It integrates knowledge from computer science, cognitive science, social sciences, and other related fields to create technologies that are intuitive, accessible, and beneficial to users in their daily lives [84].

These HCD methodologies are crucial for achieving the goals of HCAI. Interaction Design ensures that AI systems are intuitive and responsive to human needs, fostering seamless collaboration. When applied ethically, Persuasive Technology harnesses AI's potential to encourage positive behavior changes and boost user involvement. Human-Centered Computing (HCC) offers a comprehensive approach to developing AI systems that emphasize user experiences and societal benefits. Collectively, these methodologies enable the creation of software that are not only technologically sophisticated but also in harmony with human values, requirements, and abilities.

While these methodologies provide valuable frameworks for developing HCAI systems, their effectiveness relies heavily on a deep understanding of human psychology and behavior. Specifically, the ability to detect, understand, and utilize personality traits and subjectivity as contextual information is vital for creating truly adaptive and responsive AI systems. One field that has emerged to address this need is Affective Computing (AC).

AFFECTIVE COMPUTING: An interdisciplinary field that aims to develop systems capable of perceiving, recognizing, and understanding human emotions and responding intelligently, sensitively, and naturally, thus making HCI more natural [126, 127].

Understanding personality traits helps categorize broad and specific characteristics that shape an individual's thoughts, feelings, and actions. This knowledge is crucial for creating AI systems capable of engaging effectively with users in different settings. For instance, an AI system that adapts to an individual's personality traits could offer more personalized and effective interactions, improving both user satisfaction and the efficiency of the technology.

Complementing this approach, the study of human subjectivity examines how feelings, emotions, sentiments, and opinions develop and interact. By incorporating both personality traits and the nuances of subjectivity, we can design AI systems that not only adapt to individual differences but also interpret, respond to, and even predict emotions and states. This comprehensive understanding enables more sophisticated and empathetic AI applications. For example, in healthcare, AI can recognize patient distress or discomfort and adjust interactions accordingly, thereby improving patient care and outcomes. Such customization is especially valuable in digital assistants, educational tools, and therapeutic applications, where responding to individual subtleties is key to their effectiveness. Therefore, personality and affect are not just complementary, but integral to the philosophy of HCAI.

The relationship between personality and affect is complex [136]. Although they are interconnected, personality and affect are largely independent concepts. Personality captures long-term behavioral and cognitive patterns, while affect deals with short-lived emotional states. Each can be understood and studied independently, but their interaction plays a critical role in influencing human behavior in unique ways.

Personality traits and affect deeply influence each other. For instance, a person high in neuroticism often feels negative emotions like anxiety and sadness more intensely. Conversely, regular experiences of positive emotions might enhance someone's extraversion, showing how affect can also shape personality traits over time.

In the following sections, I investigate personality and affect in greater detail, which settles the goals of tasks and experiments in Part III.

2.2 HIERARCHY OF PERSONALITY TRAITS

People have unique personalities composed of different psychological traits. These traits determine how individuals think, feel, and behave consistently over time. They include perception, cognition (e.g., thoughts, attitudes), motivation (e.g., desires, needs, goals), and behaviors (e.g., verbal, nonverbal, paraverbal, extraverbal, and action sequences); personality is also linked to emotions such as affect and feelings.

Personality traits cannot be directly observed but can be inferred through people's repetitive patterns of actions and reactions. They also have an impact on individuals' personal lives, careers and social interactions. Research has shown that personality traits can predict various important outcomes in life, such as mental and physical health, lifespan, or even income level.

One important thing to note about traits and their covariance structure is that they are organized in a hierarchy. Specific traits that are correlated with each other can be grouped. These larger categories also show correlations, suggesting the presence of even broader traits. At each level of this hierarchy (except for the highest), certain factors cause groups of traits to vary together in ways described by the next higher level of the hierarchy. At the same time, other factors cause each trait to vary independently from the others. In simpler terms, all traits below the highest level of the hierarchy have both shared and unique characteristics. Studies have shown that there is distinct genetic variation associated with traits at each level of the hierarchy.

The hierarchical structure of traits also reveals another important aspect: its probabilistic nature. While lower-level traits within a higher-level trait may be correlated, they are not perfectly correlated: having a high score on a higher-level trait does not necessarily imply high scores on all lower-level traits, but rather on some of them. Different combinations of subtraits can lead to the same score on a given trait.

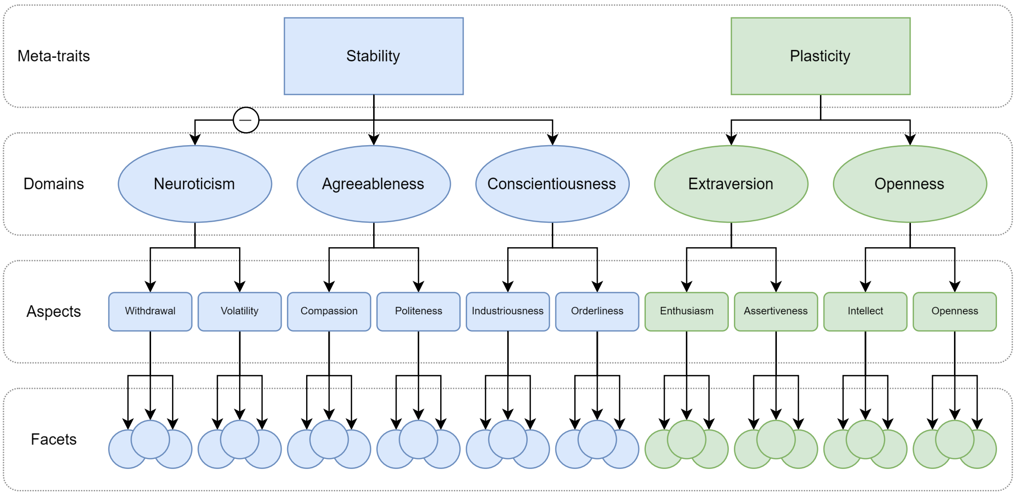

Figure 2.1: Hierarchy of Personality Traits. The minus sign indicates that Neuroticism is negatively related to Stability.

2.2.1 *Five Factor Model*

Much of the personality research conducted in the second half of the 20th century focused on organizing traits into a hierarchy based on empirical data. By the 1990s, there was a remarkable consensus that five broad factors, known as the Big Five, account for most of the relationships among more specific traits. The Five Factor Model (FFM) is the most widely used trait taxonomy [42], and is commonly represented by the acronym OCEAN (Figure 2.2). The following paragraphs introduce the Big Five personality traits.

OPENNESS: This personality trait is connected with creativity and intelligence, and it refers to how open-minded you are to new experiences and ideas. In the field of psychology, openness is also called "openness/intellect" or "openness to new experience".

People who score high on openness are generally more accepting of new ideas and experiences. They show a keen interest in exploring unfamiliar territories and are adept at linking various thoughts and ideas. This trait not only makes them more creative but also more successful in creative fields, especially the arts. Some studies suggest a correlation between high openness and greater intelligence, noting that individuals with exceptionally high intelligence often exhibit this trait.

In contrast, those with low levels of openness prefer the comfort of routine and are hesitant towards change. They are likely to avoid new opportunities that require adjusting their usual ways, even if these opportunities could benefit their educational or professional growth. Such individuals might choose to remain in a familiar job rather than pursue a potentially more fulfilling role due to their preference for familiarity.

CONSCIENTIOUSNESS: It describes how much a person embodies qualities like organization, dependability, and goal-orientedness.

Individuals high in conscientiousness are exceptionally organized, reliable, and diligent. They are often seen as highly responsible, with a strong focus on achieving their goals through meticulous planning and persistent effort. Their thorough nature makes them excel in environments that require precision and a systematic approach.

On the other side, those who score low on conscientiousness might struggle with organization and often prefer spontaneity over structured planning. They may find it challenging to adhere to schedules or meet deadlines, which can sometimes hinder their professional progress.

EXTRAVERSION: The trait measures the level at which someone is outgoing, sociable, and enjoys interacting with others.

Extraverts are typically energetic, talkative, and assertive. They thrive in social settings and are often the life of the party, drawing energy from interacting with others. Their outgoing nature helps them in roles that require networking and teamwork.

Conversely, introverts—those low on extraversion—prefer solitude and are more reserved. They tend to feel drained by large social gatherings and usually enjoy deep, meaningful one-on-one interactions rather than engaging in large, noisy settings.

AGREEABLENESS: Reflects a person's tendency to be kind, cooperative, and compassionate towards others.

Highly agreeable individuals are compassionate, kind, and cooperative. They value harmony in their relationships and are often willing to compromise to ensure a peaceful coexistence. Their empathetic nature makes them excellent team players and beloved friends and partners.

Those with lower levels of agreeableness may prioritize personal ambitions over collaborative success, often being seen as competitive or even confrontational. While this can lead to success in highly competitive environments, it might strain personal and professional relationships.

NEUROTICISM: Indicates a person's propensity to experience negative emotions, such as anxiety, worry, and sadness, more frequently than others.

People with high levels of neuroticism are more likely to perceive ordinary situations as threatening and minor frustrations as hopelessly difficult. This sensitivity can sometimes hinder their ability to handle stress effectively.

Individuals low in neuroticism tend to remain calm, even under pressure. They experience less emotional distress and are generally considered resilient, facing life's challenges with a level-headed approach.

Emotional Stability is often considered the counterpart to neuroticism—also used as reversed neuroticism. It refers to an individual's consistent ability to remain calm, composed, and untroubled by stress, thereby reflecting a resilience to negative emotions and contributing to a balanced and steady disposition. I chose *emotional stability* instead of *neuroticism* in my experiments to align it with the dimensional structure of the other Big Five traits, ensuring a uniform framework for evaluating personality.

Efforts to add more dimensions to the trait set were made. However, the five-trait model proved to be very stable in different studies, unlike the more complex models [43]. Models with fewer traits were also tested [54], but these still appeared to be linear combinations of the Big Five.

These traits are not just broad categories—they are deeply ingrained in our biology yet shaped by our surroundings. Interestingly, these traits are mostly stable across time, though they can change across the lifespan. What makes the Big Five particularly fascinating is their universality. They can be identified across different cultures and are consistent across various methods of measurement. This suggests that while our personalities are unique, the underlying structures of personality may be quite similar worldwide.

2.2.2 *Metatraits*

While it was initially assumed that the Big Five traits were independent of each other and represented the highest level of the trait hierarchy, studies have consistently found that they are actually intercorrelated. This means that there are two higher-order traits, or metatraits, that can be derived from the correlations among the Big Five [44, 39, 156, 40, 5].

STABILITY: This metatrait represents the shared variance of Emotional Stability, Conscientiousness, and Agreeableness. Each of these traits represents a

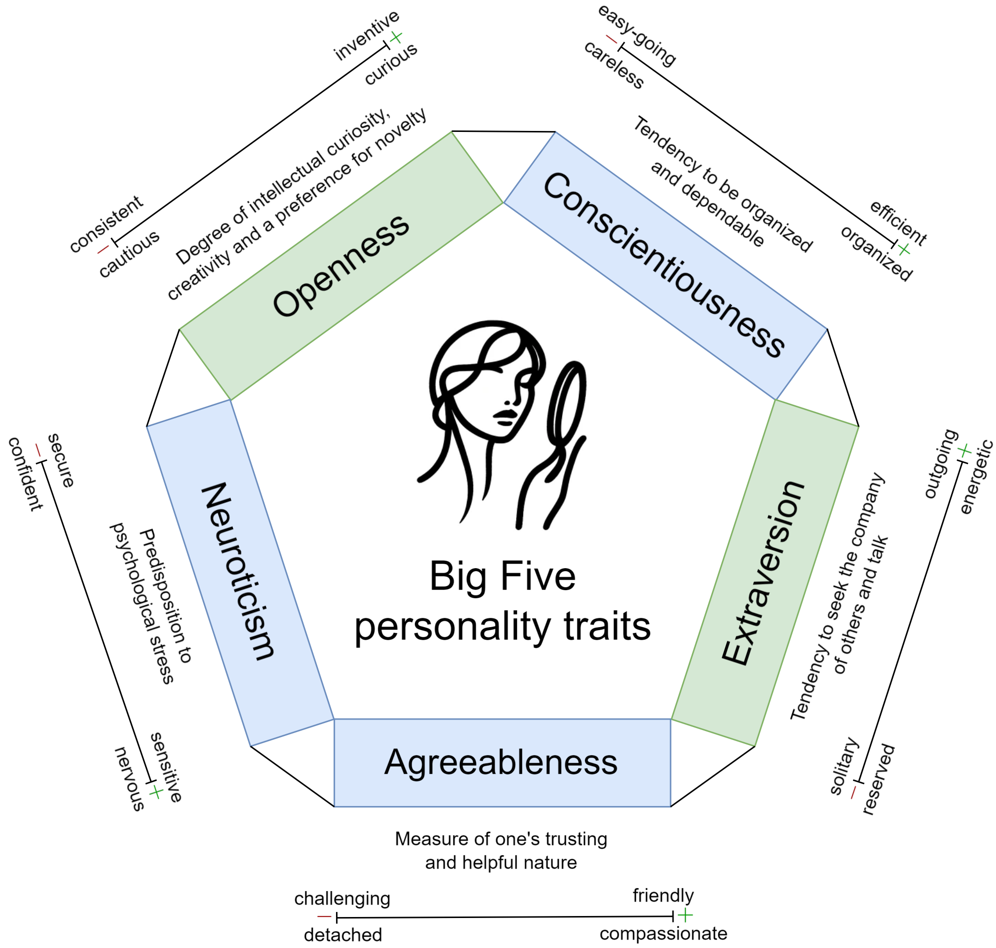

Figure 2.2: The Five Factor Model. The figure illustrates the Big Five personality traits, each accompanied by a brief description. Around each trait, the dimensionality is represented, highlighting adjectives commonly associated with low and high values within each dimension.

distinct form of steadiness: Emotional Stability pertains to emotional resilience, Conscientiousness reflects motivational consistency, and Agreeableness relates to social harmony. The metatrait is also referred to as Alpha.

PLASTICITY: This metatrait represents the shared variance of Extraversion and Openness, the general tendency toward exploration. Extraversion reflects behavioral exploration and sensitivity to specific rewards, while Openness reflects cognitive exploration and sensitivity to the reward value of information. The metatrait is also referred to as Beta.

Stability and Plasticity provide a broader understanding of how the Big Five traits relate to each other within the hierarchy. The metatraits appear to be the highest level of the personality hierarchy, with no "general factor of personality" above them.

2.2.3 *Aspects and the Facet Level*

Traditionally, directly below the Big Five traits are the facets, which are more specific attributes contributing to each of the Big Five traits. However, recent research has introduced an intermediate level known as aspects [41]. If only the facets were considered, one would expect a single genetic factor for each trait domain, but this was not the case.

For Openness/Intellect, the aspects are neatly divided into Intellect and Openness. Intellect encompasses traits such as Quickness, Ingenuity, and an aptitude for Ideas, emphasizing cognitive agility and creative problem-solving. Openness, on the other hand, includes Aesthetics, Imagination, and Fantasy, highlighting a disposition towards artistic sensitivity and a rich inner life.

Extraversion breaks down into Assertiveness and Enthusiasm. Assertiveness relates to social dominance, leadership, and the willingness to express opinions and stand one's ground. Enthusiasm refers to an individual's propensity for sociability and energetic engagement with the world around them.

For Conscientiousness, the defining aspects are Industriousness and Orderliness. Industriousness relates to a person's persistent effort and dedication, while Orderliness—preferred over the simpler Order—focuses on personal habits and the tendency to maintain a structured and organized environment.

Agreeableness comprises Compassion and Politeness as its aspects. Compassion is associated with emotional responses like Warmth, Sympathy, and Tenderness, indicative of an empathetic nature. Politeness reflects a cognitive consideration of social norms and others' needs, evident in behaviors like Cooperation, Compliance, and Straightforwardness.

Finally, the Neuroticism trait is detailed through Volatility and Withdrawal. Volatility covers emotional instability, irritability, and the challenge of controlling emotional impulses. Withdrawal, meanwhile, indicates a predisposition towards anxiety, fearfulness, and the tendency to retreat from stressors, aligning with internalizing tendencies.

The clarification of these ten intermediate factors or aspects came from further factor analyses, which examined a larger number of facets for each domain. This additional level helps provide a more nuanced structure to the Big Five, offering an empirically derived substructure that is lacking at the facet level, where the facets themselves are often derived more intuitively than empirically.

While research on these aspect-level traits is less abundant than on the Big Five or even metatraits, it is vital because it fills a structural gap in our understanding of personality layers. Practically, while we might theorize an unlimited number of facets based on the discriminant validity of trait measurements, in reality, only a few distinct facets under each aspect are likely to be significantly different.

2.3 PERSONALITY COMPUTING

The research field of Personality Computing (PC) primarily addresses three broad tasks [166, 124]: Automatic Personality Recognition (APR), Automatic Personality Perception (APP), and Automatic Personality Synthesis (APS). To differentiate these approaches, it is important to introduce key concepts that highlight the differences in personality assessment.

Personality is often viewed through two distinct lenses: *identity* and *reputation*. Identity refers to how individuals perceive themselves, reflecting their self-concept, beliefs, and values. Reputation, on the other hand, encompasses how others perceive and evaluate an individual based on their observable behaviors and interactions. These two aspects of personality do not always align, creating a complex interplay between self-perception and external evaluation.

APR and APP are two key methods that estimate personality traits based on social signals, which reflect patterns of behavior or reactions, that machines can detect. The data sources for these signals vary from text, audio, and video, to mobile phone usage, digital footprints, and wearable technology, to name a few. By analyzing these diverse inputs, machines can form assessments of an individual's personality traits automatically, offering insights without any subjective human raters.

APR systems focus on recognizing patterns in behaviors typically associated with certain personality traits, aligning with the concept of identity. For example, an APR system might analyze video footage of a person during an interview, interpreting frequent eye contact and open gestures as indicators of extraversion. These systems are trained to predict self-reported personality trait scores, reflecting how individuals view themselves.

In contrast, APP systems concentrate on how observers perceive someone's personality based on their behavior, corresponding to the concept of reputation. It does not necessarily involve an objective analysis of a person's true personality, but rather captures how others interpret and evaluate an individual's traits. For instance, if a person speaks quickly and animatedly, an APP system might rate them high on extraversion based on observers' perceptions, regardless of the person's self-perception or behavior in different contexts. These systems are typically trained on other-reported trait scores about an individual.

While APR and APP focus on assessing and interpreting personality traits based on human behaviors and perceptions, APS explores the frontier of creating artificial personalities, employing these traits in virtual or physical agents like avatars, robots, or even smart home systems. By simulating human-like personality traits, APS enhances interactions, making them more engaging, relatable, and tailored to user needs. PC terms and relations are illustrated in Figure 2.3, through Brunswik lens of processes [25].

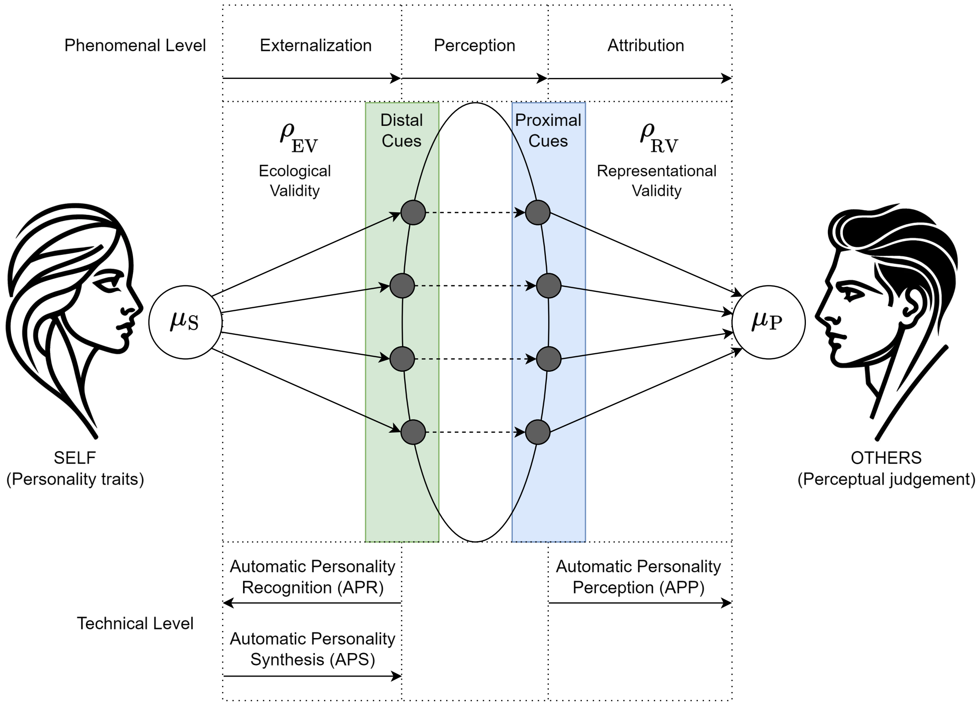

Figure 2.3: Brunswik lens of processes underlying personality computing. Adapted from [166, 124]. μ_S = inference of self-reports from distal cues; μ_P = inference of other-reports from proximal cues; ρ_{EV} = ecological validity = correlation between self-reported personality traits and distal cues; ρ_{RV} = representation validity = correlation between proximal cues and other-reported personality traits. At the Technical Level, it shows the relationship directions of APR, APP, and APS.

2.3.1 *Automatic Personality Recognition*

Automatic Personality Recognition (APR) focuses on predicting self-reported personality traits by analyzing behavioral patterns and social signals. Self-reports are one of the most common methods for assessing personality, where individuals rate themselves on scales such as the Likert-type scale (e.g., 1 = strongly disagree to 5 = strongly agree) in response to statements like "I am someone who is fascinated by art, music, or literature". These responses reflect an individual's explicit self-concept and identity, providing insight into how they perceive their own traits.

The structure and definitions of the Big Five personality traits have been extensively studied over the past 25 years, leading to significant advancements in personality assessment tools. One of the most widely used instruments is the Big Five Inventory (BFI), which measures the core qualities of each trait domain using concise and straightforward statements. The original BFI consists of 44 items, divided into five subscales representing the Big Five traits described in Section 2.2.1. While the BFI has been a reliable and valid tool in hundreds of studies, recent developments in personality research have resulted in an updated version: the Big Five Inventory-2 (BFI-2) [153].

The BFI-2 builds upon its predecessor by offering a more refined approach to measuring the Big Five traits. It includes 60 items, allowing participants to rate statements that describe their behavior or thoughts on a scale from one to five, which enables a nuanced capture of individual differences. This enhanced tool has become widely used in both clinical and research settings, supporting psychologists in crafting tailored interventions and enabling APR systems to predict personality traits with greater accuracy.

For contexts where time is limited, the Big Five Inventory-10 (BFI-10) provides a concise alternative [135]. This short version consists of 10 items, with two items for each of the Big Five traits, allowing for quick assessment of personality in situations where a full-scale evaluation is impractical. Despite its brevity, the BFI-10 offers acceptable reliability and validity, making it suitable for quick personality assessments.

2.3.2 *Automatic Personality Perception*

In contrast to APR, Automatic Personality Perception (APP) focuses on how others perceive an individual's personality based on observable behaviors. Other-reports provide insight into a person's reputation—how they are evaluated by external observers—which often differs from their self-perception. For example, observers might watch a small-group social interaction and rate a target person using tools like the BFI-2 questionnaire. These ratings reflect how others interpret the target's behavior rather than their internal self-concept.

The distinction between self-perception and others' perceptions is crucial because these perspectives do not fully overlap. While correlated, each offers unique information influenced by different biases. Self-reports may be subject to self-serving or social desirability biases, while other-reports can be shaped by familiarity or liking biases. For instance, an APP system might analyze nonverbal cues such as speaking style or body language to predict how observers perceive traits like extraversion or agreeableness.

By leveraging other-reports as benchmarks for accuracy, APP systems provide valuable insights into reputation-based assessments of personality. This perspective is particularly useful in contexts where understanding external perceptions is critical—for example, in team dynamics, customer service interactions, or leadership evaluations. Combining APP with APR can yield a more accurate representation of an individual's true personality.

2.3.3 *Involuntary Reactions*

Besides self- and other-reports, there are numerous other, more behavior-linked data sources. For example, some bio-physiological data (e.g., pulse, heart rate, skin conductance) may be indicative of people's transient affective states but also their enduring traits. Involuntary and automatic reactions offer a unique window into the subtle dimensions of human traits.

Among these reactions, blinking stands out as a particularly interesting behavior. Although primarily a physiological response meant to lubricate the eyes, the frequency and pattern of blinking can inadvertently reveal more than basic biological functions. Studies in psychophysiology suggest that blinking rates can vary with emotional states and cognitive load, potentially indicating traits such as stress reactivity, alertness, and even sociability. For instance, individuals who blink more frequently during conversations may be experiencing heightened nervousness or excitement, traits that align with higher Neuroticism or Extraversion. I will discuss blink detection in more depth in later chapters.

Furthermore, paralanguage in speech (prosody, spectral properties, intonation, etc.) is part of involuntary behavior as well as nonverbal visual clues (facial expressions, gestures, etc.).

2.3.4 *First Impressions*

People often form quick impressions of others based on minimal information, such as facial expressions and brief verbal interactions. Studies have shown that brief glimpses of a person's face can lead to judgments about traits like attractiveness and trustworthiness, while emotional expressions like a genuine smile can make someone appear friendlier [162, 173]. Similarly, the sound of a person's voice and their way of speaking influence perceptions of their intelligence, competence, and warmth. Research also indicates that even subtle language patterns, like the use of collective pronouns and fewer filler words, can significantly impact first impressions, affecting outcomes positively in contexts such as job interviews [117].

To further enhance the importance of first impressions, as minimal an exposure time as 100 milliseconds is sufficient for people to infer specific traits from facial appearance, like attractiveness, likeability, trustworthiness, competence, and aggressiveness [173]. Although longer exposure allows individuals to feel more confident and refine their judgments, these assessments tend to remain influenced by the initial snap impression. This demonstrates how effortlessly and rapidly our minds form first impressions based on minimal visual information.

APR and APP systems aim to determine the big five traits based on multimodal clues, such as smile intensity, facial mimics, or eye contact, during an initial interaction. When only such a short time interval is available, the effectiveness of these frameworks can be challenging, primarily due to the complexity and subtlety of personality traits. While 100 milliseconds of data contribute to preliminary trait inferences, more extended interactions provide a more robust and precise prediction. In this thesis, I present a method to predict personality traits from short clips.

2.4 HIERARCHY OF HUMAN SUBJECTIVITY

Human subjectivity is a feature of the mind that cannot be directly observed or objectively verified [171]. Subjective experiences are shaped by personal perspectives, reflecting the desires, beliefs, and feelings of the individual. Affect, feeling, emotion, sentiment, and opinion are often used interchangeably in the literature, with their definitions varying based on the emotion theory they align with. In the following paragraphs, I introduce the taxonomy described in [115], which represents the different aspects of human subjectivity in a hierarchy, with each layer revealing more about our mental states. Affect is the most basic level, followed by feelings and emotions, which are influenced by affect. Sentiments are more complex as they involve enduring emotional dispositions, and opinions are influenced by these emotions and sentiments, providing the most specific layer of subjectivity.

AFFECT: It represents the most abstract layer of human subjectivity, being both elusive and challenging to fully capture in language due to its foundational and non-conscious nature. It is a predecessor to feelings and emotions.

FEELINGS: They are conscious affective phenomena, shaped by personal experiences, making them inherently biographical. Each individual draws from a unique set of past experiences when labeling and interpreting their feelings.

EMOTIONS: They are preconscious social expressions of feelings and affect, shaped by cultural influences. The most popular description of emotions is based on Ekman's research, which highlights six basic emotion categories: happiness, sadness, fear, anger, disgust, and surprise. His cross-cultural studies show that these emotions are universally recognized through facial expressions. The advantage of using categories is that they align with how people intuitively label emotional expressions in daily life. However, a discrete list of emotions does not capture the entire range of emotions experienced in natural communication. Consequently, alternative descriptions emerged [150]. The dimensional emotion model defines emotional states using two dimensions, valence (evaluation) and arousal (activation), instead of discrete categories. Valence represents how positive or negative an emotion is. Positive emotions, like happiness or joy, have high valence, while negative emotions, like sadness or anger, have low valence. Activation measures the intensity or energy level of an emotion. High activation or arousal emotions include excitement and anger, while low activation or arousal emotions include relaxation and boredom. Dimensional representation allows for a broader range of emotions to be labeled. However, reducing high-dimensional emotional states to a simpler 2D space leads to a loss of some information.

SENTIMENTS: They are enduring and partly social constructs of emotions that develop over time. They represent emotional dispositions toward specific objects and evolve gradually, playing a role in decision-making. It is often conceptualized in a simplified, one-dimensional space that reflects polarity and intensity.

OPINIONS: They are personal interpretations of information that may or may not be influenced by emotions.

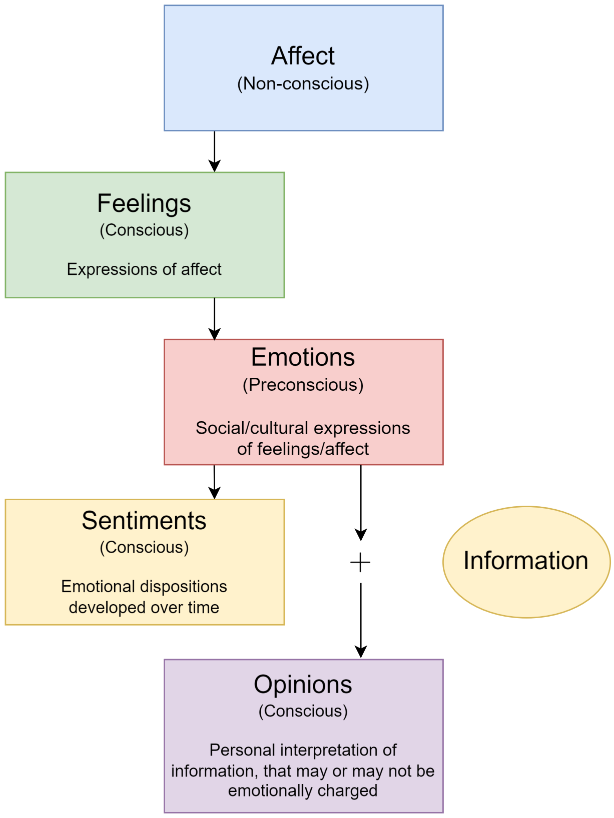

Figure 2.4: Hierarchy of human subjectivity that illustrates the conceptual relationships among affect, feelings, emotions, sentiments, and opinions. The image is based on [115].

2.4.1 *Sentiment Analysis*

Automatic Sentiment Analysis (SA) is the computational understanding of one's position, attitude, or opinion towards an entity, person or topic [150]. SA generally classifies sentiments as positive, negative, or neutral to capture the polarity of the expressed feeling. Within this dimension, the intensity of sentiment varies, showing how strongly positive or negative it is. For example, "happy" and "ecstatic" both indicate positivity but vary in intensity.

SA in language [123] is now used commercially to aggregate reviews and customer opinions at scale, providing immediate, low-cost feedback. Previously, companies relied on costly and time-consuming surveys or focus groups to gauge public sentiment. Now, with opinions expressed in multimedia formats on social media, such as spoken reviews on YouTube, SA has become a more accessible and affordable tool through crowd-sourced data.

Because of the importance of face in emotion and sentiment expression and perception, most of the vision-based affect recognition studies focus on facial expression analysis [178, 7]. To analyze facial behavior, researchers use the Facial Action Coding System (FACS) [47], which measures facial movements in terms of Action Units (AUs). AUs, being independent of interpretation, can be used for any high-level decision-making process, such as recognizing basic emotions according to Emotional FACS (EMFACS) rules [61], affective states according to the FACS Affect Interpretation Database (FACSAID) [48], and complex psychological conditions like depression or pain. The thousands of possible facial expressions can be broken down into combinations of 27 AUs and descriptors. Therefore, many studies on spontaneous facial behavior now use automatic AU recognition.

Analysis of speech to detect emotional and affective cues has a long-standing tradition [15, 1]. The boundary between sentiment and emotion analysis is often blurred. SA on spontaneous, natural speech data can be effectively realized even when word recognition rates are low. Pitch alone can provide insight into sentiment without any textual cues [108]. The sentiment we hold toward a person or object affects our interpersonal relationships and interactions. Thus, SA can significantly enhance both human-machine and human-human interactions.

Recently, the rise of a multimodal social web has shifted this landscape: vloggers share their opinions on YouTube, and user posts often include photos. Multimodal Sentiment Analysis (MSA) is the process of extracting and combining information from multiple sources, such as text, audio, and video, to understand the sentiments expressed in multimodal data [178, 63, 73]. Several challenges persist in MSA, including ambiguity in interpreting emotions due to sarcasm, humor, and cultural references, which machines struggle to comprehend. Additionally, the fast-paced nature of online opinion videos makes it difficult to segment and identify different sentiments. Further challenges lie in effectively fusing multiple modalities, as a single modality may not capture all nuances of an emotion.

2.4.2 *Group Affect*

Group Affect (GA) extends the concept of individual affect (basic emotions, feelings) to the collective experiences of groups [65, 17]. This aligns with the understanding that while affect and emotions are initially personal, they are heavily influenced by and can influence social interactions and environments. Just as individual subjectivity can change and evolve over time, GA is also dynamic. It reflects the shifting emotional states of a group based on interactions and circumstances, much like how individual emotions can fluctuate with changing personal contexts. It can impact higher layers of subjectivity such as sentiments and opinions within the group context. For instance, a pervasive positive GA can lead to more favorable opinions or sentiments about group goals or tasks. By understanding GA, we can better understand how collective emotions and moods influence group processes such as cooperation, conflict, creativity, decision-making, and overall group performance.

GA is conceptualized through two primary approaches [17]. The "top-down" approach views GA as a whole entity that impacts the emotions of individual members. Conversely, the "bottom-up" approach sees it as emerging from the cumulative affective states and traits of group members. These two perspectives help explain the dynamic and complex nature of group emotions, which can manifest in various forms such as convergence and divergence in GA, emotional culture, and GA as a dynamic process that changes over time.

Affective convergence refers to the alignment of emotional states among group members. Research suggests that several factors contribute to this phenomenon. For example, Totterdell et al. [163] found that team members who were more committed and had a positive perception of their team environment showed greater emotional alignment. Moreover, group diversity in traits like Extraversion and Neuroticism can inhibit this convergence [157].

The convergence of affect within a group significantly affects its performance. Positive GA has been linked to improved team performance, as it encourage better intragroup interactions and positive attitudes towards group tasks [29]. Another research supports this by showing that teams with higher average positive affect exhibit superior performance, indicating that group emotions play a vital role beyond individual capabilities and early planning [92].

Several antecedents like group leadership, member attributes, and interpersonal relationships influence GA. Charismatic leaders can inspire positive affect, leading to emotional convergence and enhanced group cooperation. Similarly, emotional mimicry and contagion contribute to a homogenous emotional state within the group, enhancing cooperation and collective decision-making. Conversely, affective diversity can cause conflict but also promotes creativity and problem-solving if managed effectively. Moreover, there is a dynamic and reciprocal relationship between GA and the personalities of its members. While individual traits influence the initial emotional tone of the group, the evolving GA can also impact how individuals perceive their environment and behave, potentially reinforcing or moderating their inherent traits over time. In sum, GA not only influences individual members but also shapes the group's collective behavior, ultimately affecting its effectiveness and outcomes.

3 DEEP LEARNING FOUNDATIONS

Machine Learning (ML) has emerged as a critical subfield of artificial intelligence, focusing on the development of algorithms that enable computers to learn from and make predictions based on data. Deep Learning (DL), a further specialization within machine learning, utilizes Deep Neural Networks (DNNs) to interpret data through a hierarchy of increasing complexity, from the simplest features at one level to more abstract features at higher levels. These networks excel in tasks ranging from image classification to speech processing, often achieving or even surpassing human-like performance due to advancements in computing power and dataset sizes. This section provides a high-level overview of DL concepts, focusing on key architectures and their applications, especially in HCAI. More information can be found in references [67, 22].

DL models a transformation from a set of input samples ($\mathcal{X}$) to a set of outputs ($\mathcal{Y}$), denoted as $\mathcal{X} \mapsto \mathcal{Y}$. In this thesis, I will cover the following categories:

SUPERVISED LEARNING: In supervised learning, the model predicts outcomes based on known input-output pairs. There are two main types of tasks:

- Classification tasks: Here, outputs are categorical, and the goal is to assign input to one of K predefined classes. Formally, $\mathcal{Y} = \{1, 2, \ldots, K\}$, where K is the number of classes.

- Regression tasks: In these tasks, outputs are continuous. The model predicts a real-valued output for each input. Formally, $\mathcal{Y} = \mathbb{R}^M$, where M is the number of output variables.

UNSUPERVISED LEARNING: In unsupervised learning, the goal is to discover underlying patterns and structures within the data without predefined labels or outcomes. Unlike supervised learning, there are no explicit input-output pairs to guide the learning process. Instead, the algorithm works directly with the input data $\mathcal{X}$ to uncover hidden patterns or representations. Common tasks include:

- **Clustering**: Data points are grouped into clusters that share similar characteristics.

- **Dimensionality reduction**: High-dimensional input data is compressed into a lower-dimensional representation while retaining essential information.

- **Density estimation**: The underlying probability distribution of the data is estimated.

- **Anomaly detection**: Unusual patterns or outliers in the data are identified.

In these tasks, the *output* is typically a transformed or interpreted version of the input data, rather than a separate $\mathcal{Y}$ space. The nature of this output varies depending on the specific unsupervised learning task and algorithm used.

SELF-SUPERVISED LEARNING: Self-supervised learning is a subset of unsupervised learning where the data provides the supervision. The model is trained to predict any part of its input from any other part, effectively creating its own labels from the input data itself. For instance, the model might learn to predict the next word in a sentence or to fill in missing parts of an image or a waveform. This method helps in learning useful representations of the data without requiring explicit external labels, leveraging the inherent structure of the data to learn.

In this thesis, I primarily focus on supervised learning approaches, as they align with the tasks of personality perception, sentiment estimation, and blinking detection where labeled data is available.

For training such models, the input-output pairs (samples of a dataset) can be defined as:

$$\begin{aligned} \mathbf{X} &= \begin{bmatrix} \boldsymbol{x}^{(1)} & \boldsymbol{x}^{(2)} & \dots & \boldsymbol{x}^{(N)} \end{bmatrix}^\top, \\ \mathbf{Y} &= \begin{bmatrix} \boldsymbol{y}^{(1)} & \boldsymbol{y}^{(2)} & \dots & \boldsymbol{y}^{(N)} \end{bmatrix}^\top, \end{aligned} \tag{3.1}$$

where N is the number of samples, and each $\boldsymbol{x}^{(i)}$ represents an individual input sample, while each $\boldsymbol{y}^{(i)}$ represents the corresponding label.

The structure of $\boldsymbol{x}^{(i)}$ varies depending on the nature of the data:

- **For feature vectors**: $\boldsymbol{x}^{(i)} \in \mathbb{R}^F$, where F is the number of features. Feature vectors are represented as row vectors.

- **For image data**: $\boldsymbol{x}^{(i)} \in \mathbb{R}^{H \times W \times C}$, where H is height, W is width, and C is the number of channels.

- **For sequence data**: $\boldsymbol{x}^{(i)} \in \mathbb{R}^{T \times F}$, where T is the sequence length and F is the number of features per time step. Note that T denotes the maximum sequence length; variable-length sequences are handled via zero-padding at the batch level.

Consequently, when these samples are stacked, the resulting dataset tensors have dimensions $\mathbf{X} \in \mathbb{R}^{N \times F}$ for feature vectors, $\mathbf{X} \in \mathbb{R}^{N \times H \times W \times C}$ for image data, and $\mathbf{X} \in \mathbb{R}^{N \times T \times F}$ for sequence data, where N represents the number of samples.

Similarly, the structure of $\boldsymbol{y}^{(i)}$ depends on the task:

- **For binary classification**: $y^{(i)} \in \{0, 1\}$.
- **For multi-class classification**: $y^{(i)} \in \{1, 2, ..., K\}$, where K is the number of classes.
- **For multivariate regression**: $\boldsymbol{y}^{(i)} \in \mathbb{R}^M$, where M is the number of output variables.

The training set is used to teach the model, allowing it to learn patterns and relationships within the data. Models are typically evaluated using validation and test sets. The validation set helps to fine-tune the model's hyperparameters and prevent overfitting, offering an unbiased performance evaluation. The test set, completely separate from the training and validation data, serves as the final assessment of the model's generalization ability. It simulates real-world scenarios by exposing the model to entirely new, unseen data.

The learning process is done by finding the $\boldsymbol{\theta}^*$ optimal parameter of the $f(\boldsymbol{\theta}, \cdot)$ parametric transformation on a given training set. This transformation maps the input data $\mathbf{X}$ to predicted outputs $\hat{\mathbf{Y}}$:

$$\hat{\mathbf{Y}} = f(\boldsymbol{\theta}, \mathbf{X}). \tag{3.2}$$

The goal is to minimize the difference between the predicted outputs $\hat{\mathbf{Y}}$ and the true outputs $\mathbf{Y}$ on a given training set. This is typically achieved through an optimization process that involves defining a loss function $\mathcal{L}(\mathbf{Y}, \hat{\mathbf{Y}})$ that quantifies the discrepancy between true and predicted outputs. The choice of loss function depends on the specific task. For classification tasks, Cross-Entropy (CE) is commonly used, while for regression tasks, Mean Squared Error (MSE), Mean Absolute Error (MAE), or Bell loss may be employed:

$$\mathcal{L}_{\text{CE}}(\hat{\mathbf{Y}}, \mathbf{Y}) = -\sum_{i=1}^{N} \mathbf{Y}_i \log(\hat{\mathbf{Y}}_i), \tag{3.3}$$

$$\mathcal{L}_{\text{MSE}}(\hat{\mathbf{Y}}, \mathbf{Y}) = \frac{1}{N}\sum_{i=1}^{N} (\mathbf{Y}_i - \hat{\mathbf{Y}}_i)^2, \tag{3.4}$$

$$\mathcal{L}_{\text{MAE}}(\hat{\mathbf{Y}}, \mathbf{Y}) = \frac{1}{N}\sum_{i=1}^{N} |\mathbf{Y}_i - \hat{\mathbf{Y}}_i|, \tag{3.5}$$

$$\mathcal{L}_{\text{Bell}}(\hat{\mathbf{Y}}, \mathbf{Y}) = \frac{1}{N}\sum_{i=1}^{N} \gamma \left(1 - e^{-\frac{(\mathbf{Y}_i - \hat{\mathbf{Y}}_i)^2}{2\sigma^2}}\right), \tag{3.6}$$

where $\hat{\mathbf{Y}}$ is the predicted outputs from the model, $\mathbf{Y}$ stands for the actual or true outputs in the data, N denotes the number of samples in the current set (training/batch), while σ is the deviation parameter, and γ is a scale parameter for Bell loss.

The objective function $J(\boldsymbol{\theta})$ aggregates this loss over the entire training set:

$$J(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}(\boldsymbol{y}^{(i)}, f(\boldsymbol{\theta}, \boldsymbol{x}^{(i)})) + \lambda R(\boldsymbol{\theta}), \tag{3.7}$$

where N is the number of samples, $R(\boldsymbol{\theta})$ is a regularization term, and $\lambda \geq 0$ is its weight. The optimization process aims to find the parameters $\boldsymbol{\theta}^*$ that minimize this objective function:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}). \tag{3.8}$$

The parametric function $f(\boldsymbol{\theta}, \cdot)$ can be structured as an Artificial Neural Network (ANN), which transforms input data into outputs through a series of computations organized into layers. An ANN consists of three types of layers:

- **Input layer**: Receives the raw data or pre-calculated input features and passes it to the next layer.
- **Hidden layers**: One or more intermediate layers where the main computations occur by applying weights, biases, and activation functions to learn patterns from the input data.
- **Output layer**: Generates predictions or classifications based on the processed information from the hidden layers.

When $f(\boldsymbol{\theta}, \cdot)$ includes multiple hidden layers, it is referred to as a Deep Neural Network (DNN).

To introduce non-linearity and enable the model to learn complex relationships, each layer typically includes at least one activation function. These functions, denoted as $g(\cdot)$, are applied element-wise to the output of each layer's linear transformation. Common activation functions used in this thesis include:

$$\begin{aligned}
\mathrm{linear}(v) &= v\,, \\
\sigma(v) &= \frac{1}{1+e^{-v}}\,, \\
\mathrm{ReLU}(v) &= \max(0, v)\,, \\
\mathrm{ELU}(v) &= \begin{cases} v & v > 0 \\ \alpha(e^{v} - 1) & v \leq 0 \end{cases}\,,
\end{aligned} \tag{3.9}$$

where $v \in \mathbb{R}$ and α is typically set to 1.

The choice of activation function can significantly impact the network's ability to learn and generalize. For instance, ReLU is often used in hidden layers due to its simplicity and effectiveness in mitigating the vanishing gradient problem, while sigmoid or softmax functions are commonly employed in the output layer for classification tasks.

A Feed-Forward Network (FFN) is a type of ANN where information flows in one direction: from the input layer, through one or more hidden layers, to the output layer. There are no cycles or loops in the network, and each layer transforms its input into an output using a set of learnable parameters and an activation function. Formally, FFN can be expressed as:

$$\mathbf{H}_0 = \mathbf{X}_b, \tag{3.10}$$

$$\mathbf{H}_d = f_d(\boldsymbol{\theta}_d, \mathbf{H}_{d-1}), \quad d = 1, \ldots, D\,, \tag{3.11}$$

where $\mathbf{H}_0$ is the input matrix, $\mathbf{X}_b$ denotes either a single sample or a batch of inputs fed through the network, $\mathbf{H}_d$ represents the output (activations) of the d-th layer, and $f_d(\boldsymbol{\theta}_d, \cdot)$ is the transformation applied at layer d, parameterized by $\boldsymbol{\theta}_d$, which includes the weights and biases of that layer. The set of all parameters in the network is denoted as $\boldsymbol{\theta} = \{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \ldots, \boldsymbol{\theta}_D\}$.

A Fully-Connected Network (FCN), also known as a dense neural network, is a specific type of FFN where each neuron in one layer is connected to every neuron in the next layer. A fully-connected layer is defined as:

$$f_d(\boldsymbol{\theta}_d, \mathbf{H}_{d-1}) = g_d(\mathbf{W}_d \mathbf{H}_{d-1} + \mathbf{b}_d)\,, \tag{3.12}$$

where $\mathbf{W}_d$ is the weight matrix for layer d, with dimensions determined by the number of neurons in layers $d-1$ and d, $\mathbf{b}_d$ is the bias vector for layer d, and $g_d(\cdot)$ is the activation function applied element-wise.

Backpropagation is a method used to compute the gradients of the objective function $J(\boldsymbol{\theta})$ with respect to the network parameters $\boldsymbol{\theta}$, enabling optimization algorithms (e.g., gradient descent) to iteratively update these parameters. It consists of two main phases: the forward pass and the backward pass.

In the forward pass, input data $\mathbf{X}$ is propagated through the network layer by layer (according to Equation 3.10-Equation 3.12) to compute the predicted outputs $\hat{\mathbf{Y}}$. Then, the $\mathcal{L}(\mathbf{Y}, \hat{\mathbf{Y}})$ loss function is calculated.

The backward pass computes gradients of the objective function $J(\boldsymbol{\theta})$ with respect to each parameter in the network using the chain rule. The chain rule allows us to decompose these gradients into a series of partial derivatives that can be computed efficiently layer by layer.

The gradient of $J(\boldsymbol{\theta})$ with respect to the parameters of layer d, denoted as $\boldsymbol{\theta}_d = \{\mathbf{W}_d, \mathbf{b}_d\}$, is computed as:

$$\frac{\partial J}{\partial \boldsymbol{\theta}_d} = \frac{\partial J}{\partial \mathbf{H}_d} \cdot \frac{\partial f_d(\boldsymbol{\theta}_d, \mathbf{H}_{d-1})}{\partial \boldsymbol{\theta}_d}. \tag{3.13}$$

where $\frac{\partial J}{\partial \mathbf{H}_d}$ is the gradient of the loss with respect to the activations at layer d, which is propagated from subsequent layers, and $\frac{\partial f_d(\boldsymbol{\theta}_d, \mathbf{H}_{d-1})}{\partial \boldsymbol{\theta}_d}$ is the local gradient of the transformation at layer d with respect to its parameters.

Using the chain rule, backpropagation propagates gradients backward through layers. The gradient of $J(\boldsymbol{\theta})$ with respect to the activations of layer $d-1$ is given by:

$$\frac{\partial J}{\partial \mathbf{H}_{d-1}} = \frac{\partial J}{\partial \mathbf{H}_d} \cdot \frac{\partial f_d(\boldsymbol{\theta}_d, \mathbf{H}_{d-1})}{\partial \mathbf{H}_{d-1}}, \tag{3.14}$$

where $\frac{\partial f_d(\boldsymbol{\theta}_d, \mathbf{H}_{d-1})}{\partial \mathbf{H}_{d-1}}$ is the local gradient of layer d with respect to its input. This recursive computation starts at the output layer (D-th) and proceeds backward through all layers until reaching the input layer (0-th).

At the output layer, the activations $\mathbf{H}_D$ correspond to the predicted outputs $\hat{\mathbf{Y}}$. The gradient of the objective function $J(\boldsymbol{\theta})$ with respect to these activations is computed directly from the loss function:

$$\frac{\partial J}{\partial \mathbf{H}_D} = \frac{\partial \mathcal{L}(\mathbf{Y}, \hat{\mathbf{Y}})}{\partial \hat{\mathbf{Y}}}\,, \tag{3.15}$$

where $\mathcal{L}(\mathbf{Y}, \hat{\mathbf{Y}})$ is the loss function. This serves as the starting point for backpropagation.

The gradients are then propagated backward through each layer using the chain rule. For a hidden layer $d < D$, we compute:

$$\frac{\partial J}{\partial \mathbf{H}_d} = g'_d(\mathbf{H}_d) \odot \left(\frac{\partial J}{\partial \mathbf{H}_{d+1}} \cdot \mathbf{W}_{d+1}^\top \right), \tag{3.16}$$

where $g'_d(\mathbf{H}_d)$ is the derivative of the activation function at layer d applied element-wise, $\odot$ denotes the element-wise multiplication, $\frac{\partial J}{\partial \mathbf{H}_{d+1}}$ is the gradient propagated from the next layer, and $\mathbf{W}_{d+1}^\top$ is the transpose of the weight matrix from layer $d+1$. This recursive process continues until gradients are computed for all layers, including the input layer.

Once all gradients are computed, they are used to update the parameters of the network during training. The general update rule for gradient descent is defined as:

$$\boldsymbol{\theta}_{i+1} \leftarrow \boldsymbol{\theta}_i - \alpha \left. \frac{\partial J}{\partial \boldsymbol{\theta}} \right|_{\boldsymbol{\theta}=\boldsymbol{\theta}_i}, \tag{3.17}$$

where $\boldsymbol{\theta}_i$ represents the parameters at iteration i, $\alpha > 0$ is the learning rate, and $\frac{\partial J(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$ is the gradient of the objective function with respect to the parameters.

The classic gradient descent has its flaws, it can get stuck in saddle points, increased sensitivity to hyperparameters, or slow convergence speed to name a few. Several improvements have been proposed, such as the Adam optimizer [91], which uses an adaptive learning rate and momentum. The AdamW variant decouples weight decay from gradient updates for improved training stability [104]. The RAdam optimizer improves upon Adam by rectifying its variance in the adaptive learning rate, leading to more stable and reliable training [102]. Additionally, Stochastic Gradient Descent with Restarts (SGDR) [105] utilizes a cyclical learning rate strategy with cosine annealing, helping the model escape local minima and potentially find better solutions.

The specific form of these updates depends on the type of layer. For the sake of completeness, in the case of a FCN, for a given layer d, with parameters $\boldsymbol{\theta}_d = \{\mathbf{W}_d, \mathbf{b}_d\}$, gradient descent updates are performed as follows:

$$\mathbf{W}_d \leftarrow \mathbf{W}_d - \alpha \frac{\partial J}{\partial \mathbf{W}_d}, \tag{3.18}$$

$$\mathbf{b}_d \leftarrow \mathbf{b}_d - \alpha \frac{\partial J}{\partial \mathbf{b}_d}, \tag{3.19}$$

where $\alpha > 0$ is the learning rate, $\frac{\partial J}{\partial \mathbf{W}_d} = \frac{\partial J}{\partial \mathbf{H}_d}(\mathbf{H}_{d-1})^\top$ is the gradient of the objective function with respect to weights, and the $\frac{\partial J}{\partial \mathbf{b}_d} = \frac{\partial J}{\partial \mathbf{H}_d}$ is the gradient with respect to biases.

These updates are applied iteratively across all layers and iteratively update the parameters over multiple training iterations (epochs) until convergence or a stopping criterion is met.

During the training of machine learning models, two common challenges can arise:

UNDERFITTING: It occurs when the model is too simple, capturing too little of the complexity of the data, and missing the broader patterns. As a result, the model performs poorly both on the training and test data.

OVERFITTING: It occurs when the model is too complex, and learns not only the underlying patterns but also the noise and random fluctuations in the training data. This leads to excellent performance on the training set but poor generalization to unseen test data.

The solution lies in finding a balance: ensuring the model is complex enough to learn the essential patterns without adhering too closely to the specific data it was trained on. To achieve this balance by guiding the model towards generalization rather than memorization, many techniques were developed:

EARLY STOPPING: We halt the training of a model when it no longer shows improvement on the validation set after several epochs. This prevents the model from learning the noise in the data, which can lead to overfitting.

L1 REGULARIZATION: This regularization technique adds the sum of the absolute values of the model's weights to the loss function. This penalty promotes sparsity by driving some weights to exactly zero, effectively performing feature selection. In Equation 3.7, $R(\boldsymbol{\theta})$ is defined as:

$$R(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_1 = \sum_j |\theta_j| \,, \tag{3.20}$$

where $\|\boldsymbol{\theta}\|_1$ is the L1 norm, and θ_j represents individual parameters in $\boldsymbol{\theta}$.

L2 REGULARIZATION: This regularization technique adds the sum of the squared values of the model's weights to the loss function. This penalty encourages smaller but non-zero weights, distributing importance more evenly across features. In Equation 3.7, $R(\boldsymbol{\theta})$ is defined as:

$$R(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_2 = \sum_j \theta_j^2 \,, \tag{3.21}$$

where $\|\boldsymbol{\theta}\|_2$ is the L2 norm, and θ_j represents individual parameters in $\boldsymbol{\theta}$.

DROPOUT: This technique [155] randomly ignores certain neurons during training, forcing the network to learn more robust features that are not dependent on any small set of neurons.

BATCH NORMALIZATION: This method normalizes the inputs for each layer within the network. It helps stabilize the learning process by ensuring that each input layer has a similar data distribution, which can make training faster and more stable.

DATA AUGMENTATION: A technique used to artificially expand the size and diversity of a training dataset by applying transformations such as rotations, translations, scaling, flipping, or noise addition to the original data.

CROSS-VALIDATION: This technique involves dividing the data into several smaller sets. The model is trained on each set and tested on the remaining data. This helps ensure that the model works well not just on one data chunk but across various subsets of the data.

Modern DL frameworks, such as PyTorch and TensorFlow, leverage autograd (automatic differentiation) to simplify the computation of gradients during backpropagation. Autograd automates the differentiation process by dynamically constructing a computational graph as operations are performed in the forward pass. Each operation is tracked, and its gradient is computed in the backward pass using the chain rule. This eliminates the need for manually deriving and implementing gradient calculations, even for highly complex functions or architectures. As a result, researchers and practitioners can focus on designing innovative models and loss functions without worrying about the intricacies of gradient computation. In the upcoming sections, only the forward pass will be explicitly described to maintain clarity and simplicity, while relying on autograd to handle gradient computations in an automated fashion.

Next, I will briefly introduce convolutional layers (Section 3.1), essential for analyzing audio and visual data effectively. Then, I will show, how these DNNs handle time-based data by introducing transformers (Section 3.2), crucial for tasks like speech and language processing.

3.1 CONVOLUTIONAL NEURAL NETWORKS

Convolutional Neural Networks (CNNs), also known as ConvNets, share a foundational structure with ordinary DNNs but introduce localized connectivity between neurons. This setup allows for the sharing of parameters, significantly reducing the number of parameters that need to be estimated. These shared weights act as filters within the network. This local connection strategy enables CNNs to excel at extracting higher-level features from data that has temporal or spatial relationships.

A Conv1D (one-dimensional convolution) is a type of convolutional layer designed to process sequential data, such as time series or text, where the input has one spatial dimension (e.g., time steps or sequence length) and multiple features per step. The convolution operation slides a kernel (filter) across the input along this single spatial dimension to extract local patterns. The Conv1D layer can be formally defined as:

$$\mathbf{H}_d = \text{Conv1D}(\mathbf{H}_{d-1}, \{\mathbf{W}_d, \mathbf{b}_d\}) = g_d((\mathbf{W}_d * \mathbf{H}_{d-1}) + \mathbf{b}_d), \qquad (3.22)$$

where $\mathbf{H}_{d-1} \in \mathbb{R}^{T \times F}$ is the input to the Conv1D layer (the activations of the previous layer, or the input tensor $\mathbf{H}_0 = \mathbf{X}_b$), $\mathbf{H}_d \in \mathbb{R}^{T' \times K}$ is the output (activations) of the Conv1D layer, where T is the sequence length and F is the number of features per time step. $\mathbf{W}_d \in \mathbb{R}^{S \times F \times K}$ is the kernel (filter) tensor, where S is the kernel size (number of time steps covered by each filter), F matches the number of input features, K is the number of filters (output channels). $\mathbf{b}_d \in \mathbb{R}^K$ is the bias vector for each filter and is added to each position in the output sequence along the filter dimension. $T' = \frac{T-S}{\text{stride}} + 1$ is the length of the output sequence, where stride determines how far the kernel moves along the sequence at each step, and $*$ denotes the convolution operation, which slides each filter across the input sequence. Note that activation functions (e.g., ReLU) are not part of the convolution operation itself but are typically applied after the convolution as part of the layer definition.

In the field of CV, 2D convolutions are applied to images. It involves sliding a matrix of weights, known as a kernel, across the height and width of an image. This operation transforms the spatial data by filtering and emphasizing certain features, such as edges or textures, making it essential for tasks like image classification, object detection, and image segmentation. The efficiency of 2D convolutions in capturing the local dependencies within images allows DL models to learn complex hierarchical visual features, which are also utilized in HCAI. The Conv2D layer can be formally defined as:

$$\mathbf{H}_d = \text{Conv2D}(\mathbf{H}_{d-1}, \{\mathbf{W}_d, \mathbf{b}_d\}) = g_d((\mathbf{W}_d * \mathbf{H}_{d-1}) + \mathbf{b}_d), \qquad (3.23)$$

where $\mathbf{H}_{d-1} \in \mathbb{R}^{H \times W \times C}$ is the input to the Conv2D layer (the activations of the previous layer, or the input tensor $\mathbf{H}_0 = \mathbf{X}_b$), $\mathbf{H}_d \in \mathbb{R}^{H' \times W' \times K}$ is the output (activations) of the Conv2D layer, where $H' = \frac{H - S_h}{stride_h} + 1$ and $W' = \frac{W - S_w}{stride_w} + 1$ are the height and width of the output feature map. $\mathbf{W}_d \in \mathbb{R}^{S_h \times S_w \times C \times K}$ is the kernel (filter) tensor, where S_h and S_w are the height and width of each filter, C matches the number of input channels, K is the number of filters (output channels). $\mathbf{b}_d \in \mathbb{R}^K$ is the bias vector for each filter and is added to each position in the output feature map along the filter dimension. Strides ($stride_h$, $stride_w$) control how far the kernel moves along the height and width dimensions, and $*$ denotes the 2D convolution operation.

In CNNs, pooling layers are strategically placed between the convolutional layers to streamline the data. These layers work by condensing the data, reducing its complexity. This not only speeds up the processing but also prevents the network from learning irrelevant details. The architecture typically concludes with one or more fully connected layers. These layers take all the features gathered and refined by the network and synthesize them into the final outputs, such as identifying objects in an image.

CNNs have evolved into various specialized architectures, each designed to tackle different challenges in image processing. Some of the well-known implementations include Residual Network (ResNet) [74] and Dense Convolutional Network (DenseNet) [79].

RESIDUAL NETWORK: It is designed to solve a common problem in very deep networks called the vanishing gradient, where layers deeper in the network train very slowly because the gradient of the loss function decreases exponentially with depth, making the network hard to train. The architecture of ResNet is built around residual blocks, which are the fundamental building units of the network. These blocks are combined with shortcut connections which allow gradients to flow through the network directly, bypassing several layers.

Instead of forcing layers to approximate a complex function $f(\boldsymbol{\theta}, \cdot)$, residual blocks reformulate this as:

$$f(\boldsymbol{\theta}, \boldsymbol{x}) = f_{\text{residual\_block}}(\boldsymbol{\theta}, \boldsymbol{x}) + \boldsymbol{x}, \tag{3.24}$$

where $f_{\text{residual\_block}}(\boldsymbol{\theta}, \boldsymbol{x})$ is the residual mapping learned by the block, and $\boldsymbol{x}$ is the input passed through a shortcut connection. The shortcut connection bypasses one or more layers, directly adding the input $\boldsymbol{x}$ to the output of $f_{\text{residual\_block}}(\boldsymbol{\theta}, \boldsymbol{x})$. This design ensures that gradients can flow backward through the network without vanishing, even in very deep architectures.

The ResNet architecture begins with an initial convolutional layer followed by max pooling to reduce spatial dimensions. The network then stacks multiple stages, each containing several residual blocks. The number of blocks per stage varies depending on the depth of the network (e.g., ResNet-18, ResNet-34, ResNet-50). As the network progresses through these stages, downsampling is performed using convolutions with strides or pooling operations. After passing through all residual blocks, Global Average Pooling (GAP) aggregates spatial information across feature maps. Finally, a fully connected layer maps these features to class probabilities for classification tasks.

This innovation has made ResNet—and its variants—a preferred choice in many applications requiring DL because it enables the construction of much deeper networks that can learn more complex patterns without a significant increase in training difficulty.

DENSE CONVOLUTIONAL NETWORK: Unlike traditional CNNs, where layers are connected sequentially, DenseNet introduces a novel connectivity pattern where each layer is directly connected to all preceding layers within a block. This dense connectivity reduces redundancy in feature learning and alleviates gradient flow issues. The architecture of DenseNet alternates between two key components: dense blocks and transition layers.

Within a dense block each layer receives the concatenated outputs of all previous layers as its input and the output of each layer is passed forward to all subsequent layers in the block. For a layer d in a dense block:

$$\mathbf{H}_d = f_{dense\_block}(\boldsymbol{\theta}, [\mathbf{H}_0, \mathbf{H}_1, \ldots, \mathbf{H}_{d-1}]), \tag{3.25}$$

where $[\mathbf{H}_0, \mathbf{H}_1, \ldots, \mathbf{H}_{d-1}]$ represents the concatenated outputs of all layers up to $d-1$ along the feature dimension, and f_{dense_block} is a composite function of operations like batch normalization, ReLU activation, and convolution. This design ensures efficient feature propagation and encourages feature reuse throughout the block.

Between consecutive dense blocks, DenseNet incorporates transition layers to manage computational complexity and control the growth of feature maps. These transition layers consist of a batch normalization layer followed by a 1×1 convolution to reduce the number of channels (dimensionality reduction) and a pooling operation (typically 2×2 average pooling) to halve the spatial dimensions of the feature maps.

The overall architecture begins with an initial convolutional layer to extract low-level features. Multiple dense blocks are then stacked sequentially, with transition layers inserted between them. After the final dense block, GAP aggregates spatial information across feature maps. Finally, a fully connected layer maps the pooled features to class probabilities for classification tasks.

DenseNet is highly efficient due to its ability to reuse features through dense connections, reducing redundancy and requiring fewer parameters compared to traditional architectures like ResNet. Its compact design improves gradient flow and enables better feature learning, resulting in higher accuracy on tasks such as image classification.

3.2 TRANSFORMER ARCHITECTURE

Transformers [165] have revolutionized NLP by overcoming limitations seen in earlier models like Recurrent Neural Networks (RNNs) and Long Short-Term Memorys (LSTMs), especially regarding parallelization and managing long-range dependencies. Unlike traditional models that process sequences step-by-step, transformers handle entire sequences simultaneously. This approach significantly enhances their efficiency in complex NLP tasks such as translation, summarization, and text generation.

The core of a transformer is its encoder, built from multiple identical layers. Each layer has two main components: a multi-head self-attention mechanism and a position-wise fully connected feed-forward network. Positional encodings are added to the input to retain the sequence order, an essential aspect not directly captured by the attention mechanism. The self-attention mechanism allows the encoder to consider different positions of the input simultaneously, enhancing the model's understanding and contextual awareness.

Normalizations and residual connections around these sub-layers aid in effective and efficient training by stabilizing the learning process. The encoder outputs feature representations that are rich and contextually aware, useful for varied downstream tasks, or feeding into a decoder for generating new sequences. The decoder, mirroring the encoder's structure, adds another layer of attention to focus on relevant parts of the encoder's output, essential for generating accurate sequences.

A pivotal example of a transformer application is Bidirectional Encoder Representations from Transformers (BERT) [37], which uses a bidirectional approach to preprocessing text, gaining insights from both the left and right contexts of each token across all layers. This methodology allows BERT to achieve a deeper understanding of language context, dramatically enhancing performance across diverse NLP tasks.

In this thesis, I focus specifically on the encoder component of the transformer architecture illustrated in Figure 3.1, investigating its ability to create rich contextual embeddings. This is particularly beneficial for tasks that involve classification, regression, and information extraction.

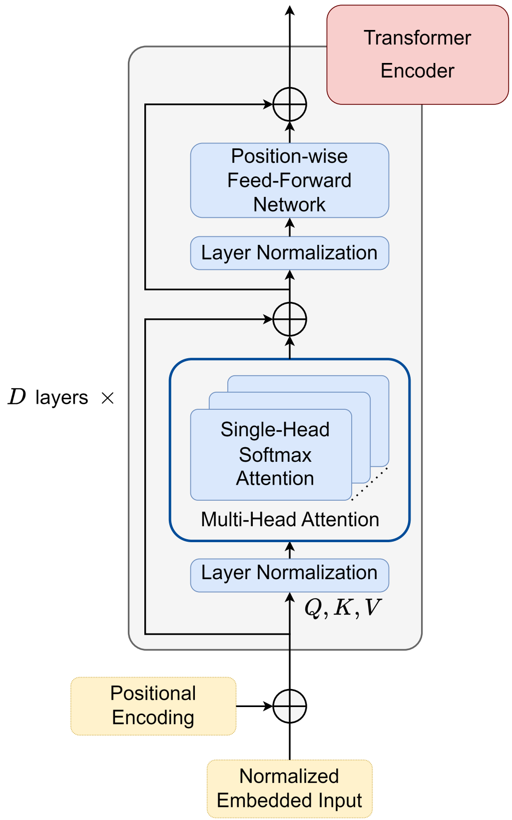

Figure 3.1: Transformer Encoder architecture.

3.2.1 *Input Projection and Positional Encoding*

In the Transformer architecture, the first crucial step is to transform the input data into a format that the network can effectively process. While Transformers are often highlighted for their impact on text processing—turning input tokens into embeddings—this concept is broadly applicable to other types of data as well. For example, inputs could be feature representations derived from audio signals or sequences of hidden states from DNNs.

In these cases, linear projection or $1D$ convolution is used to transform these various inputs into a uniform format. This transformation adjusts the dimensions and scales of the inputs to match what the Transformer's subsequent layers expect. Linear projection is used, for example in Visual Transformer [45], where the input image is split into fixed-size patches, then flattened and linearly projected into embeddings. This approach turns the spatial image data into a form that the standard transformer model can process.

However, the $1D$ convolutional operator can be used when the input sequences themselves carry spatial or temporal patterns that a simple linear projection might not capture effectively, e.g., when dealing with text, sound, or outputs from other neural networks. It can help in extracting local and positionally invariant features before these are fed into more complex architectures like Transformers. Let $\boldsymbol{x}_t$ be the projected feature vector at time step t that will be fed into subsequent layers of the Transformer.

If the input is a sequence of tokens:

$$\boldsymbol{x}_t = \boldsymbol{E}_{\text{token}}[\text{token}_t] + \boldsymbol{E}_{\text{position}}, \tag{3.26}$$

where $\boldsymbol{E}_{\text{token}}$ is the token embedding matrix, token_t is the token at position t, and $\boldsymbol{E}_{\text{position}}$ is the positional encoding vector.

In case of linear projection:

$$\boldsymbol{x}_t = \mathbf{W}_{\text{proj}}\boldsymbol{f}_t + \mathbf{b}_{\text{proj}}, \tag{3.27}$$

where $\mathbf{W}_{\text{proj}}$ is the weight matrix of the linear projection, which transforms the dimensionality of the input features to match that of the Transformer's internal architecture. $\boldsymbol{f}_t$ is the original feature vector at time step t, and $\mathbf{b}_{\text{proj}}$ is the bias vector of the linear projection, which is added to the linearly transformed feature vector to ensure the transformed data is appropriately offset in the embedding space.

Alternatively, for more complex input features:

$$\boldsymbol{x}_t = \text{Conv1D}(\boldsymbol{f}_t, \mathbf{W}_{\text{conv}}, \mathbf{b}_{\text{conv}}) + \boldsymbol{E}_{\text{position}}, \tag{3.28}$$

where $\text{Conv1D}(\boldsymbol{f}_t, \mathbf{W}_{\text{conv}}, \mathbf{b}_{\text{conv}})$ represents the convolution operation applied to the input feature vector $\boldsymbol{f}_t$ at time step t. $\mathbf{W}_{\text{conv}}$ denotes the weights of the convolutional filter, and $\mathbf{b}_{\text{conv}}$ represents the bias.

Since transformers do not process data sequentially, positional encodings are added to provide a sense of sequence order, which helps the model to understand the sequence context. The positional encoding vector can be expressed as:

$$\boldsymbol{E}_{\text{position},t}[i] = \begin{cases} \sin(t/10000^{i/d}) & \text{if } i \bmod 2 = 0 \\ \cos(t/10000^{i/d}) & \text{if } i \bmod 2 \neq 0\,, \end{cases} \tag{3.29}$$

where i is the dimension and d is the total number of dimensions in the model.

3.2.2 *Multi-Head Attention Mechanism*

At the core of the encoder is the Attention mechanism, also known as Self-Attention, that calculates a set of attention weights for each step of the output sequence. These weights assess the significance of each segment of the input, regardless of their positional distance, focusing the computational resources of the model on the areas most relevant to each output segment.

Attention mechanism can be defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}\,, \tag{3.30}$$

where **Q**, **K**, and **V** are the queries, keys, and values, respectively, computed from the input, and d_k is the dimensionality of the keys.

Multi-Head Attention (MHA) is an extension of the attention mechanism where the attention function is run in parallel across different "heads". Unlike traditional single attention mechanisms that focus on one aspect of the data, MHA allows the model to explore different subspace representations of the input simultaneously. This approach means that the Transformer can attend to different positions of the input sequence, capturing a broader range of dependencies within the data. It can be defined as:

$$\begin{aligned} \text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}^O, \\ \text{head}_i &= \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V)\,, \end{aligned} \tag{3.31}$$

where Concat refers to the concatenation of the output from each attention head. This operation combines the diverse perspectives gathered by each head into a single output vector, which enriches the model's understanding and representation capabilities. $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V$ are the weight matrices that project the queries, keys, and values in each head. $\mathbf{W}^O$ is a linear transformation weight matrix applied to the concatenated output. This matrix integrates and projects the diverse information captured by each head back into a consistent space suitable for subsequent layers or output purposes.

3.2.3 *Encoder Sublayers*

The standard encoder layer combines the MHA with two further components that are present in every layer of the models used in this thesis: a position-wise FFN and a residual connection with layer normalization:

$$\text{FFN}(\boldsymbol{x}) = \text{ReLU}(\boldsymbol{x}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2\,, \tag{3.32}$$

$$\text{LayerNorm}(\boldsymbol{x} + \text{Sublayer}(\boldsymbol{x}))\,, \tag{3.33}$$

where Sublayer($\boldsymbol{x}$) is the function implemented by the sub-layer itself. The layer normalization helps stabilize the learning process by normalizing the layer inputs to have zero mean and unit variance, which combats the internal covariate shift problem (where the distribution of network activations varies during training). This leads to smoother and faster convergence during training. Residual connections allow the input of each layer to be added to its output, mitigating the vanishing gradient problem by providing an alternative pathway for gradients during backpropagation.

3.2.4 *Linear Complexity Attention Mechanism*

Linear Attention reformulates and generalizes the softmax attention mechanism by introducing the notation $\mathrm{sim}(q,k)$, the similarity function between the query and the key as the exponentiation of the dot product:

$$\mathrm{sim}(q,k) = \exp\left(\frac{q^\top k}{\sqrt{d_k}}\right). \tag{3.34}$$

Equation 3.30 implements a specific form of self-attention called softmax attention where the similarity score is the exponential of the dot product between a query and a key. Given that subscripting a matrix with i returns the i^{th} row as a vector, [89] generalized attention equation for any similarity function as follows:

$$\mathbf{V}_i' = \frac{\sum_{j=1}^{N} \mathrm{sim}\left(\mathbf{Q}_i, \mathbf{K}_j\right) \mathbf{V}_j}{\sum_{j=1}^{N} \mathrm{sim}\left(\mathbf{Q}_i, \mathbf{K}_j\right)}. \tag{3.35}$$

The only constraint for the $\mathrm{sim}(\cdot)$ function is its non-negativity. In [89], the feature map $\phi(\cdot) = \mathrm{ELU}(\cdot) + 1$ (using the ELU activation from Equation 3.9) is used to rewrite Equation 3.35 as:

$$\mathbf{V}_i' = \frac{\phi(\mathbf{Q}_i)^\top \sum_{j=1}^{N} \phi(\mathbf{K}_j)\mathbf{V}_j^\top}{\phi(\mathbf{Q}_i)^\top \sum_{j=1}^{N} \phi(\mathbf{K}_j)}. \tag{3.36}$$

In a vectorized form it can be expressed as:

$$\left(\phi(\mathbf{Q})\phi(\mathbf{K})^\top\right)\mathbf{V} = \phi(\mathbf{Q})\left(\phi(\mathbf{K})^\top \mathbf{V}\right). \tag{3.37}$$

Considerable speed gain arises since one may compute $\sum_{j=1}^{N} \phi(\mathbf{K}_j)\mathbf{V}_j^\top$ and $\sum_{j=1}^{N} \phi(\mathbf{K}_j)$ only once and reuse these quantities in every query, making it $\mathcal{O}(N)$ complexity regarding time and memory.

4 MULTIMODAL APPROACHES IN AI

I explore the integration of multimodal information by combining audio, video, and text data. This approach significantly enriches the analysis by incorporating a variety of perspectives and details, which is particularly advantageous for tasks requiring deep contextual understanding. By merging these distinct yet complementary sources, I enhance the robustness and precision of the models, thereby improving their performance across various applications.

The next section of the thesis will delve into the specific features extracted from each modality, and I will briefly describe the techniques as well.

4.1 ACOUSTIC MODALITY

EGEMAPS: Audio frames are short, fixed-length segments of digital audio, typically 20-40 milliseconds long, used in signal processing to divide continuous sound into discrete units for analysis and manipulation. For acoustic features, I used a de-facto standard feature set called extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) [53]. The audio signals are extracted from the videos using FFmpeg with 44100 sampling frequency. Then, eGeMAPS Low-Level Descriptors (LLDs) are generated using OpenSMILE [52], resulting in 25 features for every audio frame. This feature set contains the F0 semitone, loudness, spectral flux, MFCC, jitter, shimmer, F1, F2, F3, alpha ratio, Hammarberg index, slope V0 features. Furthermore, many statistical functions are applied to these LLDs considering voiced and unvoiced regions, resulting in 88 features for every sample. Standardization is applied as a preprocessing step.

WAV2VEC: A multi-layer CNN called Wav2Vec [142] is used as a deep audio feature extractor. This model was pre-trained on 960 hours of audio data from the Librispeech dataset [121], which consists of read English speech. The training process employed a self-supervised approach, utilizing Contrastive Predictive Coding (CPC) to learn representations from unlabeled audio data. CPC enables the model to predict future audio samples based on past context, allowing it to capture meaningful acoustic features without requiring labeled data. The Wav2Vec model generates a 512-dimensional context representation for each audio segment, effectively encoding the complex patterns and characteristics of the raw audio waveform into a compact, high-level feature vector. These learned representations can then be used for various downstream audio processing tasks, potentially improving performance compared to traditional hand-crafted features.

4.2 VISUAL MODALITY

FRAMES: One of the input features for some models includes raw video frames, which are utilized in their RGB (Red, Green, Blue) format. This format is a standard method for representing color in digital images, where each color image is composed of pixels. Each pixel in these images is defined by three intensity values, which represent the red, green, and blue components. These components are essential for color construction and detail in visual data. Each RGB frame is structured as $\boldsymbol{F} \in \mathbb{R}^{H \times W \times C}$, where $\boldsymbol{F}$ denotes the frame, H is the height of the frame, W is the width, and C represents the channels. Note that C equals to three for RGB. The use of raw frames allows the model to leverage spatial and color information directly from the video.

ACTION UNITS: Action Units (AUs) are used to avoid introducing unnecessary sources of bias to the model by excluding the background and other appearance-based information that could be retrieved from raw visual data such as gender, age, or ethnicity. AUs [13] construct facial expressions encoded in the Facial Action Coding System (FACS). These AUs can be described in two ways: presence (indicating whether a particular AU is detected in a given time frame) and intensity (indicating how intense an AU is at a given time frame). For extracting AUs, I utilized OpenFace [14], an open-source toolkit, which provides a comprehensive set of AUs. Additionally, I also employed OpenGraphAU [107], a framework designed to enhance the analysis of facial expressions by integrating graph-based representations. Table A.1 shows the available AUs in OpenFace that I used for the experiments. I extracted 35 AU presence and intensity values per frame, which were standardized before further usage. A possible alternative to OpenFace is FACET (used in [73]), however, this module from the commercial iMotion software is not publicly available for research purposes.

FAB-NET: Facial Attributes-Net (FAb-Net) [172], a visual deep feature extractor, trained on VoxCeleb1 and VoxCeleb2 video datasets. The network learns an embedding that encodes facial attributes like landmarks, poses, and emotions without any labels in a self-supervised manner. It is used on all the frames of a given video (30 fps) where the speaker's face is detected. The extracted embedding has a size of 256 for each frame.

4.3 TEXTUAL MODALITY

BERT: Bidirectional Encoder Representations from Transformers (BERT) [37] is a powerful transformer architecture with self-attention that learns contextual relations between words (or sub-words) in a text. It is bidirectional, which exploits the context from both left and right to extract patterns or representations during training. BERT is applied for extracting high-level, context-dependent, 768-dimensional representations from textual data.

ROBERTA: Robustly optimized BERT approach (RoBERTa) [103] is pre-trained on raw texts using a set of large English-language datasets without human annotations. As tokenization, Byte-Pair Encoding (BPE) is applied to the transcript, then a 1024-dimensional feature vector represents context-aware semantic information of each word.

4.4 MULTIMODAL FUSION STRATEGIES

Multimodal fusion is the process of combining data from various sources for analysis tasks. It has gained increasing attention across numerous research areas because of its broad application potential, which includes, but is not limited to personality computing, sentiment analysis, emotion recognition, and eye blink detection. Data integration can provide surplus information with an increase in prediction accuracy.

Multimodal fusion strategies can be classified into the following categories: feature-level fusion, decision-level fusion, hybrid fusion, and model-level fusion [180, 10, 35, 129, 158]. Each of these strategies offers different advantages and is suitable for different scenarios depending on the characteristics of the data and the specific requirements of the task.

FEATURE-LEVEL FUSION In the case of feature-level fusion, also known as early fusion, data integration happens at the earliest possible stage, typically by combining feature vectors before any model processing occurs. This approach allows the model to directly learn interactions between features from multiple modalities from the very beginning. However, it can be less flexible when dealing with modalities that are not always present or are asynchronous.

DECISION-LEVEL FUSION Decision-level fusion, also known as score or late fusion, happens at the output stage of the processing. Different models process their respective modalities independently, and their outputs or decisions are combined using techniques such as voting, averaging, or more complex decision-making algorithms. This strategy is advantageous when modalities are best processed with distinct methods or when synchronization between modalities is not feasible. Recently, researchers tend to prefer decision-level fusion over feature-level fusion.

HYBRID FUSION Data from different sources often comes in various forms, making it important to consider when and how to combine this data effectively. In the past decade, researchers mainly focused on two methods: feature-level fusion and decision-level fusion. Some researchers also mix these methods in what is known as hybrid fusion. This strategy utilizes the raw data integration from the early stages with the late-stage decision scores. However, the primary disadvantage of hybrid fusion is its potential lack of flexibility in integrating multimodal data at various stages of the processing pipeline; it might not exploit the intermediate relationships and patterns that could emerge among different data types.

MODEL-LEVEL FUSION The main advantage of using model-level fusion, or middle fusion, lies in the flexibility that allows for more dynamic and adaptive processing of data, enabling the model to capture and leverage intermediate insights that might be lost in strictly early or late fusion approaches [129].

For example, when analyzing human behavior from video, assuming conditional independence between audio and visual data streams in decision-level fusion can lead to substantial information loss. This is particularly problematic because expressions in audio and visual forms are not only complementary but often also redundant. They work in tandem to convey emotions, intentions, and reactions. When decision-level fusion treats these streams as independent, it fails to capture the intricate correlations between them, which are crucial for a full understanding of the expressed nuances. This oversight can result in a less accurate or incomplete interpretation of the data, especially in scenarios where synchronized audio-visual cues are critical, such as in emotion recognition or social interaction analysis.

Model-level fusion is versatile and allows the model to exploit both the independent characteristics of each modality and their interactions. It can vary widely in implementation, from simple concatenation of intermediate features to more complex interactions facilitated by specialized architectures, like multimodal transformers that utilize attention-based multimodal fusion [77], or cross-modal attention mechanisms [164, 146, 116]. These systems integrate features from different modalities at multiple points within the transformer architecture. Cross-modal attention allows the model to dynamically focus on relevant information from any modality depending on the context provided by other modalities. This makes it highly effective for complex tasks where the relevance of information might shift dramatically based on subtle cues from multiple sources.

4.4.1 *Cross-modal Attention Mechanism*

The cross-modal attention mechanism enables a model to integrate and interpret information from different data types, such as acoustic, visual, and textual content. This capability is crucial for tasks that require an understanding of how different forms of data relate to each other, such as in multimedia content analysis or in scenarios where visual data accompanies acoustic signals or transcripts.

When translating information from modality β to modality α, the inputs are first projected into α-queries, β-keys, and β-values using trainable matrices as follows:

$$\begin{aligned} \mathbf{Q}_\alpha &= \mathbf{X}_\alpha \mathbf{W}_{Q_\alpha}, \\ \mathbf{K}_\beta &= \mathbf{X}_\beta \mathbf{W}_{K_\beta}, \\ \mathbf{V}_\beta &= \mathbf{X}_\beta \mathbf{W}_{V_\beta}, \end{aligned} \tag{4.1}$$

where $\mathbf{X}_\alpha$ and $\mathbf{X}_\beta$ are the input feature matrices for modalities α and β, respectively, and $\mathbf{W}_{Q_\alpha}$, $\mathbf{W}_{K_\beta}$, and $\mathbf{W}_{V_\beta}$ are trainable projection matrices.

The formulation of cross-modal attention is similar to that used in traditional transformer models:

$$\mathbf{V}'_\alpha = \text{softmax}\left(\frac{\mathbf{Q}_\alpha \mathbf{K}_\beta^\top}{\sqrt{d_k}}\right)\mathbf{V}_\beta. \tag{4.2}$$

While the original transformer uses self-attention to focus on different parts of a single input sequence, cross-modal attention enables the model to attend to specific aspects of one modality (e.g., an image) based on information from another modality (e.g., accompanying text). This integration allows the model to make richer, context-aware inferences that would be challenging if each data type were processed independently. Figure 4.1 illustrates a transformer encoder architecture with cross-modal attention units replacing the original self-attention mechanism.

4.4.2 *Multimodal Transformer*

The Multimodal Transformer (MulT) [164] is an end-to-end model that extends the standard Transformer network [165] to a multimodal setting.

The input sequences are first passed through a 1D temporal convolution layer to preserve the local structure, followed by the addition of positional encodings to capture temporal information implicitly.

The Transformer network was first introduced for Neural Machine Translation (NMT), and building on this idea the motivation of MulT is to implement modality translation by latently adapting elements across modalities with cross-modal attention. Unlike the encoder-decoder structure of traditional Transformers, MulT is composed of multiple stacks of pairwise and bidirectional multi-head cross-modal attention blocks. For three input sequences (textual, visual, and acoustic), the architecture includes six cross-modal transformers in total (Fig. 4.2).

In the original paper [164], the multi-head attention and FFN layers are postprocessed with "dropout, add residual, layer normalization" in that order. However, the implementation I used adopts a modified sequence for more robust learning: "layer normalization" is applied as preprocessing, followed by "dropout and add residual" as postprocessing.

The MulT implementation I used in experiments applies layer normalization first as preprocessing, prior to computing the outputs of multi-head cross-

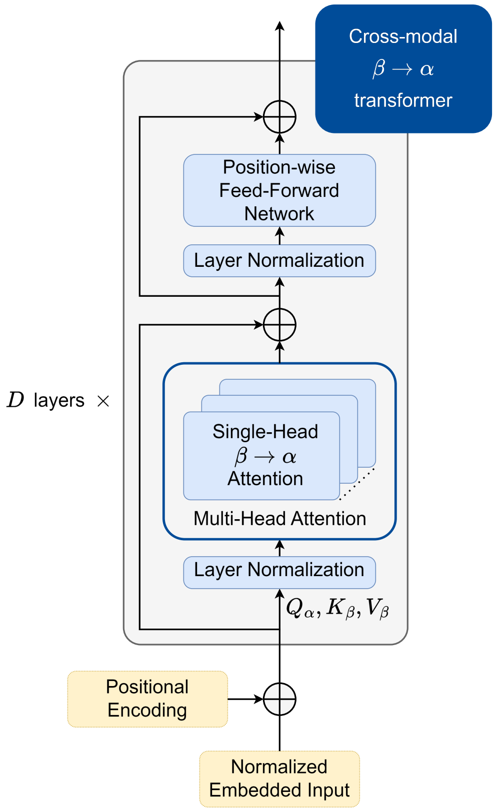

Figure 4.1: Cross-modal Transformer Encoder architecture.

modal attention. Dropout is then applied, and these outputs are combined with their inputs via skip connections. Subsequently, layer normalization is applied again before passing through a position-wise FFN with dropout layers, followed by another residual addition. This module, referred to as a cross-modal transformer, extracts temporal information across different time instants and is repeated multiple times.

The outputs are concatenated from the pair-wise bidirectional cross-modal transformer layers that share the same target modality. Then, each of them is passed through self-attention transformers with similar structure. The last elements of the sequences are extracted and passed to a FFN to make task-dependent predictions.

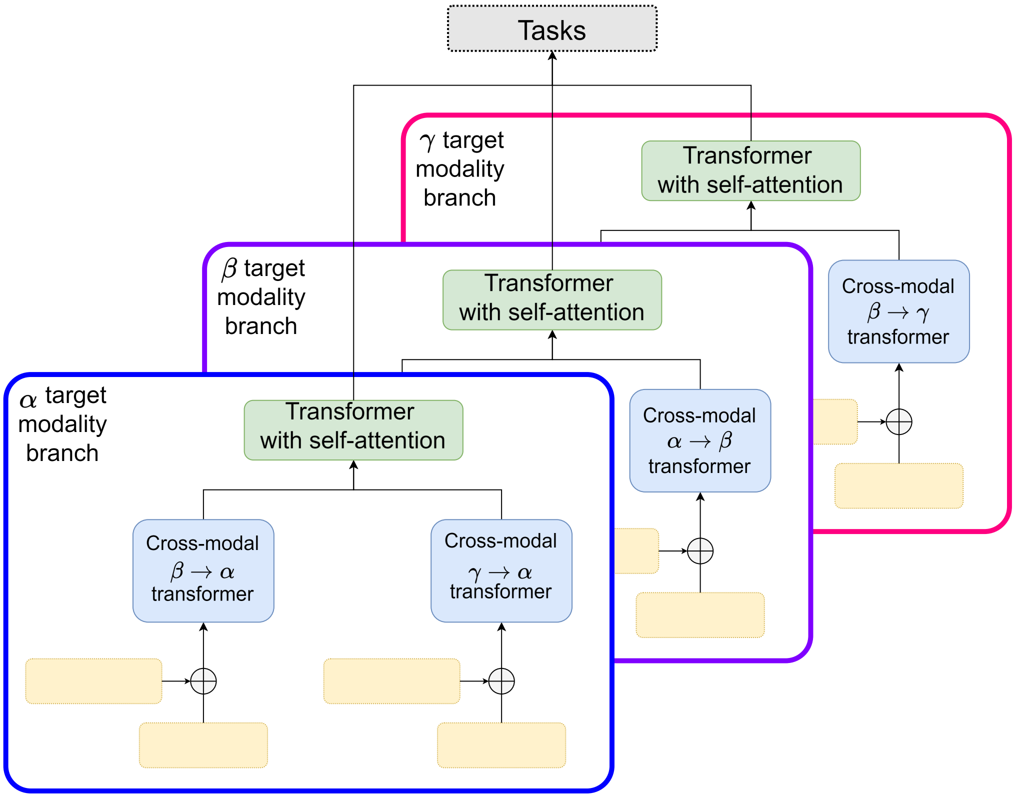

Figure 4.2: Multimodal Transformer (MulT) architecture for 3 input modalities. First, source modalities are transformed to a target modality by cross-modal transformer encoders. Two cross-modal transformer units are combined by another transformer network that utilizes self-attention to fuse the information pieces to form a branch within the multimodal network. This module is repeated for each target modality, and then outputs from all 3 branches are used for the final prediction.

Part III

CONTRIBUTIONS

5 PERSONALITY AND SENTIMENT ESTIMATION UNDER CONSTRAINTS

In social interactions, sentiment and personality influence how we behave and connect with others. Accurately perceiving these aspects is essential for building strong relationships and understanding each other effectively.

Sentiment analysis offers valuable insights into emotional states and attitudes, which are crucial for understanding human behavior and interactions. Similarly, predicting personality traits is important because it helps us anticipate decision-making patterns in individuals with stable traits across various situations. Additionally, it helps detect changes in behavior due to stress, drinking, and drugs, among others. One of the most widely studied frameworks for understanding personality is the Big Five personality traits model, which is detailed in Section 2.2.1.

Given the significance of sentiment and personality in social interactions, it is no surprise that there is a growing interest in their automatic perception within the ML and CV communities [166, 150, 180, 86, 12]. This interest has been further fueled by advancements in DL techniques and the availability of large public datasets, which have expanded the scope of research into PP [124].

Beyond the desire for competitive performance, interactive applications require at least near-real-time responsiveness while satisfying privacy-related constraints.

Cloud-based computation carries risks of data exposure, which can be addressed through techniques like de-identification. Drawing inspiration from previous research on robotic voice transformation [P7], this work aims to extend privacy protection to the visual domain by employing AUs to analyze facial expressions without compromising identities. Shifting away from cloud computing, local processing provides a compelling alternative that enhances privacy because data remains on the local device. Additionally, computational efficiency can be achieved using linear complexity variants of transformer architectures, which offer fast processing with low memory consumption.

One of my goals is to develop a method that operates in near-real-time for interactive applications. Specifically, I focus on tasks that process short videos to predict sentiment and personality traits for either a single person or multiple individuals in group conversations. In such HCAI applications, real-time processing of every frame is highly redundant and unnecessary; instead, near-real-time performance is achieved by analyzing the information from the previous e.g. 10–15-second window and providing predictions with minimal delay. For this processing pipeline, latency is critical, and the system must return results within less than one second, ensuring responsiveness while maintaining computational efficiency.

My contributions are summarized as follows:

- I propose LinMulT, a Multimodal Transformer with a Linear Complexity Attention Mechanism, designed both for the Multimodal Sentiment Analysis (MSA) and the Automatic Personality Perception (APP) tasks. LinMulT effectively handles longer sequences and multiple input modalities, delivering competitive performance with reduced computational complexity.

- To mitigate bias and improve generalization across scenarios, for visual features I use AUs and FAb-Net features, which focus on facial muscle movements and expressions rather than identity-specific attributes. This approach ensures privacy preservation and helps avoid biases related to appearance-based features such as skin tone or facial structure.

- LinMulT achieves performance comparable to MulT on the CMU-MOSI and CMU-MOSEI datasets for the MSA task. On the FI dataset for the APP task, LinMulT slightly outperforms MulT across all metrics, with an average accuracy of 0.909. Furthermore, a significant performance improvement is observed when switching from the OOB to the OOWFR feature sets, bringing LinMulT close to state-of-the-art results reported in the literature.

- I conducted a detailed computational complexity analysis, measuring inference time on both GPU and CPU, memory utilization, and Floating Point Operations (FLOPs) across varying batch sizes and window durations. The results highlight LinMulT's suitability for near-real-time applications, with an inference time of 134 ms on GPU and 234 ms on CPU for single-person interactions (batch size 1), and significant speedups in batched inference, achieving up to a 2.2x speedup on GPU and a 16.2x speedup on CPU with a batch size of 8. Scalability analysis shows up to 85.7% GPU memory and over 80% FLOPs reduction for 30-second windows, highlighting the proposed LinMulT's training and inference efficiency compared to the baseline MulT.

My approach demonstrates that near-real-time sentiment and personality trait predictions are feasible using LinMulT. Its efficiency in handling multiple relatively long input sequences makes it ideal for resource-constrained environments.

5.1 RELATED WORK

In this section, I briefly outline the related works for both multimodal sentiment analysis and automatic personality perception tasks.

5.1.1 *Sentiment*

The ML and CV communities have widely studied the analysis and understanding of people's affective state [150, 180]. The majority of works found in the literature and publicly available datasets on the topic deal with instantaneous expression categorization [70], where the task is to classify a discrete affective state using features from different modalities [150].

Over the past few years, however, the research community started to pay attention to the design and development of novel multimodal datasets and annotation protocols, taking into account how people communicate and express ideas and opinions through verbal content as well as visual and vocal features, such as facial expressions, head gestures, and voice quality [178].

In the work of [178], the Multimodal Corpus of Sentiment Intensity (CMU-MOSI) dataset was introduced, a video corpus with opinion-level sentiment intensity annotations that can be used for sentiment, subjectivity, and multimodal language studies. On the basis of interaction patterns between words and facial gestures, the authors presented a simple representation model that jointly accounts for words and gestures in each opinion segment. As a baseline, they trained prediction models using Support Vector Regression, showing that the proposed multimodal dictionary can yield better results in sentiment intensity analysis compared with common fusion methods. Later, [177] introduced the Multimodal Opinion Sentiment and Emotion Intensity (CMU-MOSEI) dataset, which is currently the largest dataset of multimodal sentiment analysis and emotion recognition. They also presented a Multi-attention Recurrent Network (MARN) model for understanding human communication. According to these authors, the main strength of MARN comes from discovering interactions between modalities over time using a Multi-attention Block and storing them in the hybrid memory of a recurrent component called the Long-short Term Hybrid Memory, which is an extension of the Long-short Term Memory by reformulating the memory component to carry hybrid information. Similarly to my work, Action Units are used as indicators of facial muscle movements. However, a set of visual features including per-frame basic and advanced emotions are also considered in their work.

Self-Supervised Embedding Fusion Transformer (SSE-FT) [146] represented text, acoustic, and visual modalities with features extracted independently from pre-trained Self-Supervised Learning (SSL) models, applied to multimodal emotion recognition. Given the high dimensional nature of SSL-based features, they introduced a novel Transformer and Attention-based fusion mechanism to efficiently combine and train on multimodal embeddings. They achieved state-of-the-art results for sentiment and emotion recognition on different datasets.

For the visual modality, they used a pre-trained FAb-Net [172] model to obtain embeddings for each frame in the video. Their work was partially inspired by MulT using unaligned language sentences [164].

The current state-of-the-art sentiment recognition method on CMU-MOSI/MOSEI is the work of [73]. In their work, a Bi-Bimodal Fusion Network (BBFN) is proposed, an end-to-end network that performs fusion (relevance increment) and separation (difference increment) on pairwise modality representations. For the visual modality, MulT and BBFN use the FACET module of the commercial iMotion software and make the extracted features available for research purposes. The FACET module estimates 35 AUs. Thus, this also mitigates possible sources of bias coming from appearance-based features.

The literature review on sentiment perception revealed that state-or-the-art methods on the topic are considering the use of visual inputs based on Action Units [177, 73], that could alleviate possible sources of bias coming from appearance-based features. This is not the typical route in automatic personality perception since features like hairstyle, piercings, and makeup are excellent tools for self-expression. Furthermore, to the best of my knowledge, I am the first to evaluate an efficient linear transformer for the task of audio-visual sentiment perception.

5.1.2 *Personality*

Personality Computing [166] is receiving high attention from different research communities due to its applicability to emerging HCAI scenarios. It covers APP, APR, and APS, which may guide the machine or the robot in interactive, collaborative scenarios.

In both automatic personality recognition and perception, supervised learning methods are commonly used, with the key difference being the origin of labels. APR typically relies on self-reported questionnaires [119], whereas APP uses external observer labels gathered through crowd-sourcing [86, 128, 51]. What these references have in common is that they focus on the ChaLearn: First Impression Challenge. This challenge employs one of the largest datasets available for APP, known as the First Impressions dataset. Despite some biases related to attributes such as gender, age, and ethnicity [51, 132, 87], this dataset is widely used in the field.

The winner of the ChaLearn First Impression Challenge developed a system that integrated visual, acoustic, and textual features into a unified framework, which was trained end-to-end to achieve competitive results [90]. Other researchers conducted ablation studies to further investigate the impact of combining these modalities, implementing a trimodal stacked CNN architecture that processed visual, acoustic, and textual inputs together [82]. The feasibility of merging apparent personality and emotion estimations within a single deep neural network in a multi-task learning framework is also studied [181]. Another notable study employed a pre-trained CNN to extract visual features related to facial expressions and scene information [71]. These features were then combined and fed into a Kernel Extreme Learning Machine (ELM) regressor, showcasing an alternative approach to integrating visual data.

Although AU-based features are not a standard when it comes to personality recognition or perception, [175] have already proposed to exploit it. In addition to using acoustic and textual information, they proposed to extract a set of visual features including facial AUs, facial landmarks, head pose, gaze tracking, and histogram of oriented gradients, proposing a many2many interaction scheme to perform time-dependent interactions explicitly inside a single modality and across different modalities.

More recently, [119] introduced UDIVA, a non-acted dataset comprising 90.5 hours of face-to-face dyadic interactions, annotated with both self-reported and apparent personality labels. As a baseline, they proposed a transformer-based method for regressing the self-reported personality traits of a target individual by leveraging multimodal data from both participants in the interaction, extracted from 3-second video segments. This approach incorporated contextual information from the other participants to improve predictions for the monitored individual.

The work was further extended in [34] by introducing variable time windows, enabling the modeling of longer-term interdependencies across interactions. Additionally, they incorporated a cross-subject layer that explicitly modeled interactions between the two individuals through attentional mechanisms. To enhance the system's ability to capture dyadic dynamics, they proposed a two-stream cross-attentional Transformer architecture. This model processed data from both participants simultaneously, allowing it to model their behaviors interactively and predict their personalities jointly.

In contrast to previous studies that rely on raw visual data or deep features extracted from it to predict apparent or self-reported personality traits, I focus on using appearance-independent features, such as AUs and FAb-Net features, as visual inputs. This approach prioritizes fine-grained facial movements over appearance-based characteristics, making it more generalizable and less prone to biases associated with factors like hairstyle, makeup, or piercings. This approach is particularly relevant and preferred for future applications in controlled environments, such as clinical settings, where subjects are typically observed against plain or empty backgrounds, and the mentioned appearance-related attributes are minimal or absent. By emphasizing subtle facial dynamics rather than self-expression through appearance or background context, this method aligns better with scenarios requiring objective and unbiased analysis.

5.2 PROPOSED METHOD

I propose a Multimodal Transformer with a Linear Complexity Attention Mechanism, called LinMulT, as an enhanced version of MulT (Section 4.4.2), specifically designed to improve both the efficiency and flexibility of multimodal data processing. LinMulT, shown in Figure 5.2, transforms input from each source modality into a target modality using cross-modal attention mechanisms that operate with linear complexity in relation to sequence length, significantly reducing computational overhead. Once the source modalities are transformed, the outputs are fused using a self-attention mechanism, forming distinct branches within the network. The outputs of these branches are then global average pooled, and the resulting vectors are concatenated along the feature dimension. This combined representation is passed through two dense layers with ReLU activations, incorporating dropout between them for regularization. Finally, the prediction is made using a linear fully-connected output layer.

A key contribution of LinMulT is the introduction of a linear complexity attention mechanism, which directly addresses the inefficiencies of the original MulT's softmax-based attention. In traditional transformers, the quadratic complexity with respect to sequence length imposes significant computational and memory overheads, especially when processing long sequences. LinMulT replaces this mechanism with a more efficient alternative, reducing the computational cost from $\mathcal{O}(L^2D)$ to $\mathcal{O}(LD^2)$, where L is the sequence length and D the hidden dimensionality. This improvement not only accelerates the model's training and inference times but also enables it to handle longer sequences more effectively, making it well-suited for near-real-time applications.

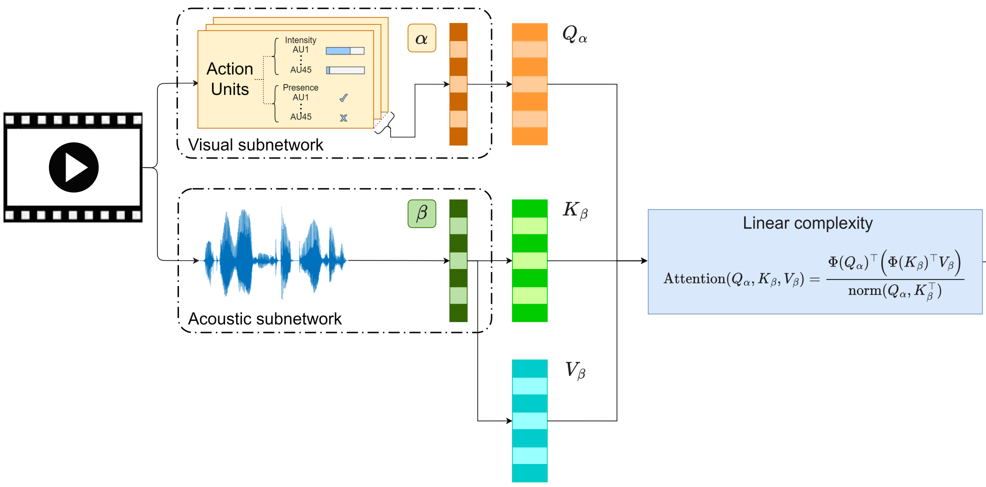

Figure 5.1: Cross-modal linear complexity attention module in LinMulT, illustrating how attention is computed between two input sequences: action units (α) and acoustic features (β).

Figure 5.1 illustrates the cross-modal linear complexity attention module in LinMulT. In this mechanism, transformers convert information from one modality (e.g., β, representing the acoustic features) into another modality (e.g., α, representing the visual features). The darker striped columns represent embeddings, which may be features derived from raw data or outputs of pre-trained models. The keys (K_β) and values (V_β) from modality β, along with the queries (Q_α) from modality α, are processed within this linear complexity cross-modal attention mechanism.

While the original MulT architecture, introduced in Section 4.4.2, was limited to handling only three input sequences, my goal was to build a more scalable and adaptable multimodal transformer that can accept up to N input sequences, where the only limitation is the available GPU memory. In my experiments, LinMulT was trained with up to 5 input sequences. This expansion is possible due to the model's capability to automatically create cross-modal attention units between pairs of input streams, which now requires less memory thanks to the new linear complexity attention mechanism.

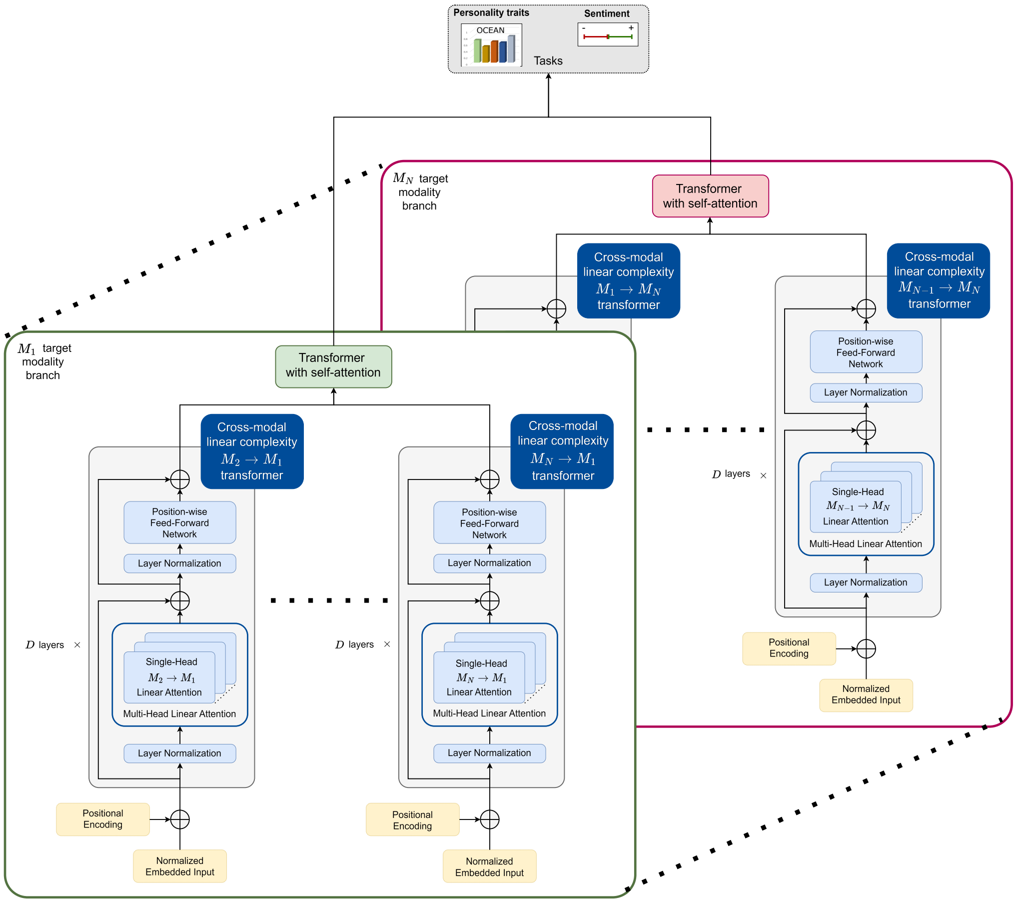

Figure 5.2: LinMulT: A Multimodal Transformer with a Linear Complexity Attention Mechanism designed for efficient and flexible multimodal data processing. It transforms source modalities into target modalities using cross-modal attention and fuses the outputs through self-attention, enabling efficient handling of multiple input modalities for final prediction.

In addition to optimizing the attention mechanism and supporting more input features to be applied during the fusion, I constrained the visual inputs to appearance-invariant features, using AUs and FAb-Net, which focus on facial muscle movements and expressions. By excluding identity-related information and background context, I prioritized generalization across datasets and scenarios, ensuring that the model remains robust and applicable in a wide range of tasks without relying on appearance or scene-specific features.

Ultimately, LinMulT's design addresses the challenges of both scalability and efficiency, making it an ideal choice for near-real-time multimodal applications, as well as for researchers requiring rapid experimentation and iteration due to the low computational requirements during both training and inference. Its ability to process multiple input streams, handle long sequences efficiently, and utilize constrained visual inputs positions it as a flexible, powerful solution for complex multimodal tasks.

5.3 METHODOLOGY

The following section provides a detailed explanation of the experimental setup to ensure the reproducibility of the proposed model and results. Section 5.3.1 introduces the datasets, Section 5.3.2 outlines the extracted features and preprocessing techniques, followed by the description of experiments and used evaluation metrics in Section 5.3.3. Section 5.3.4 covers the training process in detail.

5.3.1 *Datasets*

The baseline MulT and the proposed LinMulT architectures are trained and evaluated on 3 databases.

CMU-MOSI: The Multimodal Corpus of Sentiment Intensity (CMU-MOSI) [178] dataset is a widely-used benchmark for multimodal sentiment analysis. It consists of 2199 opinion video clips, where each clip represents a single utterance expressing an opinion. These clips are extracted from 93 unique videos featuring various speakers, and the total duration of the dataset is approximately 3 hours and 58 minutes. Each utterance is annotated with sentiment scores on a continuous scale ranging from -3 (strongly negative) to +3 (strongly positive). The original training, validation, and test subsets were used in the experiments with 1284, 229, and 686 utterances, respectively. It is approximated as 6:1:3 ratio for simplicity. This split ensures speaker independence across subsets, preventing overlap of speakers between training, validation, and testing.

CMU-MOSEI: The Multimodal Opinion Sentiment and Emotion Intensity (CMU-MOSEI) [7] dataset is one of the largest multimodal datasets for sentiment analysis and emotion recognition. It contains over 23453 sentence-level utterances from 1000 speakers and 3228 videos, covering 250 topics such as reviews, debates, consulting, and speeches. The total duration of the dataset is 65 hours and 53 minutes, making it significantly larger than CMU-MOSI. Each utterance is annotated for sentiment on a continuous scale ranging from -3 (strongly negative) to +3 (strongly positive). The dataset includes presence/intensity annotations for six Ekman emotions—happiness, sadness, anger, fear, disgust, and surprise—on a $[0, 3]$ Likert scale. However, I only consider sentiment annotation in this work. I used the original train, validation, and test split provided by the CMU-Multimodal-SDK, which is approximately a 9:1:3 ratio.

FIRST IMPRESSION V2: The revised version of the First Impression database (FI for short) is the largest publicly available in-the-wild dataset in the APP subfield, which contains 10000 clips extracted from 3060 different YouTube high-definition videos, with each video corresponding to a unique speaker mostly facing and speaking to a camera. The dataset contains 15-second-long samples, which are collected automatically. Out of the 3060 original videos, 721 were not split into multiple clips, while 2339 were split at least once. On average, each video was split into 3.27 clips ($\pm$1.8), with a maximum of 6 splits per video.

The dataset is divided into training, validation, and test sets in a 3:1:1 ratio. The total duration of all videos in the dataset is approximately 42.5 hours. The training set consists of 6000 video samples with a total duration of 25.5 hours, while the validation and test sets each contain 2000 video samples, both with a total duration of 8.5 hours. While some clips generated from the same original video are distributed across these subsets, they are taken from different time intervals within the video. As a result, there is significant variation in the content of each clip due to changes in camera angles, poses, motion, and behavior. This diversity ensures that the presence of clips from the same video in different subsets does not lead to data leakage. Moreover, studies comparing setups with and without overlapping instances have found minimal differences in performance, confirming that this distribution strategy does not compromise model evaluation or generalization [51].

Transcripts of the video clips are generated by the cloud transcription service Rev. The clips are annotated by Amazon Mechanical Turk (AMT) workers using a special interface [128]. They registered annotations using pair-wise comparisons, and then they converted the votes to cardinal values by fitting a Bradley-Terry-Luce [27] model with maximum likelihood estimation. Values are scaled, so every video sample has five continuous trait scores between 0 and 1. Each video sample in the dataset will have a continuous value associated with each trait dimension, which can be used by any supervised learning method in a classification or regression task. Notably, Emotional Stability ($\overline{\text{N}}$) is used instead of Neuroticism. In a nutshell, the labels correspond to human annotators' impressions. Therefore, the Big Five trait scores reflect perceived personalities based on first impressions, rather than self-reported personality traits. It is a different task than evaluating real personality traits with experts, but equally useful in the context of human interaction.

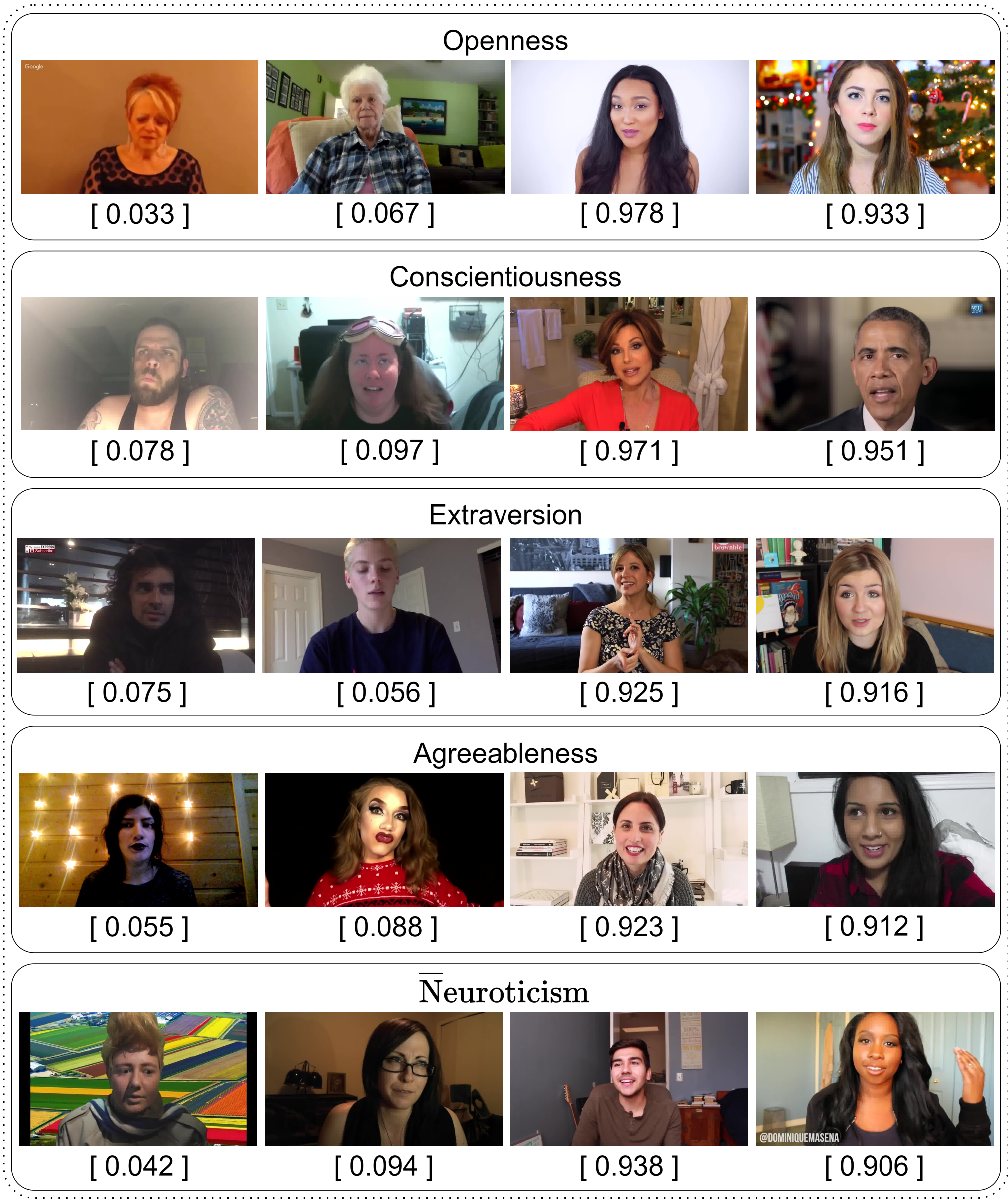

Figure 5.3: Examples of the First Impression V2 dataset. For each video, the ground truth Big Five scores are provided. For each trait, the first two samples instantiate the low extremes, and the last two examples demonstrate the high extremes of a given trait.

Each trait represents a range bounded by two extremes. For example, in the case of extraversion, the two polar ends are extreme extraversion and extreme introversion, which can be described with the words *friendly* and *reserved*, respectively. A few examples from the dataset focusing on the extreme poles per trait are depicted in Figure 5.3. Later, the FI dataset was extended [50] with the inclusion of transcripts, as well as metadata such as gender and ethnicity information, aiming to advance research on explainable machine learning. Additionally, a *job-interview variable* was introduced, indicating whether a subject in a video should be invited to a job interview. This variable was annotated—similarly to the Big Five annotations—through AMT using a pairwise ranking approach, where annotators compared two videos and selected which individual appeared more suitable for an interview. However, I excluded this annotation from my analysis as it was not the focus of my research.

One notable characteristic of this dataset is that the target variables exhibit an unbalanced data distribution. This is a common challenge in many HCAI tasks, such as emotion recognition, where the majority of samples often belong to a neutral class. Similarly, in this dataset, the scores follow a Gaussian distribution, with most values concentrated around the mean. As a result, the *regression-to-the-mean* problem is emphasized because the optimization process likely produces predictions near the mean of ground truth values to minimize the loss [170]. This behavior can lead to reduced accuracy in predicting extreme values, which are less represented in the data.

5.3.2 *Feature Extraction and Preprocessing*

Taking into account computational and memory restrictions, I use different sets of features (Table 5.1) for the experiments based on the adopted architecture, detailed in Section 5.4.

In the experiments, I utilized two primary multimodal feature sets to evaluate performance. The first, labeled "OOB", consists of features extracted using OpenSMILE for calculating the eGeMAPS acoustic feature set, OpenGraphAU for predicting AUs from detected face images, and BERT for extracting embeddings from the transcripts. The second feature set, labeled "WFR", is composed of Self-Supervised Embeddings (SSE) extracted from pre-trained SSL models, specifically Wav2Vec2 for audio, FAb-Net for visual, and RoBERTa for textual features. I used RetinaFace to detect the monitored person's face in each video frame and tracked the individual using a simple intersection-over-union bounding box tracker. FAb-Net features were extracted solely from the cropped face area, excluding any background information. For more details about the features used, please see Section 4. No further finetuning was applied to these models as I utilized their publicly available pre-trained weights. Additionally, I introduced a combined feature set, "OOWFR", which integrates hand-crafted features from OOB with deep features from WFR, allowing for a more comprehensive representation of the multimodal data.

Set	Features
OOB	eGeMAPS, AUs, BERT
WFR	Wav2Vec2, FAb-Net, RoBERTa
OOWFR	eGeMAPS, AUs, Wav2Vec2, FAb-Net, RoBERTa

Table 5.1: Multimodal feature sets. OOB is shorthand for the **O**penSMILE (for the eGeMAPS features), **O**penGraphAU (for the Action Units based features), and **B**ERT feature combination, and WFR is for the **W**av2Vec2, **F**Ab-Net, and **R**oBERTa feature combination. OOWFR is used as a combination of hand-crafted and deep features.

For CMU-MOSI and CMU-MOSEI, I set a maximum sequence length of 10 seconds for audio-visual modalities. For the FI dataset, I consider 15 seconds as the sequence length. Sequences shorter than 10 (or 15) seconds are padded with zeros and longer ones are truncated.

5.3.3 *Experiments*

The experiments were designed to evaluate the performance of the baseline MulT, and the proposed LinMulT model in both Multimodal Sentiment Analysis (MSA) and Automatic Personality Perception (APP) tasks, utilizing various feature sets (OOB, WFR, OOWFR).

For MSA, the models were trained on extracted input features from the CMU-MOSI and CMU-MOSEI datasets to predict sentiment values per video. The performance was assessed using a combination of regression and classification metrics. First, I calculated the Mean Absolute Error (MAE) to measure the average prediction error and the Pearson Correlation (Corr) to assess the linear relationship between predicted and true sentiment scores. Then, I discretized the sentiment scores into binary categories by considering values greater than 0 as positive and values less than or equal to 0 as negative. Using these binary labels, I calculated Binary Accuracy (Acc2) to measure the prediction accuracy for distinguishing between positive and negative sentiments. Additionally, the weighted Binary F1 Score (F1) was computed which accounts for balancing precision and recall. Finally, I discretized the sentiment scores again, this time into seven classes by rounding each value to the nearest integer within the range of -3 to +3. Based on these seven discrete classes, I calculated the 7-Class Accuracy (Acc7) to evaluate how accurately the predictions match the true sentiment labels across all seven categories.

To further examine the performance differences between LinMulT and MulT across all their variants, I visualized the predictions of each model using a Q–Q plot to assess the normality of their prediction errors. Many statistical tests, including the paired t-test, assume that the data follows a normal distribution. The Q–Q plot helps verify this assumption by comparing the quantiles of the observed data against those of a theoretical normal distribution. After confirming normality, I conducted two-sided paired t-tests to determine whether the observed differences in predictions between LinMulT and MulT were statistically significant. The null hypothesis (H_0) for these tests states that there is no difference in the mean predictions of LinMulT and MulT, while the alternative hypothesis (H_a) posits that there is a significant difference in their means. The calculated p-values from these tests quantify the strength of evidence against H_0, with smaller p-values indicating stronger evidence for rejecting H_0. To account for multiple comparisons and control the family-wise error rate, I applied Bonferroni correction to adjust the p-values. Based on these corrected p-values, decisions were made regarding whether to reject or fail to reject H_0.

For the APP task, the models were trained on features extracted from the FI dataset to predict perceived OCEAN personality traits. The performance was measured using the trait-wise $\mathrm{R}^t_{\mathrm{acc}}$. This metric is first calculated between the predicted and ground truth values independently for each personality trait as follows:

$$\mathrm{R}^t_{\mathrm{acc}}(\mathbf{Y}_t, \hat{\mathbf{Y}}_t) = 1 - \frac{1}{N} \sum_{i=1}^{N} |y_t^{(i)} - \hat{y}_t^{(i)}|, \quad t = 1, \ldots, 5, \tag{5.1}$$

where N is the number of samples, $\mathbf{Y}_t$ and $\hat{\mathbf{Y}}_t$ are the ground truth and predicted values for the t-th trait, respectively, both represented as vectors of length N. Each $y_t^{(i)}$ and $\hat{y}_t^{(i)}$ corresponds to the ground truth and predicted value for the i-th sample and t-th trait.

To compute the overall performance metric across all five personality traits, the trait-wise values are averaged as follows:

$$\mathrm{R}_{\mathrm{acc}}(\mathbf{Y}, \hat{\mathbf{Y}}) = \frac{1}{5} \sum_{t=1}^{5} \mathrm{R}^t_{\mathrm{acc}}(\mathbf{Y}_t, \hat{\mathbf{Y}}_t). \tag{5.2}$$

The R^2 metric, or the coefficient of determination, was also calculated for FI. It quantifies how well the predicted values explain the variance in the target variable. Values range from 0 to 1, with higher values indicating better model performance. Negative values may occur if the model performs worse than a simple mean-based prediction.

Similar to the MSA task, Q–Q plots were visualized to determine normality. Then, two-sided paired t-tests were applied to evaluate whether the observed performance differences between different versions of LinMulT and baseline models were statistically significant.

In addition to performance evaluation, the computational complexity of both the baseline MulT and the proposed LinMulT models was analyzed. The theoretical complexity of LinMulT's linear attention mechanism was briefly detailed, highlighting its advantages over traditional softmax-based attention in terms of computational efficiency.

To measure practical computational efficiency, inference times were calculated across different batch sizes and analysis window sizes. These measurements are crucial because they reflect the models' ability to scale efficiently in real-world applications, particularly in scenarios requiring near-real-time processing or handling multiple inputs simultaneously. Both GPU and CPU were used for these calculations to provide a comprehensive view of the models' performance across different hardware configurations.

Furthermore, GPU memory utilization was assessed during inference across various batch sizes and window sizes. This analysis aimed to quantify the memory efficiency of LinMulT compared to MulT, which is essential for applications where memory constraints are significant.

Finally, the number of FLOPs required for both training and inference was measured. This metric provides a hardware-independent measure of computational complexity, allowing for a direct comparison of the models' efficiency in terms of energy consumption and processing speed.

Overall, these experiments aimed to comprehensively evaluate LinMulT's performance, computational efficiency, and scalability, making it suitable for resource-constrained environments and near-real-time applications.

5.3.4 *Training Details*

Following the work of [164], I used 1D convolution to capture temporal information and reduce dimensionality for further computations. I chose 32 as the number of kernels and 8 as the number of attention heads. I consistently used 5 stacked transformer layers with dropout (0.2) throughout the experiments related to different tasks. The bfloat16 AI datatype was utilized in the experiments to enhance computational efficiency and precision during model training and evaluation [75].

Both models were trained for 30 epochs using Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1e-8$) and a learning rate of $1e-3$. The learning rate is modified according to the 1cycle learning rate policy [149].

I used the Bell loss (Equation 5.3) to alleviate the regression-to-the-mean problem. I set the parameters of Bell loss $\sigma = 9$ and $\gamma = 300$. As empirical results showed in [101], the $\mathcal{L}_{\text{Bell}}$ had difficulties at the beginning of the optimization and shined at later optimization stages. To avoid this issue, the sum of $\mathcal{L}_{\text{MSE}}$ and $\mathcal{L}_{\text{MAE}}$ guided the stochastic gradient descent algorithm in the earlier stages by producing a higher gradient:

$$\mathcal{L}(\mathbf{Y}, \hat{\mathbf{Y}}) = \mathcal{L}_{\text{Bell}}(\mathbf{Y}, \hat{\mathbf{Y}}) + \mathcal{L}_{\text{MSE}}(\mathbf{Y}, \hat{\mathbf{Y}}) + \mathcal{L}_{\text{MAE}}(\mathbf{Y}, \hat{\mathbf{Y}}). \tag{5.3}$$

5.4 RESULTS

I present results on MSA in Section 5.4.1. I show and discuss the obtained results on the APP task in Section 5.4.2. State-of-the-art results on these tasks and the corresponding datasets are also reported and briefly discussed. I show a brief excerpt of the t-test table after each experiment.

5.4.1 *Multimodal Sentiment Analysis*

First, I compare LinMulT and MulT on both CMU-MOSI and CMU-MOSEI and demonstrate performance scaling with respect to dataset size. The obtained results on CMU-MOSI and CMU-MOSEI are shown in Table 5.2 and Table 5.3, respectively.

In the case of CMU-MOSI (Table 5.2), the results show that the performance difference between MulT and LinMulT is minimal when using the same feature sequences as inputs. The results demonstrate that WFR features consistently outperform OOB features, and most metrics (Acc7, MAE, Corr) show further improvement when switching to OOWFR features.

With OOWFR features, the two models remain very close in performance, though MulT slightly outperforms LinMulT in most metrics. Specifically, MulT achieves an Acc2 of 0.831 and an F1 of 0.832 compared to LinMulT's Acc2 of 0.822 and F1 of 0.824. Both models achieve identical Corr values (0.745), while MulT has a slight edge in MAE (0.856 vs. 0.875).

Method (Feature)	Acc7 ↑	Acc2 ↑	F1 ↑	MAE ↓	Corr ↑
MulT (OOB)	0.299	0.741	0.743	1.079	0.621
LinMulT (OOB)	0.303	0.736	0.738	1.082	0.620
MulT (WFR)	0.324	0.827	0.826	0.895	0.714
LinMulT (WFR)	0.329	0.828	0.828	0.897	0.712
MulT (OOWFR)	0.360	0.831	0.832	0.856	0.745
LinMulT (OOWFR)	0.353	0.822	0.824	0.875	0.745
MARN [177]	0.347	0.771	0.770	0.968	0.625
MulT [164]	0.401	0.833	0.829	0.832	0.745
BBFN [73]	0.450	**0.843**	**0.843**	**0.776**	0.755
SSE-FT [146]	**0.465**	0.839	0.835	**0.776**	**0.768**

Table 5.2: Results for multimodal sentiment analysis on CMU-MOSI. The rows on top: evaluated methods. The rows at the bottom: state-of-the-art results obtained from the literature. Notations: ↓ (↑) shows that lower (higher) values are better. The best results are highlighted in bold.

LinMulT (OOWFR) outperforms MARN across all metrics. However, there remains a noticeable gap between LinMulT and the state-of-the-art models BBFN and SSE-FT in metrics like Acc7 (0.353 vs. 0.450 for BBFN and 0.465 for SSE-FT) and MAE (0.875 vs. 0.776 for both). LinMulT matches or comes close to these models in binary metrics such as Acc2 (0.822 vs. 0.843 for BBFN and 0.839 for SSE-FT) and F1 (0.824 vs. 0.843 for BBFN and 0.835 for SSE-FT), as well as correlation (0.745 vs. 0.755 for BBFN and 0.768 for SSE-FT). These results highlight LinMulT's competitive performance compared to the more complex models in the literature.

The results for CMU-MOSEI (Table 5.3) show that LinMulT and MulT perform comparably when using the same feature sets, with minimal differences between them. With the larger dataset and use of OOWFR features, both models improved across all metrics compared to their OOB counterparts, similar to the trends observed in the smaller CMU-MOSI dataset. The results indicate that WFR features improve performance over OOB features and demonstrate competitive outcomes compared to OOWFR. LinMulT remains highly competitive with MulT, showing nearly identical metrics—LinMulT maintains similar predictive accuracy while offering computational advantages.

Compared to the literature, LinMulT with OOWFR features performs competitively with state-of-the-art models on several key metrics. For example, LinMulT achieves a higher correlation score (0.784 *vs.* 0.767) and a slightly lower MAE (0.530 *vs.* 0.529) compared to BBFN [73]. While LinMulT's Acc7 (0.524 *vs.* 0.548) and binary classification metrics (Acc2: 0.751 *vs.* 0.862; F1: 0.746 *vs.* 0.861) fall short of BBFN, the results remain competitive given LinMulT's reliance on more constrained visual features.

Method (Feature)	Acc7 ↑	Acc2 ↑	F1 ↑	MAE ↓	Corr ↑
MulT (OOB)	0.514	0.744	0.740	0.564	0.745
LinMulT (OOB)	0.508	0.741	0.737	0.571	0.738
MulT (WFR)	0.538	0.737	0.731	0.533	0.775
LinMulT (WFR)	0.531	0.753	0.750	0.539	0.774
MulT (OOWFR)	0.522	0.749	0.744	0.533	0.783
LinMulT (OOWFR)	0.524	0.751	0.746	0.530	0.784
MulT [164]	0.511	0.845	0.845	0.570	0.758
BBFN [73]	0.548	0.862	0.861	**0.529**	0.767
SSE-FT [146]	**0.557**	**0.873**	**0.870**	**0.529**	**0.792**

Table 5.3: Results for multimodal sentiment analysis on CMU-MOSEI. The rows on top: evaluated methods. The rows at the bottom: state-of-the-art results obtained from the literature. Notations: ↓ (↑) shows that lower (higher) values are better. The best results (per metric) for both the evaluated methods and those found in the literature are highlighted in bold.

LinMulT also demonstrates comparable performance to SSE-FT [146] on several metrics, such as MAE (0.530 *vs.* 0.529) and Corr (0.784 *vs.* 0.792). However, SSE-FT outperforms LinMulT in binary classification metrics, achieving much higher Acc2 (0.873 *vs.* 0.751) and F1 scores (0.870 *vs.* 0.746). This superior performance can be attributed to SSE-FT's use of additional self-attention transformers for visual and acoustic modalities, which increase its capacity (and computational complexity).

Unlike SSE-FT, which uses the entire video frame (including background information) to extract richer features, LinMulT relies on more limited visual inputs: it uses AUs and FAb-Net features extracted from tightly cropped facial regions, which focus only on the face and exclude broader contextual details. This difference explains why SSE-FT achieves better performance.

To ensure the validity of statistical tests, Q–Q plots were visualized to assess the normality of the data (Figure A.1-A.2). The results showed that normality could be assumed, allowing for the application of parametric tests like two-sided paired t-tests to evaluate the significance of performance differences between models.

The t-test results for both CMU-MOSI and CMU-MOSEI datasets are summarized in Tables A.2 and A.3, respectively. Notably, the comparison between LinMulT (OOWFR) and MulT (OOWFR) on CMU-MOSI revealed a statistically significant difference, indicating that the observed performance difference was not due to chance. On CMU-MOSEI, the same comparison showed a significant difference as well, though the effect size was smaller. Other comparisons, such as between LinMulT (OOWFR) and its other feature variants, also showed significant differences, highlighting the impact of feature selection on model performance. Full tables with all comparisons are provided in the appendix for further reference.

5.4.2 *Automatic Personality Perception*

Table 5.4 shows the results for APP on the FI dataset, obtained from distinct architectures and feature sets.

First, it should be noted that all evaluated methods presented a similar performance on the FI dataset, as shown in Table 5.4. This is a well-known problem related to the First Impressions dataset, where most personality scores are centered in a very small region close to the mean. A Prior model that predicts the average value per trait achieves an R_{acc} of 0.8813 at the test stage [51], due to the highly centralized distribution. While predicting the average is safe, it struggles with values at the distribution tails, making even small differences, like changes in the third digit, significant.

Surprisingly, the worse results in the case of personality perception were obtained using the WFR features. Following this tendency and the same training strategy, LinMulT (OOB) obtained overall better results than LinMulT (WFR). These results were not observed for the case of sentiment analysis, previously discussed. The OOWFR feature set, which combines the strengths of both OOB and WFR features, yields the best results, offering a richer and more effective representation.

The comparison between MulT and LinMulT highlights that LinMulT offers almost identical performance using each feature set, with only small differences in individual traits. However, LinMulT shines in terms of efficiency, requiring less computational power and memory. This efficiency becomes even more valuable with complex feature sets like OOWFR, where MulT's quadratic attention is too resource-intensive. LinMulT (OOWFR) achieved the highest average R_{acc} of 0.909 and an R^2 of 0.372. Overall, LinMulT strikes a great balance, delivering strong results while being more resource-friendly.

Method (Feature)	O ↑	C ↑	E ↑	A ↑	$\overline{N}$ ↑	R_{acc} ↑	R^2 ↑
Prior	0.883	0.875	0.878	0.894	0.877	0.881	0.000
MulT (OOB)	0.905	0.902	0.908	0.904	0.902	0.904	0.310
LinMulT (OOB)	0.905	0.901	0.908	0.905	0.901	0.904	0.310
MulT (WFR)	0.901	0.900	0.895	0.900	0.896	0.898	0.228
LinMulT (WFR)	0.900	0.899	0.894	0.901	0.896	0.898	0.233
MulT (OOWFR)	0.908	0.908	0.910	0.907	0.908	0.908	0.369
LinMulT (OOWFR)	0.910	0.907	0.911	0.908	0.908	0.909	0.372
CR-Net [101]	**0.9195**	**0.9218**	**0.9202**	**0.9177**	**0.9146**	**0.9188**	–

Table 5.4: Personality perception (trait-wise and average) R_{acc} performance results on the First Impressions V2 dataset. ↑ shows that higher values are better. The best results for the evaluated methods and those found in the literature are highlighted in bold.

In comparison with the state-of-the-art, CR-Net [101] still gives better results compared to the evaluated architectures, feature sets, and training strategies. However, CR-Net works under a particular and different condition I am trying to avoid, i.e., it uses raw visual data (from the entire scene and the person's face) that can be sensitive to appearance-based features. In contrast, the evaluated approaches use AUs and FAb-Net [172] features as visual inputs that do not capture identity-specific information according to the literature [3]. Additionally, the evaluated approaches and features have the potential to enhance generalization across different datasets, domains, and scenarios.

The Q–Q plots were used to verify the normality of the data, ensuring the validity of the subsequent statistical tests (Figure A.3). Two-sided paired t-tests were conducted to assess the significance of performance differences between LinMulT and MulT models. The results, detailed in Table A.4, show statistically significant differences between LinMulT (OOWFR) and MulT (OOWFR) for each trait.

5.4.3 *Computational Complexity*

I measured the computational complexity differences between MulT and LinMulT by comparing inference time on both GPU and CPU, memory utilization (including both allocated and reserved memory), and the number of floating point operations (FLOPs) per sample as a hardware-independent complexity metric. My findings present that LinMulT requires significantly less memory and computational power compared to MulT, while the performance remains the same or shows only minimal differences. This advantage stems from LinMulT's linear attention mechanism, which reduces the number of computations, leading to faster training times and improved scalability.

The code used in these experiments is written in PyTorch. I utilized an NVIDIA Titan RTX GPU with 24GB of VRAM and an AMD Ryzen Threadripper 1920X CPU. I used OOWFR features of the First Impression V2 dataset. Data loading was handled using 3 parallel workers, as increasing the number of workers did not result in any further reduction in batch generation time.

THEORETICAL COMPLEXITY The linear attention mechanism in LinMulT offers significant computational advantages over the traditional softmax-based attention in MulT, particularly by achieving linear complexity with respect to sequence length. While the softmax attention mechanism in MulT has a complexity of $\mathcal{O}(L^2D)$—where L represents the sequence length and D is the dimensionality of the queries, keys, and values—LinMulT reduces this to $\mathcal{O}(LD^2)$ by leveraging feature map-based dot product attention. This approach approximates attention by applying a feature map function $\Phi(.)$ to the queries and keys, computing $\boldsymbol{V}' = \text{normalize}(\Phi(\boldsymbol{Q})\Phi(\boldsymbol{K})^T)\boldsymbol{V}$ instead of the traditional $\boldsymbol{V}' = \text{softmax}(\boldsymbol{Q}\boldsymbol{K}^T)\boldsymbol{V}$. As a result, the attention mechanism scales linearly with sequence length L, resulting in significantly lower memory consumption and computational overhead for long sequences. This design ensures that LinMulT is more efficient in environments where near-real-time processing and limited resources (such as GPU units or memory) are critical constraints.

However, it is important to note that the theoretical complexity of LinMulT still retains a quadratic on the model dimensionality D due to matrix operations involving feature maps.

INFERENCE TIME ACROSS DIFFERENT BATCH SIZES An important consideration is that while single-sample performance is often the main focus in real-time applications, the ability to scale efficiently with higher batch sizes allows interactive systems to handle aggregated data processing or multiple inputs simultaneously, ensuring consistent responsiveness. In group interactions or multiplayer interactive applications, participants' inputs can be efficiently processed when batched, making it important to assess how well the system scales in these situations.

The bar plots on Figure 5.4 compare the inference times of MulT and LinMulT across different batch sizes (1, 2, 4, 8) on both GPU and CPU. The analysis window is 15 seconds. The left plot shows inference time on a GPU, while the right plot focuses on CPU performance.

By showing both the mean inference times and the associated standard deviations across different batch sizes, the plots emphasize the stability and reliability of LinMulT under various conditions. I used 10 iterations as a warmup phase before conducting 10 trials of inference, from which I calculated the mean and standard deviation values. On top of the bars, I show the speedup from MulT to LinMulT, which is calculated by dividing MulT's inference time by LinMulT's inference time.

On the GPU, while LinMulT initially lags slightly behind MulT for smaller batch sizes due to the highly parallel nature of GPU processing, MulT performs efficiently in these cases. However, as the batch size increases, LinMulT's scaling advantage becomes evident—starting from batch size 4, LinMulT became 10% faster, further increasing to 2.2x speedup at batch size 8. This demonstrates LinMulT's ability to outperform MulT at larger batch sizes, capitalizing on its linear complexity for more efficient scaling. On the CPU, the speedup is even more pronounced, starting at 8.2x for a single sample and reaching up to 16.2x for batch size 8. The 134 ms (234 ms) inference time on GPU (CPU) makes LinMulT well-suited for near-real-time multi-actor applications, where quick responsiveness and computational efficiency are essential.

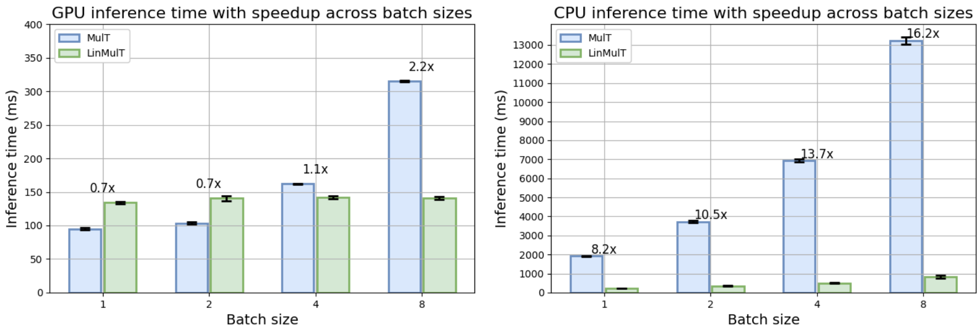

Figure 5.4: Mean inference time comparison of MulT and LinMulT across different batch sizes on GPU and CPU, with the standard deviations visualized as error bars. The speedup from MulT to LinMulT is displayed on top of the bars for each window size.

INFERENCE TIME ACROSS DIFFERENT WINDOW SIZES Figure 5.5 displays the inference time for both MulT and LinMulT across varying analysis windows (7.5s, 15s, 22.5s, and 30s) on both GPU and CPU. The batch size is 1. These windows represent different temporal lengths for a single input sample, allowing us to observe how the models scale with larger input durations. The high level of GPU parallelization makes the scaling difference between MulT and LinMulT less noticeable for smaller windows. On the CPU, however, the difference is more pronounced: MulT experiences a significant rise in inference time as the window size increases, while LinMulT remains relatively more stable. This analysis demonstrates how LinMulT's performance scales better for longer windows, particularly on the CPU, where its linear complexity with respect to sequence length becomes a key advantage. The speedup ratios steadily increase as the window size grows, starting at 2.0x for a 7.5-second window, reaching 8.2x for the 15-second clip, and peaking at 19.8x for the 30-second window. LinMulT's mean inference time on GPU (CPU) is 134 ms (336 ms) even with a 30-second analysis window, which is still acceptable for near-real-time applications.

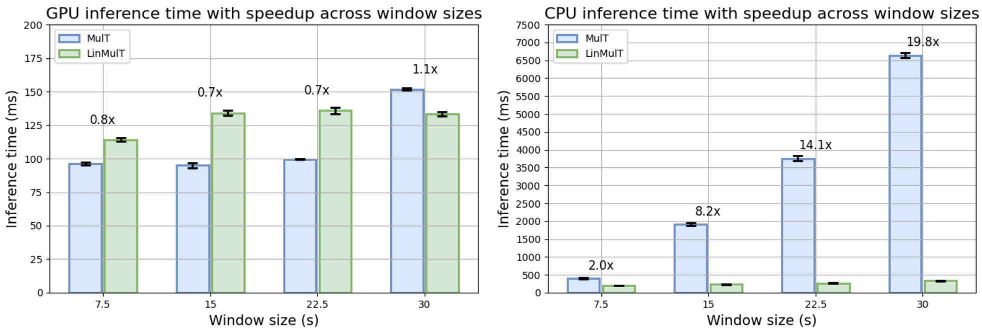

Figure 5.5: Mean inference time comparison of MulT and LinMulT across different window sizes on GPU and CPU, with the standard deviations visualized as error bars. The speedup from MulT to LinMulT is displayed on top of the bars for each window size.

gpu memory usage during inference In PyTorch, *allocated memory* refers to the actual amount of GPU memory used during the forward pass to store the tensors and perform computations. *Reserved memory*, on the other hand, is the total amount of GPU memory that PyTorch has reserved, which is typically higher than the allocated memory. This reserved memory helps optimize runtime by minimizing the need for frequent memory re-allocations and deallocations, thus improving efficiency during model training and inference.

Figure 5.6 compares the GPU memory utilization of MulT and LinMulT across different batch sizes and analysis window durations. The left plot visualizes both peak allocated and reserved memory for both models for a fixed 15-second window duration, illustrating a significant reduction in memory usage for LinMulT. LinMulT reduces allocated memory by 65.5% to 85.3% compared to MulT, depending on the batch size, and similarly reduces reserved memory by 66% to 85.1%. The reductions become more pronounced as the batch size increases.

The right plot displays the GPU memory utilization across different analysis window durations with a fixed batch size of 1. LinMulT consistently outperforms MulT in terms of memory efficiency, reducing allocated memory by 59.7% to 84.8%, and reserved memory by 55.8% to 86%, as the window size increases. These memory savings highlight LinMulT's scalability, especially for applications that require long temporal input windows, such as video processing or behavioral analysis. The ability to maintain low memory usage even with longer input sequences makes LinMulT highly efficient in resource-constrained environments.

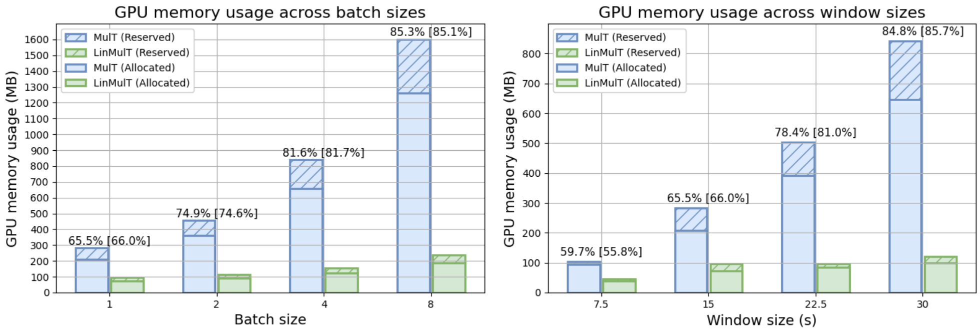

Figure 5.6: GPU memory utilization comparison of MulT and LinMulT across different batch sizes (left) and window sizes (right). The left plot shows memory usage with varying batch sizes (1, 2, 4, and 8) and a fixed analysis window of 15 seconds, while the right plot shows memory usage across different analysis windows (7.5s, 15s, 22.5s, and 30s) with a fixed batch size of 1. Both peak allocated and reserved memory are visualized. LinMulT demonstrates substantial memory savings over MulT. Percentage reductions for both allocated and [reserved] memory are shown on top of the bars.

FLOATING POINT OPERATORS Figure 5.7 compares the Floating Point Operations (FLOPs) of MulT and LinMulT during both inference and training across varying window sizes. The bar plot illustrates the FLOPs required for processing a single sample in both models. LinMulT demonstrates a significant reduction in FLOPs, with the inference FLOPs decreasing by 51% to 80.9% compared to MulT, depending on the window size. Similarly, training FLOPs are reduced by 51.4% to 81.1%. These reductions become more pronounced as the window size increases, making LinMulT more computationally efficient for larger temporal windows, especially during training.

The FLOPs reduction highlights the substantial computational advantage LinMulT offers over MulT. For 15-second samples, LinMulT achieves a 67.8% reduction in inference FLOPs from 2.1e+10 to 6.8e+9 and a 68.1% reduction in training FLOPs from 6.4e+10 to 2e+10. For a 30-second analysis window, LinMulT reduces inference FLOPs from 7.15e+10 to 1.36e+10, an 80.9% decrease, and training FLOPs from 2.17e+11 to 4.09e+10, an 81.1% reduction. This dramatic decrease not only speeds up processing but also reduces overall energy consumption, making LinMulT highly favorable for resource-constrained environments such as IoT applications.

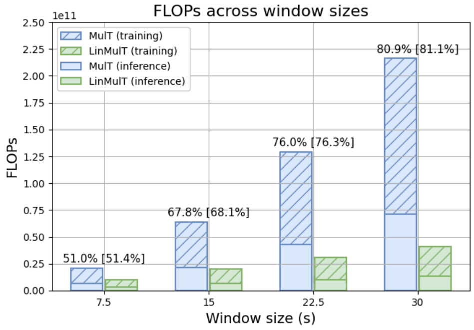

Figure 5.7: Floating point operations (FLOPs) comparison of MulT and LinMulT during inference and training across different window sizes (7.5s, 15s, 22.5s, and 30s). Percentage reductions for both inference and [training] FLOPs are displayed on top of the bars.

5.5 SUMMARY

In this work, I introduced LinMulT, a variant of the MulT model that employs a linear complexity attention mechanism to better handle long sequences. Additionally, I extended both MulT and LinMulT to support more than three input sequences, allowing for more diverse multimodal data integration. To evaluate the models, I conducted experiments on two tasks—MSA and APP—using three datasets: CMU-MOSI, CMU-MOSEI, and the FI dataset.

For the MSA task, LinMulT demonstrated significant improvements when transitioning from the OOB to the OOWFR feature set on all datasets. For example, on CMU-MOSI, LinMulT (OOWFR) achieves an Acc7 of 0.353 and an F1 of 0.824, which are very close to MulT's Acc7 of 0.360 and F1 of 0.832. Correlation values are identical (0.745). Similarly, on CMU-MOSEI, LinMulT (OOWFR) achieves an Acc7 of 0.524 and an F1 of 0.746, comparable to MulT's respective scores of 0.522 and 0.744. Despite these minimal differences in performance, LinMulT offers significant computational advantages.

For the APP task, LinMulT showed a slight performance edge over MulT on the ChaLearn First Impressions V2 (FI) dataset. For instance, using the OOWFR feature set, LinMulT achieves an average R_{acc} of 0.909 and an R^2 of 0.372, slightly surpassing MulT's R_{acc} of 0.908 and R^2 of 0.369. While the differences are small, LinMulT's consistency across metrics underscores its effectiveness in personality perception tasks.

I also contextualized these results by comparing them to state-of-the-art methods from the literature. Although top-performing models like SSE-FT and CR-Net surpassed both MulT and LinMulT, this can be attributed to their use of visual features that include identity-specific information and background context. In contrast, I restricted the visual inputs to appearance-invariant features (AUs and FAb-Net), which likely contributed to the performance gap. My approach offers better generalization across datasets and scenarios by avoiding identity-specific information.

I verified that the predictions exhibited normality through Q-Q plots, and subsequently, two-sided paired t-tests revealed statistically significant differences in performance between the models.

Beyond performance, LinMulT exhibited significant computational advantages, particularly in near-real-time scenarios. For single-person interactions with a batch size of 1, LinMulT's inference time was 134 ms on GPU and 234 ms on CPU, making it highly suitable for near-real-time applications. When handling larger batch sizes for multi-person group sessions, LinMulT showed its strength in batched inference, achieving a 2.2x speedup over MulT on GPU with a batch size of 8, and an impressive 16.2x speedup on CPU.

In terms of scalability, LinMulT demonstrated remarkable efficiency gains. For a single 15-second sample, LinMulT reduced reserved GPU memory by 66% compared to MulT. When the batch size was increased to 8, the reserved memory reduction reached up to 85.1%. Additionally, as the clip duration increased from 7.5 seconds to 30 seconds, the reserved memory reduction also grew, ranging from 55.8% to 85.7%, thanks to LinMulT's advantageous scaling properties. For a 15-second analysis window, LinMulT reduces inference FLOPs by 67.8% (from 21.2 billion to 6.8 billion) and training FLOPs by 68.1% (from 64.1 billion to 20.4 billion). These reductions become even more pronounced with longer windows: for a 30-second duration, LinMulT achieves an 80.9% reduction in inference FLOPs (from 71.5 billion to 13.6 billion) and an 81.1% reduction in training FLOPs (from 216.6 billion to 40.9 billion).

The results demonstrate that LinMulT is well-suited for environments with limited computational resources, where optimizing speed, memory, and processing power is crucial. This study highlights its performance and efficiency, showcasing its low memory usage, reduced FLOPs, and scalability. These qualities make it an excellent choice for near-real-time applications and iterative research in resource-constrained settings.

6 GROUP AFFECT DURING DYADIC AND SMALL GROUP INTERACTIONS

Personality expression is shaped by various factors, including social context and interpersonal relationships. When describing the personality of an individual, it is common to utilize a trait-based description [111], in which an individual's personality is described by the FFM (Section 2.2.1). In AC and PC, personality traits can be assessed through self-reporting (APR) or external observation (APP). While previous work focused on APP, this study also incorporates APR by utilizing self-reported traits.

Personality perception is influenced by a wide set of sources [49], including specific attributes of speech such as speech rate [147], voice quality, and intonation [113]. Non-verbal cues, such as gaze [81], back channels [23], and body language [138], also play a significant role in how personality is perceived [167].

The relationship between personality and performance is supported by studies both at the individual [183, 130] and group levels [131, 118] across various tasks. Personality traits generally exhibit long-term stability but are not fixed [6, 30, 106]. Studies have shown that within-person variation of traits can show significant short-term change [56]. However, the resolution at which this variation can be examined is limited by the frequency of questioning which may be used in current experience sampling methods [31].

Additionally, groups often display convergent behavior in emotions [122, 66] and affect [65, 17, 16]. A conceptual model by Roberge [137] combines individual and collective personality within the Big Five framework, suggesting dynamic trait behavior during group interactions. This model suggests that personality traits can shift between being dominant or less expressive during group interactions. Additionally, it shows that group members' traits adapt dynamically based on the task at hand.

Similarly to group affect, I can also consider the Group Personality (GP). The link between the two is also evident in considering how I estimate the perceived personality state: I take into account information such as facial expression, smiling, and characteristics of the voice and map these to an external viewer's perception of the Big Five traits.

In considering the theoretical perspectives, it needs to be clear that in the following study, I am not able to predict an individual's long-term stable personality traits as considered in the classical sense [6]. Rather, what I extract are features relevant to expressed behaviors which can then be mapped back to features commonly associated with these traits. These are influenced by the nature of the situation [168], and the differing nature of the interpersonal relationships between peers [179]. This is in line with recent interactionist approaches to personality, particularly the view taken from trait activation theory [160].

Unlike traditional machine learning research, which typically involves benchmarked datasets and incremental model improvements, this study focuses on APP in small groups and dyads during task-oriented interactions and aims to answer specific research questions about personality expression and group dynamics, supported by a comprehensive analysis. This area is more commonly addressed in PP, but its automatic detection is a novel research area within the machine learning community.

This study focuses on dyadic and small-group interactions to address the following research questions:

- How reliably does the perceived personality estimator perform in out-of-domain sessions compared to human annotators?
- Does the perceived personality of members within a group cluster?
- Is the group-level perceived personality a stronger predictor of group performance than self-reported personality?

Personality expression is known to be influenced by the situation experienced by an individual [18, 68, 109, 152, 159]. This dynamic nature of personality suggests that individuals adapt their behavior based on the context and social environment. Additionally, in social interactions, participants often mirror their partners' behaviors, which can further shape their expression [26].

I built on these foundational concepts by examining personality dynamics in group settings. I utilized multiple datasets: the MULTISIMO dataset to validate the generalizability of LinMulT, the UDIVA dataset for dyadic interactions, and the ELEA and AMI meeting corpus for small groups. The study aims to provide a comprehensive analysis of how personality is expressed and perceived in collaborative tasks. I investigated the emergence of GP by examining time-averaged personality states for clustering effects. Finally, I compared the predictive power of perceived versus self-reported traits for group performance.

6.1 PROPOSED METHOD

To analyze the perceived personality of individuals in small groups and dyads during task-oriented interactions, I utilize video recordings of these sessions. These videos provide three modalities: visual information, speech, and transcripts. I propose using Multimodal Transformer with Linear Complexity Attention Mechanism (LinMulT) [P1], which is designed to efficiently process multimodal data by utilizing cross-modal transformer encoders to transform input from one modality to another. The outputs from each modality are then combined using a self-attention transformer encoder, forming distinct branches within the network. To alleviate the computational burden of these transformer encoders, I employed a special attention mechanism (described in Section 3.2.4), which operates with linear complexity with respect to the input sequence length. The resulting vectors are then global average pooled, concatenated along the feature dimension, and passed through dense layers for prediction.

The network is pre-trained on the ChaLearn First Impressions V2 (FI) dataset for the Automatic Personality Perception (APP) task. It processes input within a 15-second temporal context, providing perceived OCEA$\overline{\text{N}}$ (Openness, Conscientiousness, Extraversion, Agreeableness, and Emotional Stability) predictions for each window. To handle long video sessions, I employ a sliding window approach with a stride duration of 1 second. This method yields Big Five values for every second, based on the past 15 seconds, creating a time series of perceived personality estimates for each participant.

However, the personality estimator is trained on samples where participants are actively speaking. To maintain reliability, I focus only on time steps where participants speak at least 20% of the analysis window. Estimates from non-speaking intervals are ignored, ensuring that the input samples are structurally similar to those used during training.

From the OCEA$\overline{\text{N}}$ estimates, I calculate metatraits as described in Section 2.2.2. These metatraits are plasticity (PLA) and stability (STA), the former combining the traits of openness and extroversion and the latter conscientiousness, agreeableness, and emotional stability. These metatraits are expressed as linear combinations of the relevant Big Five traits with equal weights. An example of the extracted metatrait trajectories for one group is shown in Figure 6.1.

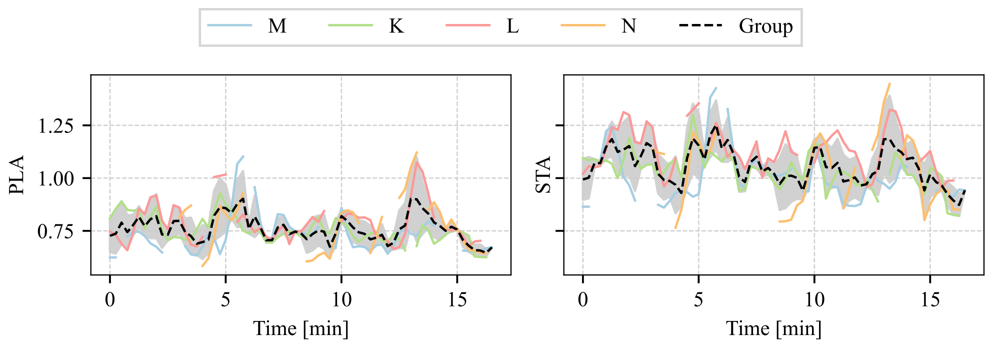

Figure 6.1: Illustrative sample of the temporal changes in the perceived personality metatraits of plasticity (left) and stability (right). Colored curves: Trait estimation for individual participants, where each participant is assigned a letter in place of a name for anonymization in the dataset. Missing line segments: non-speaking intervals. Dashed line: average over participants. The grey shaded area gives the standard deviation of the individual values. Perceived personalities tend to move together over time.

6.2 METHODOLOGY

6.2.1 *Datasets*

Four video datasets were utilized in these experiments.

MULTISIMO: MULTImodal and MULTIparty Social Interactions MOdelling (MULTISIMO) [95] is a triadic dataset, however, the majority of interactions are dyadic. It involves scenarios where two participants collaborate to solve a quiz while assisted by a facilitator. The dataset includes 18 sessions, spanning approximately 3 hours. Each session typically lasts around 10 minutes. The corpus includes audiovisual recordings, full speech transcriptions, dialogue structure annotations, feedback from the facilitator, laughter-type annotations, and partial gesture annotations. Additionally, it contains survey materials, such as personality tests and experience assessment questionnaires filled out by all participants.

For personality assessment, the dataset provides two types of annotation. It includes self-reported personality data, where all 39 participants completed the BFI-2 before the recordings, providing individual scores and percentiles on the Big Five traits. It also includes external observations, where 8 raters listened to 79 audio slices and used the BFI-10 to assess the participants' perceived personality traits.

ELEA: The Emergent Leader (ELEA) dataset [140] is a multimodal corpus designed to analyze emergent leadership in small groups. It consists of video recordings of groups with 3 to 4 participants, totaling 102 participants. The groups were formed into 21 four-person teams and 6 three-person teams. In this dataset, participants engage in a collaborative task where they rank the importance of 12 items for survival in a hypothetical winter plane crash scenario. The task involves an initial 5-minute individual ranking (not recorded), followed by a 15-minute group discussion to reach a consensus ranking. The average duration of these group discussions is approximately 14.61 minutes, ranging from 8 to 19 minutes.

The dataset includes audio and video recordings. The audio was captured using a Microcone commercial microphone array placed at the center of the discussion table, which automatically segments speakers and provides audio for prosodic cue extraction. Video recordings were made using two Logitech Webcam Pro 9000 cameras with a frame rate of 30 fps. The full corpus spans about 10 hours of audio and video content. Additionally, it includes manual transcriptions from the recordings in English.

The ELEA dataset provides annotations for self-reported and perceived personality traits, concepts related to leadership, and group performance in the survival task. Not all sessions in the dataset include all three modalities; some lack video recordings and have only audio, while others are missing transcripts.

Therefore, I focused on analyzing 17 sessions (identified as 12, 14, 16, 17, 21–26, 28, 30, 32, 34, 36, 38, and 39) that had all three types of data: audio, video, and text.

ami: The AMI meeting corpus [110] is a multi-modal dataset focused on small-group interactions. It involves a series of four sessions where groups collaborate to design a new remote control, starting with an initial kick-off meeting (A), followed by meetings on functional design (B), conceptual design (C), and finishing with a detailed design meeting (D).

In total, the dataset contains 138 sessions with a mean duration of 31.5 minutes. There are 34 meetings for task A, 35 for task B, 35 for task C, and 34 for task D. Each group would have 4 participants. After the video sessions were recorded, automatic speech-to-text was used to generate transcripts. The dataset includes a variety of annotations, such as dialogue acts, topic segmentation, abstractive and extractive summaries, named entities, individual actions, and focus of attention. However, some meetings were removed from the analysis due to issues with synchronization, audio and transcript quality.

I focused on sessions that included video of the participants, recorded speech, and corresponding transcripts. This reduced the number of usable scenario-based sessions from 138 to 131, which still represents about 95% of the original dataset—30 groups from A, 35 from B, 34 from C, and 32 from meeting D.

udiva: Understanding Dyadic Interactions from Video and Audio Signals (UDIVA) is a multimodal, multiview, non-acted dataset that consists of face-to-face dyadic interactions, totaling 90.5 hours of recordings across 188 sessions with 147 participants [119]. This dataset is multilingual, with Spanish being the most common language, followed by Catalan and English. The participants sit at a table and are individually recorded during the sessions. There are 4 different tasks: Animals, Lego, Talk, and Ghost. The average duration of these sessions is 6 minutes and 41 seconds. These tasks are designed to elicit varying levels of collaboration and competition, providing a rich context for studying social interactions. The dataset includes detailed metadata, such as sociodemographic information and self-reported personality traits

In the case of Animals task the collaboration is achieved by the process of questions and answers made by the participants to discover the participants' assigned animal. While the Lego task is collaborative, it does not require discussion or high levels of engagement between the participants. In this case, the target object to construct is given, and only the process of construction is required. The Talk task presents an open conversation with no specific goal to be achieved as a group, while the Ghost task is purely competitive, with both participants aiming to optimize their score against the other.

For my analysis, I focus on the English language component of the dataset, which involves 18 participants forming 14 dyads. Each session averages approximately 6 minutes and 40 seconds, capturing the interactions through individual recordings.

6.2.2 *Feature Extraction and Preprocessing*

To analyze the long video clips, a 15-second moving window is applied, allowing features to be extracted for each analysis window. These features are then fed into the pre-trained LinMulT personality estimator for inference. Specifically, the audio modality involves extracting eGeMAPS features using the OpenSMILE software. From the video frames, AUs are obtained using the OpenFace software to avoid influence due to visual factors such as clothing, location, and background objects visible in the video. For more details on the extracted features visit Section 4.

In addition to the multimodal features, I computed a set of group-level features to investigate the relationship between individual personality scores and group performance. These features encompass metatraits (PLA and STA) and Big Five traits (OCEA$\overline{\text{N}}$), aggregated through various statistical measures including average, standard deviation, moving standard deviation, and sum of squared differences.

6.2.3 *Experiments*

generalisability testing: To evaluate the performance of the pretrained personality trait estimator, I conducted experiments on the MULTISIMO dataset. This involved calculating inter-rater correlation factors using the Intraclass Correlation Coefficient (ICC(2,k)), which measures the consistency of ratings from multiple raters. ICC(2,k) is specifically used when raters are randomly selected and each subject is rated by the same set of raters, providing a measure of reliability for continuous or ordinal data [145]. It is formally defined as:

$$\mathrm{ICC}(2,k) = \frac{\mathrm{MSB} - \mathrm{MSE}}{\mathrm{MSB} + (k-1)\mathrm{MSE} + \frac{k}{n}(\mathrm{MSJ} - \mathrm{MSE})}, \tag{6.1}$$

where MSB is the mean squared between subjects, MSE is the mean squared error, MSJ is the mean square between raters, k is the number of raters, and n is the number of subjects.

Additionally, I employed a Two-One-Sided T-test (TOST) to assess the equivalence of errors between the trait estimation model and human raters. TOST is used to validate whether the difference between two means falls within a predefined equivalence interval, allowing us to determine if the model's errors are statistically equivalent to those of human raters.

group affect: To determine whether personality traits meaningfully converge within groups, I employed a Permutational Multivariate Analysis of Variance (PERMANOVA). This non-parametric test evaluates the null hypothesis (H_0) that there are no differences in group centroids or dispersion (i.e., groupings are random), against the alternative (H_a) that observed clusters reflect meaningful differences in personality traits—without requiring normality of the data [9]. The p-value quantifies the likelihood of observing the data under H_0, with values below a significance threshold rejecting randomness.

To validate robustness, the test was repeated 1,000,000 times. In each iteration, participants were randomly reassigned to artificial groups, simulating H_0 (no inherent structure). This creates a null distribution of the test statistic, ensuring that observed significance reflects true group differences rather than chance. A high number of permutations minimizes estimation error, yielding reliable p-values.

The pseudo F-ratio quantifies group differences as:

$$F = \frac{SS_A/(a-1)}{SS_W/(N-a)}, \tag{6.2}$$

where SS_A denotes the between-group variance, SS_W is the within-group variance, a is the total number of groups in the study, N is the total number of observations. A high F-ratio indicates that the between-group variance is much larger than the within-group variance, suggesting strong differences between groups. Conversely, a low F-ratio indicates weaker differences between groups.

GROUP PERFORMANCE: For predicting group performance, I used a Gradient Boosted Tree (GBT) model, which is a powerful ensemble learning technique that iteratively combines weak decision trees to improve predictive accuracy [60]. The model was trained on aggregated group-averaged perceived and self-reported personality trait features to determine whether perceived personality traits are better predictors of group performance compared to self-reported traits. It was configured with 20 estimators (trees), a learning rate of 0.1, and a maximum depth of 3.

Group performance prediction is conducted as a regression task. The GBT model was trained using leave-one-group-out cross-validation to address the limited dataset size. The performance metric under evaluation is the differences in ordering of the item importance determined by the group compared to an expert.

6.3 RESULTS

The findings related to each of the three research questions are presented and discussed in detail in the following subsections.

6.3.1 *Generalisability Testing*

> How reliably does the perceived personality estimator perform in out-of-domain sessions compared to human annotators?

To assess inter-rater reliability, I calculated the ICC(2,k) for 8 external observers in the MULTISIMO dataset (Table 6.1). Using guidelines from [94], agreement levels were moderate for Openness (0.550), Conscientiousness (0.624), and Emotional Stability (0.573), good for Extraversion (0.807), and poor for Agreeableness (0.426). Metatraits showed good (Plasticity: 0.797) to moderate (Stability: 0.615) agreement, confirming acceptable reliability for most traits.

	ICC(2,k)	F	df1	df2	p-val	CI95%
Openness	0.550	2.25	78	546	<.001	[0.39, 0.68]
Conscientiousness	0.624	2.91	78	546	<.001	[0.49, 0.74]
Extraversion	0.807	5.97	78	546	<.001	[0.73, 0.87]
Agreeableness	0.426	1.79	78	546	<.001	[0.22, 0.59]
Emotional Stability	0.573	2.55	78	546	<.001	[0.42, 0.70]
Plasticity	0.797	5.43	78	546	<.001	[0.72, 0.86]
Stability	0.615	2.95	78	546	<.001	[0.47, 0.73]

Table 6.1: ICC(2,k) values for big-five traits and metatraits from 8 external evaluators surveyed in the MULTISIMO dataset.

Next, I compared the model's performance to human raters using a TOST equivalence test (Table 6.2). For the Big Five traits, the model's absolute error was statistically equivalent to at least half of the human raters. For metatraits, equivalence was observed with 3 out of 8 raters for Plasticity and 7 out of 8 for Stability. The model's mean absolute error (MAE: 0.26 ± 0.11) closely matched human raters' MAE (0.19 ± 0.07), indicating comparable reliability.

While human annotators show high variability, the model achieves reliability comparable to the majority of raters. This suggests it can be confidently applied to similar datasets with video, audio, and text modalities. These results align with prior work on automated personality assessment [166], demonstrating practical utility for real-world applications.

Trait	bound	Rater							
		1	2	3	4	5	6	7	8
Openness	0.09	.006	< .001	.165	< .001	.004	.003	< .001	.010
Conscientiousness	0.10	.747	.04	< .001	.003	.987	.859	.019	.911
Extraversion	0.12	.131	.012	.002	.001	.959	.007	< .001	.457
Agreeableness	0.10	< .001	<.001	.208	.044	< .001	< .001	< .001	< .001
Emotional Stability	0.13	.063	< .001	< .001	.733	< .001	< .001	.611	.007
Plasticity	0.17	.603	.477	.037	.996	.253	< .001	< .001	.754
Stability	0.26	.013	< .001	.001	.005	.514	.022	< .001	.028

Table 6.2: P-values from the independent TOST analysis for each personality trait and metatraits. The specified bound for testing equivalence is defined as the standard deviation of the error across all human raters for the specified trait.

6.3.2 *Group Affect*

Does the perceived personality of members within a group cluster?

The distribution of participant session averaged metatraits are shown for ELEA in Figure 6.2, revealing a notable clustering of perceived traits. This visual evidence complements existing literature on group dynamics and further motivates my investigation into group-level personality convergence. The clustering observed in perceived traits, which reflect dynamic behavioral responses during collaborative tasks, suggests that these measures may capture nuances of group interaction not evident in static self-reports.

To quantify this observation, a Permutational Multivariate Analysis of Variance (PERMANOVA) was performed across datasets. The test evaluates whether group centroids (multivariate personality profiles) differ significantly from random chance. The complete results, including additional metrics, are available in the appendix Table A.5, while the key F-ratio values are highlighted in the following paragraphs.

ELEA: For the groups in the ELEA there is a statistically significant difference in the group centroids. I observe statistics for the five- and two-factor personality models of $F = 4.31$ and $F = 4.59$ respectively, both with $p < .001$. However, these differences are not observed in the case of the self-reported personality of the ELEA participants (Meta: $F = 0.55$, OCEA$\overline{\text{N}}$: $F = 0.92$). Perceived traits—shaped by interactions during collaborative tasks—better reflect shared group dynamics than self-reports.

AMI: All meetings (A-D) showed significant clustering ($F_a = 4.41$, $F_b = 3.97$, $F_c = 7.92$, and $F_d = 7.08$ for the Big Five, and $F_a = 3.26$, $F_b = 5.20$, $F_c = 6.94$, and $F_d = 5.84$ for the Metatraits) indicating personality convergence across sequential collaborative tasks. Each of these meetings has $p < .001$.

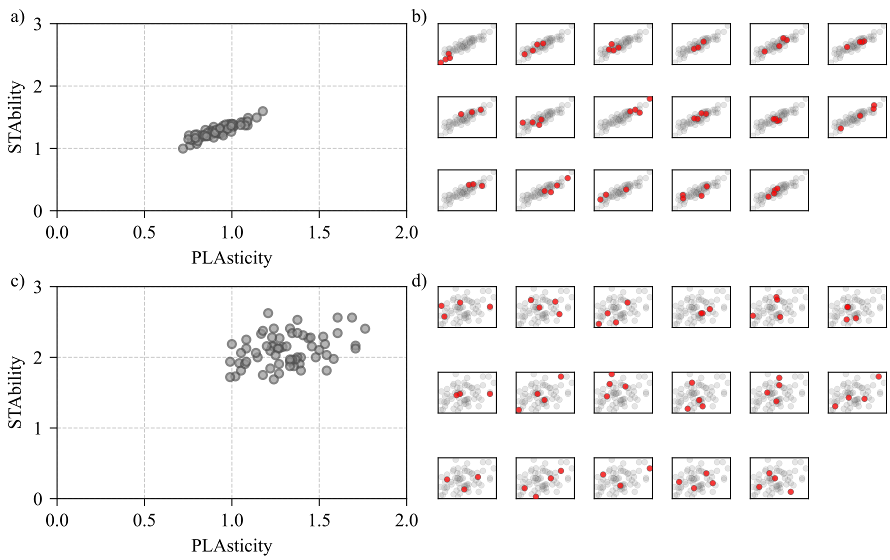

Figure 6.2: Stability and plasticity values. a) Perceived personality averaged over the full session duration. b) Each subplot shows the original averages as seen in a). c) Self-reported metatraits. d) Each subplot shows the original averages as seen in c). Traits corresponding to the members of groups are highlighted by red circles. I observe that the perceived personality of group members tends to form closer clusters than in the self-reported case.

UDIVA: Statistically significant clustering occurred only in the Animals task ($F = 6.50$ for Meta, and $F = 5.69$ for OCEA$\overline{\text{N}}$), while Ghost, Talk, and Lego tasks showed weaker effects ($F = 1.94 - 2.28$) despite the observed p-values.

The resulting histograms for the permutation test are shown in Figure 6.3, with the Big Five results in red and the Metatraits in blue.

For ELEA and AMI, sustained collaboration (e.g., winter survival or designing a remote control across meetings) likely amplifies mutual influence, as participants align behaviors to achieve shared objectives. This aligns with interactionist perspectives, where situational demands activate trait-relevant behaviors (trait activation theory, [160]).

The results support the concept of GP as a dynamic, context-dependent construct. While long-term personality traits remain stable [6], perceived traits reflect situational adaptations, modulated by task engagement and social dynamics [168]. For example, active speaking in collaborative tasks (ELEA, AMI, UDIVA-Animals) enhances observable convergence, whereas passive or competitive interactions (UDIVA-Lego/Ghost) weaken it. This supports the notion that personality states are dynamic and context-dependent [56, 168].

The observed clustering of perceived personality in collaborative tasks may be interpreted through the lens of social exclusion avoidance [38]. This suggests that individuals may adjust their expression to fit in with the group, particularly in situations that require cooperation and shared goals.

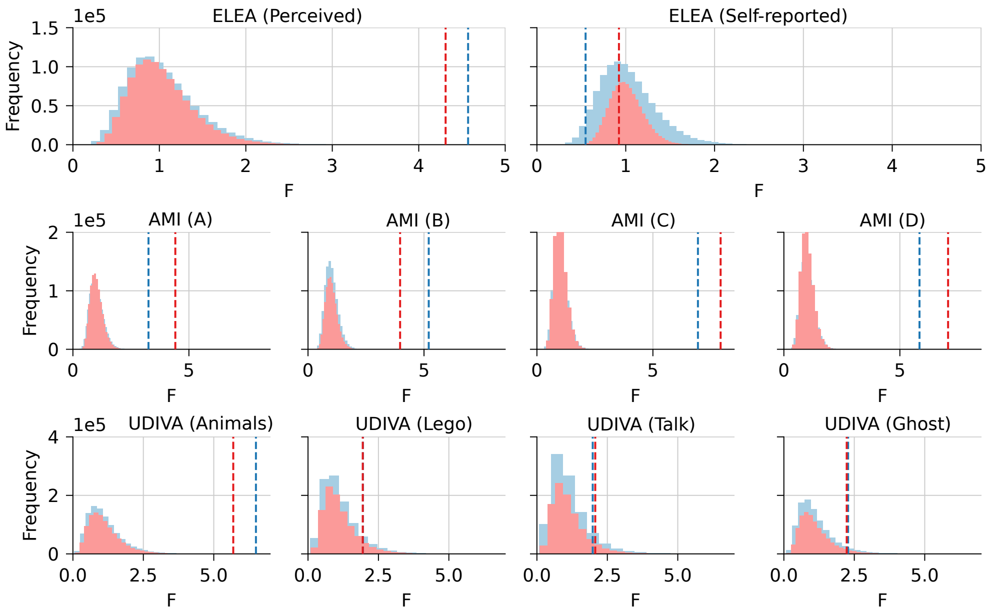

Figure 6.3: Histograms of the permutation test. For each plot, I show the F-ratio of the Metatraits in blue and the Big Five traits in red. The top row shows the ELEA perceived personality (left) and self-reported (right). The middle row presents the results for AMI (kick-off, conceptual design, functional design, detailed design), while the bottom row is for UDIVA (Animals, Lego, Talk, Ghost).

6.3.3 *Group Performance*

> Is the group-level perceived personality a stronger predictor of group performance than self-reported personality?

To investigate the relationship between personality traits and group performance, I focused on the ELEA dataset, which uniquely provides group performance annotations. I trained a GBT model using group-averaged personality traits to predict group performance scores. The model was trained on features derived from both Big Five traits and Metatraits.

I achieved a MSE of 16.17 ± 0.45 with the group average perceived Metatraits and 22 ± 1.26 for the Big Five traits. This is compared to a MSE of 43.88 ± 1.17 in the case of self-reported group average Metatraits and 40.68 ± 1.10 for the Big Five traits. The results show that the model-estimated perceived personality traits, particularly metatraits, were better predictors of group performance than the self-reported scores.

While a large body of literature exists examining the relationship between self-reported personality and performance across a variety of tasks [131, 118, 72, 97], these results add further support to the idea that self-report alone may not be sufficient and in fact, other-rated personality traits may be stronger predictors [130, 114, 148].

6.4 SUMMARY

This research provides a comprehensive analysis of perceived personality in dyads and small groups, utilizing machine learning models across multiple datasets. The study addresses three key questions: the reliability of applied APP method, the clustering of perceived personality within groups, and the predictive power of personality traits for group performance.

Firstly, I used a multimodal linear transformer model for estimating Big Five personality traits from thin slices of recorded meetings. The model's performance was comparable to human raters in the MULTISIMO dataset, demonstrating its potential as a reliable tool for personality assessment in social interactions.

Secondly, using these automated personality estimates, I investigated group dynamics through non-parametric permutation tests. Significant clustering of personality traits was observed in the ELEA and AMI datasets $p < .001$, while in UDIVA, only 1 task out of 4 showed significant results. This task-dependent clustering suggests that collaborative activities may foster a convergence of expressed traits within groups. Notably, this effect was not observed for self-reported traits in the ELEA dataset, highlighting the unique insights gained from perceived personality measures.

Finally, I compared how well perceived and self-reported traits could predict group performance. Using the winter survival task in the ELEA dataset, I found that group-averaged perceived metatraits (plasticity and stability) were superior predictors of overall group performance compared to self-reported traits.

Several factors should be considered regarding limitations. The AMI dataset's potential confound of prior work relationships among participants may influence group dynamics. Furthermore, the ELEA, AMI, and UDIVA datasets were not originally designed to assess changes in personality states. This limitation restricts the ability to identify specific factors causing the observed changes in personality expression across different tasks. Since the tasks in these datasets were not structured to elicit specific trait behaviors, it is challenging to draw clear conclusions about the drivers of these changes in expression. Lastly, while the proposed multimodal personality state estimator shows promise, further validation with human annotations across a wider variety of social interactions is necessary to fully establish its generalizability and limitations.

These results show that machine learning models can be useful in personality research: automated trait assessments are acceptably reliable and can predict group performance well. The study contributes to our knowledge of personality dynamics in group settings, highlighting how expression can adapt to situational and social contexts. Future work should focus on refining these models and testing them in diverse real-world environments, such as workplaces or healthcare settings, to further explore the impact of personality on teamwork and group outcomes.

7 EYE BLINK DETECTION

Blinking detection is important in affective behavior analysis as it can be an involuntary indication of an emotional state [93] or carry a communicative signal [76], among others. Frequent blinking may signal disagreement, stress, or heightened cognitive load. This information is useful for studying how people interact with each other and computers. In this chapter, I focus on blink presence detection and eye state recognition, to be able to utilize this subtle communication cue in future works.

Detecting eye blinks is a challenging problem that can be used to solve a number of facial analysis tasks—it can reveal deep fake manipulation [100], detect driver drowsiness [4], measure the level of attention and eye fatigue during task performance, and health disorders, just to highlight a few. Most approaches depend on face recognition and frame-wise eye state classification, even with the rise of transformer-based sequence models in many other research areas. However, these methods may struggle to handle cases where blinks occur for only a few frames or where input data are noisy or incomplete.

My contributions are listed as follows:

- I propose BlinkLinMulT, a novel transformer-based model with a linear complexity attention mechanism for eye blink detection, integrating low- and high-level features from video inputs such as RGB texture, iris and eye landmark-based features, Eye Aspect Ratio (EAR), and head pose angles. The model effectively handles challenges like lighting changes and diverse head poses while incorporating motion information.

- Extensive within-dataset and cross-dataset evaluations are performed to quantify the robustness of BlinkLinMulT on holdout samples, and it is found that a single network trained on a union of datasets improves the results obtained on all datasets separately.

- The proposed approach is validated on both blink presence detection and eye state recognition tasks across multiple public benchmark datasets, achieving results that are competitive with or surpass the current state-of-the-art, including a 3.8% improvement on EyeBlink8 and a 4.0% improvement on Researchers' Night datasets.

- I conducted a detailed feature fusion ablation study, demonstrating the individual and combined contributions of various feature sets to blink presence detection performance.

- I showed that the proposed method performs robustly even under extreme head poses, such as when participants are looking down, achieving high F1 scores in challenging scenarios.

To my knowledge, this is the first study to utilize transformer architecture for detecting eye blinks by combining visual features, such as head pose angles.

The chapter is organized as follows: the recent eye blink detection literature is briefly summarized in Section 7.1, then Section 7.2 describes the eye blink detection pipeline and the proposed model. The benchmark datasets, experiments, evaluation metrics, and training details are described in Section 7.3. Then, Section 7.4 presents how the proposed method was performed under different conditions and experimental settings. Finally, closing remarks and future work are outlined in Section 7.5.

7.1 RELATED WORK

The machine learning and computer vision communities are focused on using different visual representations to tackle the blinking detection problem. Three categories emerged in the last decade: (i) feature-based methods use predetermined higher-level features like iris and eye landmark distances, (ii) appearance-based methods utilize low-level eye representations (e.g., RGB texture) and then adopt learning algorithms to classify eye states, and (iii) motion-based methods use input representations that encode motion information or a sequence of eye-related features to determine eye states.

7.1.1 *Feature-Based Methods*

Chen et al. [28] proposed an eye blink detection and gaze estimation method. After the eyes are detected, several image preprocessing procedures are performed to eliminate the noise caused by the changes in normal-light conditions and reflections: pixel gradients, valley-peak field, fitted parabolas, iris mask, and pupil center landmark are calculated, among others. Using an adaptive Starburst extraction algorithm, their proposed technique correctly identifies both the iris and limbus features in various lighting conditions, and then the aspect ratio of the bounding box that contains the iris mask is computed over time. Large values indicate blink events while small values are determined as open-eye states.

Phuong et al. [125] presented a blink detection method based only on eye landmarks. EAR is calculated from the eye landmarks, a dynamic threshold—based on the max and current EAR values—is applied for each participant, which determines the eye blink state.

In [98], Kuwahara et al. estimated the eye fatigue sensitivity from detected spontaneous blinks. They applied a simple moving average filter over the EAR values, then applied Eye Aspect Ratio Mapping to further reduce the noise caused by irregular movements, such as looking down. First, they classified blinks, and then a strong correlation is presented in the experimental results between the median spontaneous blink rate and the time between the objective estimation of eye fatigue and the subject's awareness of eye fatigue.

Kraft et al. [96] calculated relative EAR, which is the current EAR divided by the maximum of the last $n = 2 * \text{FPS}$ values. Then, a set of conditions must be met to select blink candidates, which are classified by applying a threshold.

7.1.2 *Appearance-Based Methods*

Appearance features are extracted using two branches of DNNs in [182]. Convolutional layers extracted features from the eye patch; meanwhile, another DNN model extracted the characteristics of the eye patch vector and reduced the features by use of the fully connected layers directly. Given their different structures, the proposed DCNN and DNN are combined to build a deep integrated model, called DINN. A transfer learning strategy was applied to extract effective abstract eye features and improve the classification capability on small-sample datasets.

In the work of [36], cropped eyes from RGB texture were used as an input to a VGG16 trained from scratch for binary eye blink state classification. Their main contribution was the introduction of a new blink dataset called mEBAL. The samples are recorded with an Electroencephalography (EEG) band, Near Infrared (NIR), and RGB cameras, which further improve on previous datasets. The authors showed that using RGB texture without temporal patterns can lead to great performance if there is quality data to train on.

The RT-GENE dataset [55] is used for gaze estimation in natural environments. Then, gaze estimation and blink detection are integrated into a unified network [32]. They also published the RT-BENE dataset, which contains cropped close eye patches from RT-GENE with eye state annotations. Their dataset covers a wider range of camera–subject distances and head poses, which is beneficial for training DNNs and led to further improving the SOTA results.

EyeNet [134] is proposed for binary eye state classification. They evaluated the model under different conditions (lighting, reflection, appearances, and devices) and also show some improvements over other methods [182].

In the work of [112], a robust, real-time pipeline with auxiliary models was proposed considering rotation compensation and head pose orientations. The CNN produced consistently better results than the support vector machine method in their experiments.

Recently, an additional CNN called 4D was presented in [83]. It is based on the VGG19 architecture with modified hyperparameters. The network is evaluated on the MRL Eye dataset and presented better results than others [169, 161] in the literature.

7.1.3 *Motion-Based Methods*

Motion vectors are applied to blink detection in [46], and then the research is extended in [57]. In the latter, motion vectors are obtained by applying Gunnar–Farneback tracking in the eye region. Eyeblink states are determined by providing a normalized average motion vector with standard deviation and time constraints to a state machine. Motion information is calculated between two frames; in later works, multiple time steps' representation is used instead.

In the work of [78], appearance and motion information are simultaneously utilized. Local Binary Pattern (LBP) visual descriptor is extracted to represent the local eye region. The difference between the LBPs from two consecutive frames is used to encode the motion characteristics of eye blinks. The combination of appearance and movement features had a good impact on the final performance; they are concatenated as the input of LSTM. To deal with the multiple temporal cases, a multi-scale LSTM (MS-LSTM) model was proposed, which uses the outputs of the last few LSTM units jointly by concatenation.

More recently, [20] introduced an approach that is divided into two main phases. First, face detection and eye localization are implemented. Then, within the second phase, eye blink detection is conducted via moving-windowed SVD using pixel energy, and eye blink verification is performed via a 2D Pyramidal Bottleneck Block Network (PBBN), which produces the final predictions. They presented improved performance compared to [36, 78] by using temporal patterns from a simple energy-based feature.

Long-term Recurrent Convolutional Network (LRCN) for eye blinking detection is utilized to expose deep fake videos [100]. They estimated facial landmarks, and, after alignment, the eye region is cropped. The visual feature extractor is a VGG16, while the dynamic information was captured by an LSTM. While the mean resting blinking rate is 17 blinks/minute or 0.283 blinks/second [21], during a conversation this rate increases to 26 blinks/minute, and decreases to 4.5 blinks/second while reading. Their method is focused on the detection of eye blinking in the videos, which is a physiological signal that is not well presented in the synthesized fake videos. The output of the LRCN model is the predicted probability of the binary eye state.

Building upon LRCN, Eye-LRCN was proposed as a novel approach to blink detection and blink completeness detection [33]. The Siamese architectures have proven effective for other problems with high class imbalance [19]. Since most of the eye blink databases available today are class-imbalanced, there are several improvements over the baseline model proposed by Li et al.; e.g., they applied a Siamese architecture for CNN training and they used a bidirectional LSTM instead of an unidirectional one.

7.2 PROPOSED METHOD

Appearance-based methods can extract richer information than feature-based methods. The feature-based method cannot be applied if the quality of input data is not good enough, and face-, landmark-, or iris detectors fail. The method I present tries to complement the literature by combining the above categories.

I propose BlinkLinMulT, a fast transformer-based model designed for two tasks: blink presence detection, which determines if a blink event occurs within a short video sequence, and eye state recognition, which identifies frame-wise whether the eye is open or closed. The model combines low- and high-level features from sequences using linear complexity attention mechanisms.

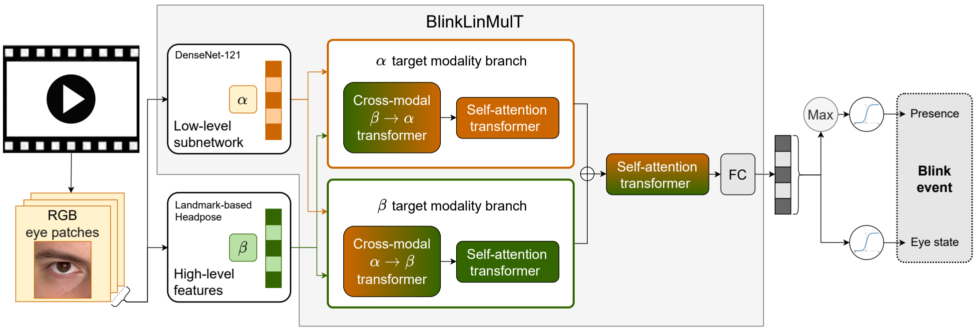

Figure 7.1: BlinkLinMulT: a multimodal transformer architecture for blink presence detection and frame-wise eye state recognition.

For low-level visual features, I use the RGB frame sequence from the video. A face detector is applied to locate the face, followed by a landmark detector to identify the eye regions. Eye state and blink presence are analyzed for each eye of the participant. The eye region is cropped from the image sequence for further processing. To extract frame-wise deep hidden representations, I use DenseNet-121 (introduced in Section 3.1), a CNN pretrained on ImageNet. DenseNet-121 is chosen due to its strong performance in feature extraction tasks and its ability to capture rich spatial information from RGB data [32]. The sequence of RGB eye patches serves as the first input to the proposed model.

I extract high-level features from the cropped eye region's RGB data. Specifically, I predict eye and iris landmarks and use them to calculate the iris diameter, eyelid-pupil distances, and Eye Aspect Ratio (EAR). Additionally, the detected face is used to estimate the person's head pose. These frame-wise high-level features—including eye and iris landmarks, iris diameter, eyelid-pupil distances, EAR, and head pose—are organized into sequences and concatenated along the feature dimension. This concatenated sequence serves as the second input to the proposed model.

The backbone of BlinkLinMulT is built on LinMulT, a fast multimodal transformer designed to process sequence data using linear complexity attention mechanisms (Section 5.2). BlinkLinMulT has two branches. Each branch starts with a cross-modal transformer that fuses information from the two input sequences. This is followed by a self-attention transformer within each branch, which enhances temporal dependencies in the fused sequences. The self-attention mechanism at this stage ensures that temporal relationships are captured independently for each input sequence.

The outputs from both branches are concatenated along the feature dimension. A self-attention transformer processes this combined output to further refine relationships across time steps. This step integrates complementary information from both branches before making the final prediction. To improve efficiency, a linear-complexity attention mechanism (Section 3.2.4) is used throughout the network, reducing computational costs in terms of time and memory. Finally, a time step-wise fully connected layer predicts the output logits for each frame in the sequence.

Frame-wise logits from both eyes are combined using max aggregation, and a sigmoid function is applied to determine the final blink presence and eye states. If individual eye evaluations are required, the aggregation step can be skipped. The proposed model is illustrated in Figure 7.1, and the full inference pipeline is visualized in Figure 7.2.

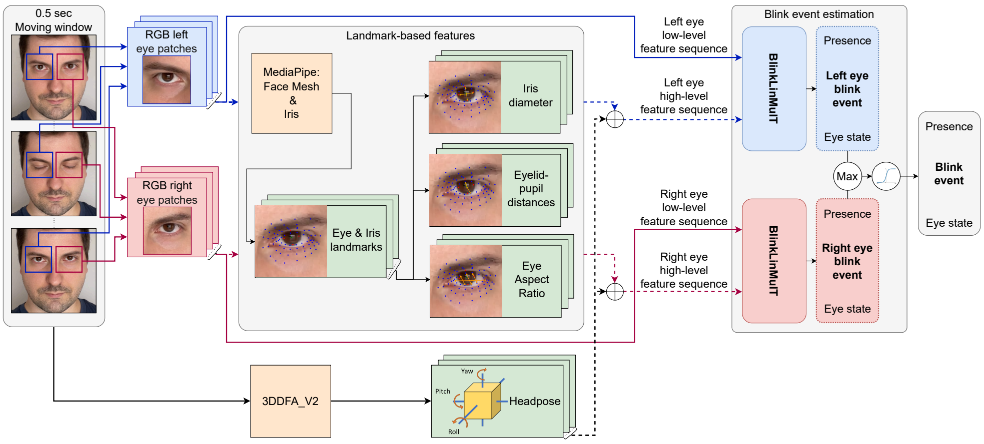

Figure 7.2: Overview of the blink estimation pipeline.

7.3 METHODOLOGY

The following sections outlines the experimental setup to support the reproducibility of models and their results.

7.3.1 *Datasets*

I used four image datasets and five video datasets in my experiments.

CEW: The Closed Eyes in the Wild (CEW) dataset was created by Song et al. [151]. Close face crops are collected from 2423 subjects; 1192 subjects with both eyes closed are collected directly from the Internet, and 1231 subjects with eyes open are selected from the Labeled Face in the Wild (LFW) database [80]. The face images have a resolution of 100×100 pixels. CEW does not have predefined training, validation, and test sets. For within-dataset evaluation and comparison with existing methods, I used 10-fold cross-validation. For cross-dataset evaluations of backbone models, I performed a random split of 70%-15%-15% for training, validation, and testing to maintain consistency with other evaluation protocols.

ZJU EYE: Song et al. [151] cropped the original ZJU Eye blink dataset [120] images to extract eye patches and expanded the initial set of images by rotation, blurring, the addition of Gaussian noise, and contrast adjustment. The original image resolution is 320 × 240 pixels, and the extracted eye patches are 24 × 24 pixels. Binary eye states (open/closed) are annotated. This already preprocessed and augmented dataset has predetermined train and test subsets. During experiments, I used 20% of the training subset as a validation set.

MRL EYE: The MRL Eye dataset was collected from 37 participants (33 men and 4 women) [62]. This dataset contains low- and high-resolution infrared images captured by different sensors like RealSense, IDS, and Aptina. The images were taken under diverse lighting conditions, and annotated as "good" or "bad" based on quality. It also features three reflection modes categorized as "none", "low", and "high", based on the size of the reflections. Additionally, annotations indicate whether the eye patches include glasses or not. The dataset provides binary labels for eye state, specifying whether the eyes are open or closed. The eye regions are pre-cropped at various resolutions. MRL Eye does not have predefined splits. I experimented with 3 different dataset splits, which are detailed in Section 7.4.1 for clarity.

RT-BENE^{img}: The videos on the RT-GENE dataset were recorded using a Kinect v2 RGB-D camera to provide RGB images at 1920×1080 resolution [55]. Eye patches are extracted from the face images and then annotated and published as RT-BENE [32]. However, to estimate landmarks around the eyes and head pose, I used the full-face images from RT-GENE and the blink annotation from RT-BENE. The eye regions are cropped at various resolutions but resized to 64×64 pixels. I denote the subset, where both the image and the annotation were available, as RT-BENE^{img} in the experiments. The dataset consists of 16 participants, with the split based on participant IDs: IDs 1 to 9 are used for the training set, IDs 10, 12, 13, and 14 for the validation set, and IDs 0, 11, 15, and 16 for the test set.

RT-BENE^{seq}: The multimodal sequences of RT-BENE^{seq} are created from RT-BENE^{img} [32]. The face images with blink states, and eye RGB patches, among other frame-wise features, are stacked along the time dimension to form sequences. I used the same participant-based split as in RT-BENE^{img}.

EYEBLINK8: The EyeBlink8 dataset contains 8 videos that were recorded in a home environment [46, 64]. Four participants are sitting in front of the camera. There are 408 eye blinks on 70992 annotated frames with a resolution of 640×480. The data is split by video IDs as follows: I used 1 to 4 for training, 8 and 9 for validation, and finally, 10 and 11 are utilized for testing.

TALKINGFACE: TalkingFace is a single 200-second-long video stream, which is created to evaluate facial landmarks detection precision [46]. The video is captured with 25 FPS and a resolution of 720×576, and it contains a single participant talking in front of the camera. Annotations for 61 eye blinks are also available, which were used in several related works besides their small size. The dataset is too small for training; to remain consistent with prior literature, it is only used for testing.

RN15 & RN30: The Researcher's Night (RN) dataset was collected during an event called Researcher's Night 2014 [57]. People were asked to read an article on a computer screen or blink while being recorded. The authors of the dataset collected 107 videos with 223000 frames of different people with various head poses and cluttered backgrounds. In addition to binary eye states, eye locations within the frames are also annotated. There are two subsets: Researcher's Night 15 (RN15) and Researcher's Night 30 (RN30) that are captured with 15 and 30 FPS with a resolution of 640×480.

7.3.2 *Feature Extraction and Preprocessing*

For the CEW and RT-BENE^{img} datasets, face images are provided, and 3DDFA_V2 is used to predict facial landmarks. Based on these landmarks, the eye regions are determined, cropped, and resized to 64×64 pixels. For ZJU Eye and MRL Eye datasets, the eye patches are already pre-cropped at various resolutions; these images are simply resized to 64×64 pixels for consistency. Head pose information is not utilized in experiments involving image datasets. From the RGB eye patches, I calculate landmark-based high-level features such as EAR, iris landmarks, iris diameters, and eyelid-pupil distances.

For the sequence datasets (RT-BENE^{seq}, EyeBlink8, TalkingFace, RN15, and RN30), either face bounding boxes or annotated landmarks are available to locate the eyes. For videos lacking this information, face detection and tracking are employed. In benchmark datasets, video durations often exceed the length that the model can process in a single pass. Considering that the average duration of a single blink is 0.1–0.4 seconds [141], I set the analyzing window to cover 0.5 seconds to ensure that each sample includes sufficient temporal context for blink detection. Since most benchmark videos have a frame rate of 30 FPS, the 0.5-second analysis window corresponds to 15 frames along the time dimension. For training, validation, and testing, samples are generated using this 15-frame moving window, applied sequentially without overlap. The generated sample and dataset statistics are summarized in Table 7.1.

Type	Dataset	# Samples	# Open Eyes (Train+Val)	# Blinks (Train+Val)	# Open Eyes (Test)	# Blinks (Test)	% Blink in Test Set
Images	CEW	2423	1050	1009	182	182	50%
	ZJU Eye	8984	5770	1574	1230	410	25%
	MRL Eye	84898	34618	33830	8334	8116	49%
	RT-BENE^{img}	107169	72281	2847	30347	1694	4%
Sequences	RT-BENE^{seq}	6969	4356	547	1869	197	10%
	EyeBlink8	4616	3422	293	786	115	13%
	RN15	6439	2830	434	2903	272	9%
	RN30	11231	5346	650	4729	506	10%
	TalkingFace	324	-	-	263	61	19%

Table 7.1: Overview of the eye blink detection datasets.

Utilizing the head pose information is important, particularly during conversations. The participants tend to lower their gaze toward the table or look down at themselves, causing the top eyelids to become visible. This subtle cue can be challenging for a CNN to differentiate as it operates on frames. Thus, utilizing sequence models becomes essential as they can benefit from temporal information. These cases can be detected or learned if the network has access to the head pose angles (yaw, pitch, roll) predicted by, e.g., 3DDFA_V2 [69]. Additionally, visualizing head pose angles could potentially provide valuable insights to better understand the dynamics of these sessions.

Additionally, 3DDFA_V2 gives coarse facial landmarks to locate the eyes. Similarly to the image data, these regions are cropped and resized to 64×64 pixels.

Fine eye landmark estimation is performed using MediaPipe Face Mesh [88], which is a residual neural network that can predict dense coordinates even if the face is partially occluded. I used it to extract 71 landmarks around the eye. Note that 3DDFA_V2 can make errors when the eye is closed, however, Face Mesh accurately predicts eye landmarks even in such cases.

Then, I calculated Eye Aspect Ratio (EAR) introduced in [154]. I used MediaPipe Iris [2], which is a tiny neural network, to predict 5 landmarks in 2D: the pupil center, and 4 points of the outer iris circle. For more high-level features, I calculated horizontal and vertical iris diameters and the distances between the pupil and the eyelid landmarks. All of these frame-wise features are flattened and concatenated, resulting in a 160-dimensional feature vector for each eye. The list of features with the corresponding dimensions is in Table 7.2.

Feature name	Source	Dimensions
RGB eye patch	RGB face crop	$\mathbb{R}^{64\times64\times3}$
Head pose angles	RGB face crop	$\mathbb{R}^{3}$
Face Mesh landmarks	RGB eye patch	$\mathbb{R}^{71\times2}$
Iris landmarks	RGB eye patch	$\mathbb{R}^{5\times2}$
Iris diameters	Iris landmarks	$\mathbb{R}^{2}$
Eyelid-pupil distances	Iris and Face Mesh landmarks	$\mathbb{R}^{2}$
Eye Aspect Ratio	Face Mesh landmarks	$\mathbb{R}$

Table 7.2: Overview of the visual features utilized in the experiments, including their sources and corresponding dimensional representations.

An input sample of the image datasets is a simple eye patch in RGB channel order ($x^{(i)} \in \mathbb{R}^{64\times64\times3}, \quad i = 1, \ldots, N$, where N is the number of samples in the image dataset). The corresponding annotation is a binary eye state ($y^{(i)} \in \{0, 1\}$).

An input sample of the video datasets consists of tuples for a single eye with a low-level feature sequence of eye patches ($x^{(i)}_{\text{low}} \in \mathbb{R}^{15\times64\times64\times3}$ RGB data) and a high-level feature sequence ($x^{(i)}_{\text{high}} \in \mathbb{R}^{15\times160}$ landmark-based features and head pose angles). The annotation is a tuple with a sequence of frame-wise binary eye states ($y^{(i)}_{\text{state}} \in \{0, 1\}^{15}$), and a single binary value, which indicates that a blink event present or not within the sequence ($y^{(i)}_{\text{presence}} \in \{0, 1\}$).

7.3.3 *Union of Datasets*

The observation that increasing data volume alone leads to diminishing returns in model performance comes from the RT-BENE study, where it was shown that expanding the training data from 25% to 75% resulted in a significant F1 score improvement (from 0.581 to 0.693). However, further increasing the volume to 100% yielded only a marginal gain (from 0.693 to 0.706). This plateau suggests that beyond a certain threshold, simply adding more data without increasing its diversity has limited impact on model performance. Motivated by this finding, I hypothesized that further performance improvements could be achieved by not only increasing the volume of training data but also enhancing its diversity. Since the datasets used in this study were created in a similar manner (e.g., eye patches extracted from RGB videos or images), they share certain characteristics but differ primarily in terms of diversity, such as variations in capture resolutions, lighting conditions, head poses, and subject demographics. By combining multiple datasets, I aimed to leverage both increased volume and diversity to improve model generalization across different conditions.

I merged the predefined training subsets from each dataset into a single training set. Similarly, the validation subsets from individual datasets were combined into a unified validation set. For testing, I retained the independent test subsets of each dataset to ensure fair and consistent evaluation. This setup allows for direct comparison with results obtained from single-dataset evaluations while ensuring that no overlap occurs between training and testing data. The experimental results where the network is trained on the merged datasets are referred to as "Union" throughout Section 7.4.

7.3.4 *Experiments*

I chose the binary F1 score as my main evaluation metric due to the unbalanced nature of the datasets, but I also calculated precision (P), recall (R), and average precision (AP).

Precision (P) is calculated as the ratio of true positives (TP) to the total predicted positives (TP + FP), indicating how many of the predicted positive samples are correct. Recall (R) is the ratio of true positives to the total actual positives (TP + FN), measuring the model's ability to identify all positive samples. The F1 score is the harmonic mean of precision and recall, and Average Precision (AP) summarizes the precision-recall curve by calculating the weighted mean of precision values at various recall levels, offering an overall measure of performance across different decision thresholds.

To determine the optimal backbone model for BlinkLinMulT, I conducted a series of experiments. Initially, I evaluated multiple frame-wise models in a within-dataset setting, where each network was trained and tested on samples from the same image dataset. This setup allowed me to assess how well the backbones generalized within their respective domains.

Following the within-dataset evaluations, I performed cross-dataset experiments. In this setup, each backbone model was tuned on one dataset's predetermined training (and validation) set, and evaluated on the test sets of each dataset. This approach enabled me to evaluate the backbones' ability to generalize across different domains.

To further enhance model performance, I combined the training (and validation) subsets from multiple datasets into a unified set. I then re-trained the models on this merged dataset and conducted cross-dataset evaluations again. I expect a consistent improvement in performance, which can be attributed to the increased diversity of the training data as described in Section 7.3.3.

To validate the robustness of the chosen backbone model, I tested it on the MRL Eye dataset, which was not included in the initial training process. This experiment assessed the model's performance on a new dataset under various conditions, such as different lighting and reflection scenarios.

Next, I trained and evaluated the proposed BlinkLinMulT model on video datasets using the pretrained backbone model. Similar to the backbone evaluations, I conducted within-dataset and cross-dataset experiments, followed by training on the union of datasets. This comprehensive approach allowed me to assess BlinkLinMulT's robustness and adaptability across different video datasets.

The performance of BlinkLinMulT was then compared to existing methods in the literature to evaluate its competitiveness and potential improvements over state-of-the-art approaches.

Additionally, I conducted a feature significance study to determine the added value of different feature sets used in the model. This analysis highlights the importance of combining low- and high-level features for improved performance.

To demonstrate the benefits of sequence modeling over frame-wise analysis, I compared the performance of the frame-wise backbone model with the proposed sequence model. I expect, that incorporating temporal context significantly enhance performance.

Finally, I investigated the impact of head pose angles on BlinkLinMulT's performance. This experiment reveals the model's robustness to extreme head poses, a common challenge in real-world scenarios.

7.3.5 *Training Details*

The models were fine-tuned using AdamW [104]. The initial learning rate was set to 1×10^{-3}, and L2 regularization to 5×10^{-4}. The learning rate is multiplied by 0.1 if no improvement is observed in the validation loss for a period of 10 epochs. Binary cross-entropy is used as the loss function between the frame-wise prediction and the corresponding ground truth for all experiments. I used 64 as the batch size. The networks are trained using a single NVIDIA RTX A4000 with 16GB VRAM. The CNN backbones were trained for 310 epochs on the image datasets, whilst BlinkLinMulT was trained for 50 epochs using samples extracted from the video datasets. To handle the unbalanced class distribution of the datasets (Table 7.1), I calculated the class weights per dataset in advance, and oversampling is achieved by weighting the loss itself.

7.4 RESULTS

I present results about backbone models in Section 7.4.1-7.4.2. The performance of the proposed method is followed by its comparison to the recent literature between Section 7.4.3-7.4.5. Ablation study and experiments focusing on robustness are presented in Section 7.4.6-7.4.7.

7.4.1 *Backbone Within-Dataset Evaluations*

My goal was to select a good feature extractor backbone for my proposed sequence model. In this section I introduce the potential backbones.

EYENET: This is a recent CNN designed for classifying eye states [134]. It comprises convolutional, max pooling, and global average pooling layers with fully-connected layers. The model performed well on datasets like CEW, ZJU Eye, and MRL Eye; therefore, I considered using it in the experiments.

RESNET-50 & DENSENET-121: I also incorporated ResNet-50 and DenseNet-121 backbones as they were found to be the best-performing models in [32]. The aforementioned CNNs are pretrained on ImageNet.

CLIP RESNET-50 & CLIP VIT-B/16: Motivated by recent literature, I included Contrastive Language-Image Pre-Training (CLIP) models in the experiments to assess their effectiveness [133]. The utilized visual backbones are a modified ResNet-50 and a Vision Transformer (ViT-B/16).

First, I trained and evaluated all 5 frame-wise backbone models on 3 of the image datasets (CEW, ZJU Eye, RT-BENE^{img}). I report the performance in Table A.6.

The CEW database is relatively small in terms of the number of samples, but it is a popular benchmark dataset as most faces are frontal, balanced by class labels, and there are also higher-resolution images. It does not have pre-defined training and test images; therefore, similar to others in the literature, I employed a 10-fold cross-validation scheme. All backbone models performed well on this dataset with only a slight 0.006 difference in F1 score between the worst and best models. The EyeNet results are successfully reproduced (0.990 ± 0.006 F1 score); however, the ResNet-50 and DenseNet-121 outperformed it under the same conditions (0.994 ± 0.003 and 0.995 ± 0.002 F1 scores, respectively). The positive effect of the better pre-training method is measured with higher-resolution images using the CLIP ResNet-50 and ViT-B/16; the models achieved the highest average scores with 0.996 ± 0.005 and 0.996 ± 0.003.

The backbones performed worse on low-resolution, augmented, and noisy images of the ZJU Eye database. The two best-performing models are the ResNet-50 and DenseNet-121 with 0.929 and 0.927 F1 scores, as well as 0.965 and 0.959 AP, respectively. CLIP backbone variants like ViT-B/16 performed worse (with 0.915 F1 score and 0.957 AP) due to there being less spatial context available to learn global dependencies and a large number of parameters can lead to overfitting on small datasets. CNNs can effectively learn local features with fewer parameters, making them a better choice for smaller-resolution images.

The RT-BENE^{img} dataset contains more samples, rich in different head poses, but it is rather unbalanced, with many more open-eye samples than blinking samples. To address occlusions caused by glasses, the dataset creators used a semantic segmentation technique to identify the frame of the glasses and then inpainted the occluded regions using a Generative Adversarial Network (GAN), which may have introduced additional noise. This added complexity is reflected in the accuracy of the models trained on this dataset; EyeNet achieved 0.868, followed by the CLIP ResNet-50 and ViT-B/16 with 0.870 and 0.871 F1 scores, respectively. DenseNet-121 performed well with 0.883, but ResNet-50 outperformed the other models by a considerable margin with a 0.912 F1 score and 0.946 AP.

EYE PATCH RESOLUTION DEPENDENCE: CLIP backbones are underperforming, and one possible reason for this is the resizing of low-resolution images from 64×64 to 224×224. While the information content is the same, the larger input makes pattern recognition more difficult. To confirm this assumption, I have trained a ResNet-50 with the same hyperparameters, the only difference being in the image dimensions, and it is denoted as ResNet-50-224px in Table A.6. The CNN trained on the 224×224 images achieved lower performance metric scores in every case versus its 64×64 counterpart: 0.929 vs. 0.932 precision, 0.837 vs. 0.893 recall, 0.881 vs. 0.912 F1 score, and 0.918 vs. 0.946 AP.

7.4.2 *Backbone Cross-Dataset Evaluations*

I conducted cross-dataset evaluations to assess the performance of the backbone models on test subsets from other datasets. Each backbone model was trained on the training and validation sets of a specific dataset and then evaluated on the test sets of the remaining datasets without any additional fine-tuning. This process was repeated for all possible dataset combinations, and the resulting metrics are presented in Table A.7.

When the networks trained on the ZJU Eye dataset and evaluated on others, the F1 score is acceptable on the test set of the ZJU Eye dataset, e.g., 0.929 for the best-performing model ResNet-50; the same network can only achieve 0.652 on the CEW and 0.502 on the RT-BENE^{img}, showing a massive performance degradation, 30%, and 46%, respectively. The networks in general learned better features when trained on the CEW images (e.g., F1 score of ResNet-50 is 0.717 on ZJU Eye, 0.672 on RT-BENE^{img}); however, the best cross-dataset performance can be achieved when the RT-BENE^{img} is used as a training dataset. ResNet-50 and DenseNet-121 outperformed the other backbones by a large margin; however, I cannot choose a clear best model based on this experiment; 0.806

and 0.851 F1 scores are achieved on ZJU Eye, while 0.894 and 0.836 F1 scores on CEW, respectively.

To address the performance degradation observed during the cross-dataset evaluations, I combined the training and validation subsets of multiple datasets to increase data volume and enhance diversity, as discussed in Section 7.3.3. I trained the models using these merged subsets, and reported their metrics in Table 7.3. DenseNet-121 utilized the extended set of samples and achieved the best F1 scores among other metrics over all backbones. Training on the union of the datasets improved on previous results measured in Section 7.4.1: the F1 scores changed from 0.995 to 0.997 on CEW, from 0.927 to 0.933 on ZJU Eye, and from 0.883 to 0.913 RT-BENE^{img}.

Model	Train DB	Test DB	P	R	F1	AP
EyeNet	Union	CEW	0.974	0.955	0.965	0.995
<u>ResNet-50</u>	Union	CEW	0.994	0.987	<u>0.990</u>	0.999
DenseNet-121	Union	CEW	0.994	1.000	**0.997**	0.999
CLIP-ResNet-50	Union	CEW	0.899	0.905	0.902	0.961
CLIP-ViT-B/16	Union	CEW	0.969	0.981	0.975	0.993
EyeNet	Union	ZJU Eye	0.912	0.861	0.886	0.948
<u>ResNet-50</u>	Union	ZJU Eye	0.920	0.900	<u>0.910</u>	0.961
DenseNet-121	Union	ZJU Eye	0.914	0.954	**0.933**	0.956
CLIP-ResNet-50	Union	ZJU Eye	0.819	0.859	0.838	0.899
CLIP-ViT-B/16	Union	ZJU Eye	0.872	0.917	0.894	0.957
<u>EyeNet</u>	Union	RT-BENE^{img}	0.913	0.901	<u>0.907</u>	0.954
ResNet-50	Union	RT-BENE^{img}	0.879	0.879	0.879	0.947
DenseNet-121	Union	RT-BENE^{img}	0.913	0.913	**0.913**	0.959
CLIP-ResNet-50	Union	RT-BENE^{img}	0.851	0.800	0.825	0.899
CLIP-ViT-B/16	Union	RT-BENE^{img}	0.939	0.850	0.892	0.944

Table 7.3: Performance of the backbone models using the union of multiple datasets. The highest F1 ratings and corresponding models are in bold. The second-highest ratings and corresponding models are underlined.

To verify the robustness of DenseNet-121, I measured its performance using the MRL Eye images under various conditions, including the presence or absence of glasses, good or bad lighting conditions, and varying levels of distracting reflections on the eye surface or glasses.

The F1 scores from 4 experiments (Exp-A to Exp-D) are summarized in Table 7.4. Below is a breakdown of each experiment and its relevance:

- **Exp-A**: DenseNet-121 was trained on the combined datasets of CEW, ZJU Eye, and RT-BENE^{img} and evaluated on the entire MRL Eye dataset. This experiment provides a baseline for evaluating DenseNet-121's generalization to the MRL Eye dataset without any specific fine-tuning.

- **Exp-B**: The same training setup as Exp-A was used, but evaluation was performed only on a predetermined test subset of MRL Eye (images from participants 35 and 36). The negligible difference in average F1 scores between Exp-A and Exp-B suggests consistent performance across different subsets of MRL Eye.

- **Exp-C**: In this experiment, samples from the MRL Eye dataset were added to the training data. The same holdout set as in Exp-B was used for evaluation and 2% improvement is measured compared to Exp-B.

- **Exp-D**: DenseNet-121 was trained and tested exclusively on the MRL Eye dataset using a 75% train, 15% validation, and 15% test split. This setup allows for direct comparison with methods in the literature. The model achieved an average F1 score of 0.9953, successfully reproducing results reported in other studies. However, a 3.8% degradation in F1 score was observed when DenseNet-121 was not fine-tuned using MRL Eye images.

I used the DenseNet-121 model trained in Exp-C as the RGB backbone within the proposed BlinkLinMulT architecture, leveraging its improved performance from incorporating MRL Eye samples into training.

Condition	Exp-A	Exp-B	Exp-C	Exp-D
With glasses	0.9518	0.9632	0.9797	0.9959
Without glasses	0.9714	0.9478	0.9742	0.9944
Good light	0.9679	0.9723	0.9838	0.9953
Bad light	0.9631	0.9469	0.9749	0.9948
No reflection	0.9727	0.9560	0.9784	0.9954
Low reflection	0.9613	0.9536	0.9934	1.0000
High reflection	0.8987	0.9574	0.9486	0.9912
Average	0.9553	0.9567	0.9761	0.9953

Table 7.4: F1 scores of DenseNet-121 under different conditions and experimental setups (Exp-A to Exp-D) on the MRL Eye dataset.

7.4.3 *BlinkLinMulT Within-Dataset Evaluations*

The proposed BlinkLinMulT is a sequence model designed to process both low-level and high-level feature sequences as inputs. As a result, the following experiments are conducted on video datasets, which provide the necessary temporal context for sequence modeling. The two tasks—eye state recognition and blink presence detection—are evaluated using the predetermined test subsets of the video datasets: RT-BENE^{seq}, EyeBlink8, RN15, and RN30.

The results in Table 7.5 demonstrate that BlinkLinMulT achieves strong performance across both tasks, particularly in blink presence detection. For blink presence detection, F1 scores are consistently high across all datasets, with values ranging from 0.921 on RN30 to 0.996 on EyeBlink8. In contrast, eye state recognition shows more variability, with F1 scores ranging from 0.723 on EyeBlink8 to 0.936 on RT-BENE^{seq}. This variability may be attributed to differences in dataset quality, annotation consistency, and challenges such as lighting conditions or motion blur present in the videos. This frame shift may affect (and lower) the frame-wise evaluation metrics of eye state recognition; however, it has less impact on event detection, as most blink presence events can still be identified even if they are shifted by a few frames.

Task	Dataset	P	R	F1
Eye state	RT-BENE^{seq}	0.940	0.931	0.936
	EyeBlink8	0.569	0.990	0.723
	RN15	0.692	0.927	0.792
	RN30	0.683	0.886	0.771
Presence	RT-BENE^{seq}	0.922	0.959	0.940
	EyeBlink8	0.991	1.000	0.996
	RN15	0.925	0.952	0.938
	RN30	0.916	0.927	0.921

Table 7.5: Eye state recognition and blink presence detection performances of the proposed method BlinkLinMulT on the RT-BENE^{seq}, EyeBlink8, RN15, and RN30 datasets.

One notable observation is that blink presence detection generally outperforms eye state recognition in terms of accuracy. This could be due to two factors: first, the start of a blink event is often annotated as "closed eye" even when the eye is still open in the given frames, leading to apparent errors when the model correctly predicts "open eye". Second, although the annotation is defined frame-wise, the decoders available today do not extract the frames in exactly the same way, especially if there are different FPS within the video. These discrepancies can result in slight frame shifts that were not manually corrected during evaluation.

7.4.4 *BlinkLinMulT Cross-Dataset Evaluations*

Similar to Section 7.4.2, I carried out cross-dataset evaluations using the proposed BlinkLinMulT model. I used the EyeBlink8, RT-BENE^{seq}, RN15, and RN30 datasets for training and evaluation, and due to its limited size, the TalkingFace is used only for testing purposes. Table A.8 is grouped by test databases for ease of reference. I observed improved performance when RN15 and RN30 datasets were used within training despite their lower resolution and in-the-wild difficulty.

I trained the BlinkLinMulT on the union of the datasets. Table 7.6 shows that performance are consistently improved compared to those experiments, where only a single dataset is used for training. BlinkLinMulT captured the unique patterns from all used datasets; the eye state recognition F1 score increased from 0.723 to 0.87 for EyeBlink8, 0.792 to 0.871 for RN15, and 0.792 to 0.836 for RN30. The blink presence detection F1 score increased from 0.938 to 0.959 for RN15 and 0.921 to 0.933 for RN30. While, in the literature, the proposed models are retrained for each dataset, my network is trained once and performs well on all the public blink benchmark datasets.

Task	Train DB	Test DB	P	R	F1
Eye state	Union	RT-BENE^{seq}	0.824	0.796	0.810
	Union	EyeBlink8	0.782	0.979	0.870
	Union	RN15	0.867	0.876	0.871
	Union	RN30	0.890	0.787	0.836
	Union	TalkingFace	0.993	0.614	0.759
Presence	Union	RT-BENE^{seq}	0.914	0.965	0.938
	Union	EyeBlink8	0.991	0.991	0.991
	Union	RN15	0.977	0.941	0.959
	Union	RN30	0.966	0.901	0.933
	Union	TalkingFace	1.000	1.000	1.000

Table 7.6: Eye state recognition and blink presence detection performance of the proposed method BlinkLinMulT on the test subsets of TalkingFace, RT-BENE^{seq}, EyeBlink8, RN15, and RN30 datasets. The model is trained on the union of the train subsets of these datasets.

7.4.5 *Comparison to the Literature*

The evaluation of my model was conducted on the same datasets, samples, and tasks as the existing methods to ensure a fair and consistent comparison. I show that the proposed method generalizes well to different scenarios, and the performance of the network is comparable to or surpasses the performance of state-of-the-art methods, in Table 7.7.

Note that TalkingFace is only used for testing, but, nevertheless, the looking-down events are successfully recognized, while other methods make mistakes. The proposed BlinkLinMulT excels in the blink presence detection task on all datasets. The per-frame evaluation metrics of the eye state recognition task are slightly lower; however, this is due to occasional frame-shifting during decoding and the fact that annotation often starts and ends with an open eye. In these cases, although the network is not flawed, it will be reflected in the frame-wise results during evaluation.

BlinkLinMulT achieved notable advancements in blink presence detection, with an F1 score of 0.991 on EyeBlink8, surpassing the previous state-of-the-art by 3.8%, and improving RN dataset scores by 4.0% (from 0.906 to 0.942). It also achieved perfect scores (1.000 F1) on TalkingFace, effectively handling challenging cases like downward gazes. For eye state recognition, BlinkLinMulT showed competitive performance on EyeBlink8 and RN datasets and significant improvement on RT-BENE^{seq}, increasing F1 scores by 12.3% (from 0.721 to 0.810).

7.4.6 *Feature Significance*

To measure the added value of the different features used in the experiments, I trained deep networks using the input features separately and then using the different combinations. The F1 scores obtained on the image test subsets are in Table 7.8, and I also report the metric using the video datasets in Table 7.9. To facilitate transparency, the following feature groups are created:

- **MediaPipe Iris landmark features (ILM)**: the feature set contains the iris landmarks, iris diameters, and eyelid–pupil distances.
- **MediaPipe Face Mesh landmark features (FLM)**: the feature set contains the eye landmarks and eye aspect ratio.
- **Head pose angles (HP)**: the head pose angles are highlighted independently.
- **Texture (T)**: RGB texture contains the most information.

Based on the dimensionality of the feature set, different models are used. I evaluated multiple hyperparameters for the networks on the validation set, and report the F1 metrics obtained on the test subset of RT-BENE^{img} dataset in Table 7.8.

Database	Task	Method	P	R	F1
TalkingFace	Presence	[125]	0.951	0.936	0.943
		[112]	-	-	0.950
		[96]	0.983	0.934	0.958
		[58]	-	-	0.971
		[33]	-	-	0.979
		Ours	1.000	1.000	**1.000**
EyeBlink8	Eye state	[8]	-	-	0.834
		[33]	-	-	0.910
		[32]	0.995	0.958	**0.976**
		Ours	0.782	0.979	0.870
	Presence	[112]	-	-	0.870
		[96]	0.909	0.904	0.905
		[58]	-	-	0.913
		[33]	-	-	0.946
		[154]	0.943	0.960	0.952
		[125]	0.953	0.958	0.955
		Ours	0.991	0.991	**0.991**
RN	Eye state	[33]	-	-	0.807
		[32]	-	-	**0.913**
		Ours	0.878	0.828	0.852
	Presence	[58]	-	-	0.879
		[33]	-	-	0.906
		Ours	0.970	0.915	**0.942**
RT-BENE	Eye state	[8]	-	-	0.529
		[33]	-	-	0.602
		[32]	0.664	0.791	0.721
		Ours	0.824	0.796	**0.810**

Table 7.7: Comparison of BlinkLinMulT to the methods in the literature in the case of two tasks: blink presence detection and frame-wise eye state recognition. The highest F1 scores per dataset and task are shown in bold.

DENSE: This network consists of a single hidden layer with 128 units, ReLU activation, and dropout ($p = 0.6$) to prevent overfitting. The hidden layer is followed by a linear output layer with one unit, which provides the binary prediction for the eye state.

For ILM and FLM, this Dense network is trained. For T, the pretrained DenseNet-121 is used from Exp-C (Section 7.4), reported in Table 7.3. T+ILM+FLM utilizes every available high-level feature. Head pose angles are excluded because ZJU Eye and MRL Eye datasets are used for training the CNN backbone; however, it contains only eye patches, and head pose angles are not available. The RGB texture is the richest in detail; therefore, increased performance is expected in this case. While the iris and eye landmark features alone show fairly low performance, combining them with the texture further improves the performance on RT-BENE^{img} compared to texture-only evaluation (0.922 vs. 0.913 F1 scores).

Modality	Model	RT-BENE^{img}
ILM	Dense	0.453
FLM	Dense	0.595
T	DenseNet-121	0.913
T+ILM+FLM	DenseNet-121+Dense	**0.922**

Table 7.8: Eye state recognition performance ablation study, presenting the impact of different features. The networks trained on the union of CEW, MRL Eye, and RT-BENE^{img} datasets. The metric is the F1 score on the individual test subsets. The highest F1 score is shown in bold.

I evaluated the effectiveness of individual and combined modalities on all video datasets. F1 score is reported in Table 7.9 for the eye blink presence task.

LINT: For sequences of high-level features, a transformer encoder with linear complexity, called LinT, is used. The input features are first projected to 32 dimensions using a 1D convolutional layer, followed by a transformer encoder consisting of 5 layers. A final linear output layer with one unit provides the binary prediction for the eye state.

BLINKLINT: The unimodal version of the proposed model, BlinkLinT, builds upon LinT. The input sequence of eye images is first processed by the pretrained DenseNet-121, and its deep hidden representations are then fed into the LinT transformer encoder to learn the dynamics.

The HP, ILM, and FLM sequences are fed to LinT. ILM and FLM landmark-based feature sets show potential when the dynamics are also utilized by a sequence model. For sequences of RGB frames, BlinkLinT is used. The model achieved high F1 scores on multiple datasets, showing the importance of low-level RGB texture over high-level features.

While the head pose angles do not hold information about the blinking event, the combination with the texture and other high-level features helps in extreme cases. The proposed model, BlinkLinMulT, utilizes both low- and high-level information sources effectively. There are negligible differences between the different versions of the proposed model, but I would like to highlight, that in the last case, where all feature sequences are used, the problematic extreme blinks when the person looks down are detected in TalkingFace, achieving a 1.0 F1 score in the blink presence detection task.

Modality	Model	RT-BENE^{seq}	EyeBlink8	RN15	RN30	TalkingFace
HP	LinT	0.267	0.227	0.303	0.276	0.529
ILM	LinT	0.814	0.978	0.869	0.860	0.992
FLM	LinT	0.761	0.974	0.925	0.867	1.000
T	DenseNet-121	0.779	0.887	0.645	0.620	0.976
T	BlinkLinT	0.922	0.991	0.946	0.920	0.992
T+HP	BlinkLinMulT	**0.962**	**0.991**	**0.973**	0.929	0.992
T+FLM+ILM	BlinkLinMulT	0.947	**0.991**	0.958	**0.939**	0.992
T+HP+FLM+ILM	BlinkLinMulT	0.938	**0.991**	0.959	0.933	**1.000**

Table 7.9: Blink presence detection performance ablation study, presenting the impact of different features. The networks trained on the union of RT-BENE^{seq}, EyeBlink8, RN15, and RN30 datasets. The metric is the F1 score on the individual test subsets. The highest F1 scores per dataset are shown in bold.

FRAME-WISE MODEL VERSUS SEQUENCE MODEL To show the importance of modeling motion dynamics, I evaluated the best backbone model, DenseNet-121, per frame on sequences of the video datasets. Then, similarly to the BlinkLinMulT pipeline, I performed max aggregation within the sequence to determine the presence of a blink event for both eyes. In Table 7.9, the transformer-based sequence modeling shows superiority over frame-wise estimations.

7.4.7 *Head Pose Angle Dependence*

I investigated the impact of head pose on the blink presence detection performance of BlinkLinMulT and report the F1 scores in Figure 7.3 and Accuracy, Average Precision (AP), and F1 scores in Table 7.10.

Orientation	Angle	Accuracy	AP	F1
Yaw	$<-25°$	0.991	0.990	0.960
	$[-25°..0°)$	0.990	0.985	0.956
	$[0°..25°)$	0.990	0.985	0.945
	$>25°$	0.984	0.954	0.917
Pitch	$<-25°$	0.986	0.996	0.933
	$[-25°..0°)$	0.985	0.973	0.930
	$[0°..25°)$	0.992	0.988	0.957
	$>25°$	0.951	0.800	0.744

Table 7.10: Blink presence prediction performance of the proposed BlinkLinMulT. Test samples from all 5 sequence datasets are used for the experiment. Yaw and pitch degrees are predicted by 3DDFA_V2. Extreme yaw and pitch angles, when the participant is looking sideways or down, might cause slightly decreased performance.

The union of RT-BENE^{seq}, EyeBlink8, RN15, RN30 test subsets, and the samples from the TalkingFace video is used to include more diverse head poses. For each sequence, the means of frame-wise head pose estimations are calculated. The result of the experiment revealed that the performance of the model drops at higher yaw angles compared to the frontal faces; the 0.917 F1 score is calculated from 623 blink samples for >25° angles compared to the 0.951 average F1 score from 10962 frontal samples. In these cases, self-occlusion makes the prediction harder for the network, which originated from the fact that, beyond the imbalanced class distribution, only a fraction of the blink patterns have a high head pose angle associated with them. In the case of positive pitch angles, the participants are looking down. Most of the errors originated from this extreme head pose because the eyelids are more visible, they move over a shorter distance, and the landmark-based features are less precise considering the available resolution. I experienced a lower F1 score compared to negative pitch angles (0.744 vs. 0.933). Higher performance metrics are expected for samples when the monitored people are looking up; it is quite common for the camera to be positioned slightly below the facial lines. Even with the low number of extreme samples (9%), the performance is fairly consistent and acceptable across head pose variations.

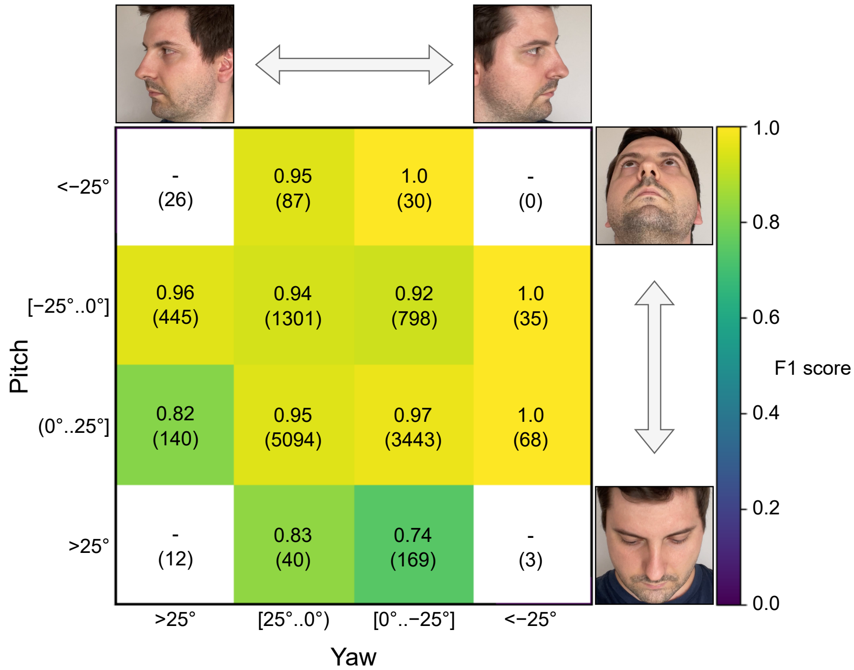

Figure 7.3: Head pose angle dependence of BlinkLinMulT in the case of blink presence detection task. The head poses are predicted by 3DDFA_V2; the colors represent the F1 score calculated for the blink presence task, which is also written within the boxes together with the number of samples (in parenthesis) considered during the metric evaluations. F1 score cannot be calculated for those extreme cases, where closed-eye samples are not available. Test samples from all 5 sequence datasets are used for the experiment. Blinks can be predicted accurately in the case of frontal faces, and while the participant is looking up. Performance slightly decreases when the monitored person is looking down.

7.5 SUMMARY

In this chapter, I proposed BlinkLinMulT, a novel transformer-based model for eye blink detection. The model extends the multimodal transformer with linear attention (LinMulT) to fuse multiple input features, including RGB texture, iris and eye landmarks, and head pose angles.

There are several contributions. First, I introduced a sequence-based method for two tasks: blink presence detection and eye state recognition, leveraging both low- and high-level feature sequences. I conducted extensive evaluations, including within-dataset and cross-dataset experiments, to assess the robustness of the proposed method. To improve generalization, I combined multiple datasets into a unified training set, which enhanced both data volume and diversity. This approach improved performance across all datasets, with DenseNet-121 (as a frame-wise backbone of the proposed model) achieving F1 scores of 0.997 on CEW, 0.933 on ZJU Eye, and 0.913 on RT-BENE^{img}.

I also performed detailed analyses to evaluate the significance of features and their combinations. The results demonstrated that the combination of low- and high-level features always yields better performance than individual features. For instance, combining RGB texture with iris and eye landmarks improved the F1 score from 0.913 (texture only) to 0.922 on RT-BENE^{img}. Similarly, in sequence evaluations, the combination of texture with landmark-based features and head pose angles improved performance over single-modality evaluations by 4.3% on RT-BENE^{seq}, 1.8% on EyeBlink8, 2.7% on RN15, 2.0% on RN30, and 0.8% on TalkingFace, highlighting the importance of multimodal feature fusion in BlinkLinMulT.

Furthermore, I compared BlinkLinMulT against state-of-the-art methods in the literature and demonstrated competitive or superior performance across public benchmark datasets. For blink presence detection, BlinkLinMulT achieved an F1 score of 0.991 on EyeBlink8, outperforming the current SOTA by 3.8% (0.955 F1 from [125]). Similarly, on RN datasets, results improved significantly, with F1 scores increasing from 0.906 to 0.942—an improvement of 4.0%. On TalkingFace, BlinkLinMulT achieved perfect scores (1.000 F1), effectively handling extreme cases like downward gazes. For eye state recognition, while evaluations on EyeBlink8 and RN datasets remain competitive with prior methods, BlinkLinMulT demonstrated substantial improvements on RT-BENE^{seq}, with F1 scores increasing from 0.721 to 0.810—an improvement of 12.3%.

Finally, I explored the dependence of blink detection performance on head pose angles and showed that while extreme poses (e.g., looking down) slightly degrade performance (0.744 F1 score for pitch angles >25°), the model remains robust even with portraits in profile (0.917 F1 score for yaw >25°).

In summary, BlinkLinMulT successfully combines motion dynamics and multimodal features to achieve accurate blink detection during challenging in-the-wild scenarios. The proposed model not only advances SOTA performance but also provides a robust framework for future applications in HCAI.

Part IV

CONCLUSION

8 CONCLUSION

In this dissertation, I explored human characterization, focusing on the assessment of perceived personality traits and sentiment. I then examined how perceived personality contributes to a better understanding of group dynamics, such as affective convergence, and studied its relation to group performance. Additionally, my research addressed the detection of non-verbal social signals, specifically focusing on eye blinking as a subtle but meaningful communication cue. The three theses are connected and situated within an interdisciplinary field that integrates psychology, artificial intelligence, and social science.

First, I presented LinMulT, a linear complexity multimodal transformer, to model automatic sentiment and personality perception (Chapter 5). I limited the visual input modality to high-level facial features such as AUs to concentrate on the face, the most critical source of affect and personality information, and to disregard the background scene texture. I enhanced the MulT model to support more input sequences, up to 5. In the proposed audio-visual-textual configuration, I also used FAb-Net features as visual input, leveraging their efficient training to encode head pose and facial affect cues, enhancing the model's performance. The results on the MOSEI dataset for sentiment analysis are very competitive, especially in MAE and correlation metrics. For personality perception, while the results do not reach state-of-the-art levels, the model offers significant computational advantages. Early on, I encountered slow processing and inference times; to address this, I incorporated linear complexity attention, which significantly sped up training and inference, preparing the model for real-world applications.

Then, using estimates on perceived personality applied to both small groups and dyads engaged in a task-oriented interaction, I examined the time-averaged personality states of group members (Chapter 6). Since the personality model was pretrained on samples where participants were actively expressing themselves, I limited the personality trajectory calculations and subsequent experiments to only those segments within the analysis window where individuals were actively speaking. To remove personal bias, I followed earlier experiences and used action units only as visual features. The experiments show that the LinMulT used in [P1] is reliable even during out-of-domain sessions, the perceived personality of members within a group cluster—which is a sign of group affect convergence—for collaborative tasks where the participants actively spoke to solve the task, and finally, that perceived personality is a stronger predictor of group performance than self-reported personality.

Last, I proposed BlinkLinMulT, a transformer-based model for blink presence detection and eye state recognition (Chapter 7). Moving from frame-by-frame eye state prediction to a sequence-based approach yielded a clear improvement, allowing the model to capture how the eyes move between frames. To address lighting variations, I employed multiple datasets together with data augmentation, mitigating challenges such as motion blur and low-quality images. Combining RGB texture with high-level features such as landmarks substantially improved accuracy. A further challenge was posed by extreme head positions—for example, when participants look down and the upper eyelids become more visible. In such cases, CNN-based models struggle because they do not account for temporal dynamics, whereas sequence models such as the proposed one handle these conditions effectively.

Although the progress made in this thesis is noteworthy, there are still many areas that require further exploration and development.

A limitation of the presented LinMulT in [P1] is its training primarily on samples where individuals are speaking directly to the camera. Including data from social interactions where people are listening and providing back-channel responses, or from sessions with richer non-verbal cues, could enable the network to better capture subtle affective and behavioral indicators, potentially enhancing overall performance. While the model is fair and matches current methods in performance, adding non-verbal inputs like body pose, blinking, and gaze patterns could significantly enhance its ability to reflect affective states and more detailed interaction dynamics. This motivated [P3], so I can reliably capture blink events and incorporate this information in my future work. Real-time evaluation imposes strict memory constraints, and although linear transformers like LinMulT are more scalable than their quadratic counterparts, I only assessed the inference time on a single GPU without considering the feature extraction modules. These modules not only need performance enhancements but also must be optimized for real-time applications to avoid becoming a bottleneck in a real-time data streaming pipeline.

For the group experiments [P2], I used three input modalities; however, the input features do not hold any information about body gestures and visual focus of attention, among others. A significant challenge in analyzing the group datasets is occlusions, with hands often not visible due to being under the table or self-occlusions where hands are near the face, sometimes blocking the view partially or entirely. Additionally, extreme head poses, particularly when a participant looks down, can cause the detector to fail to predict action units reliably. Although these sections are excluded from the analysis, employing specialized tools to address these challenges could enhance the modeling of group interactions in greater detail, potentially improving the recognition of group affect phenomena.

Aside from improving sensory quality, refining various elements of the preprocessing pipeline can substantially boost blink detection accuracy and the overall effectiveness of BlinkLinMulT [P3]. Face detection, despite the high precision of modern detectors, can still be compromised by motion blur and self-occlusions. By employing more precise 2D and 3D face landmark models, such as 3DDFA_V2 and FaceMesh, the selection of eye regions and the detection of high-level features—face-, iris-, and pupil landmarks—improves. Challenges such as self-occlusions, which are often caused by the participant's nose blocking one eye, can be addressed by considering the head pose and focusing on the visible eye only. I demonstrated that utilizing multiple blinking datasets improved overall performance; however, the research field still lacks diverse, high-quality datasets that account for different races and phenomena like incomplete blinks or extremely rapid blinks that occur within a single frame, not to mention the complications when multiple issues occur simultaneously.

In conclusion, this dissertation has made important advances in HCAI and DL, and it sets the stage for more work to be done. The LinMulT model has been adapted to estimate gaze, categorical classes, dimensional emotion (valence and arousal), and intensity levels, extending its capabilities beyond its original focus. Additionally, the BlinkLinMulT model has been utilized to detect blink events in clinical sessions, helping to identify behavioral differences between typically developing children and those with autism spectrum disorder. The knowledge and experience I gained from this work are instrumental in driving ongoing research, which is crucial for understanding complex behaviors, applying AI in healthcare, and enhancing social and human-computer interactions.

Part V

APPENDIX

A FIGURES AND TABLES

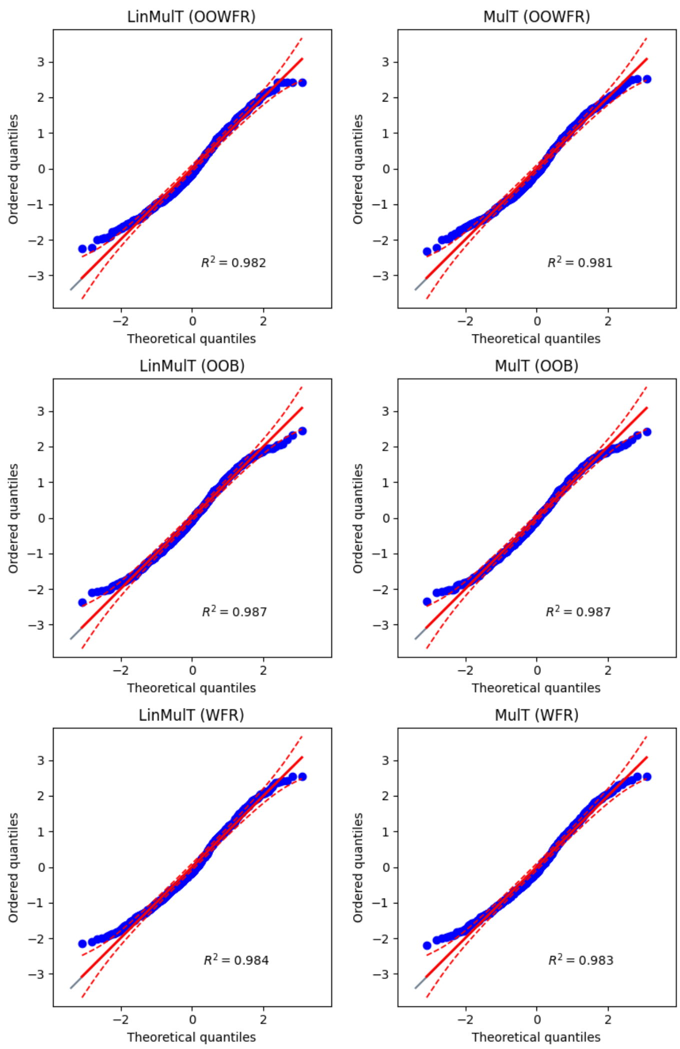

Figure A.1: Q–Q plots for the MOSI dataset. Columns represent LinMulT and MulT models, and rows correspond to three feature sets: OOWFR, OOB, and WFR. Each plot compares predicted sentiment quantiles to theoretical quantiles to evaluate residual normality.

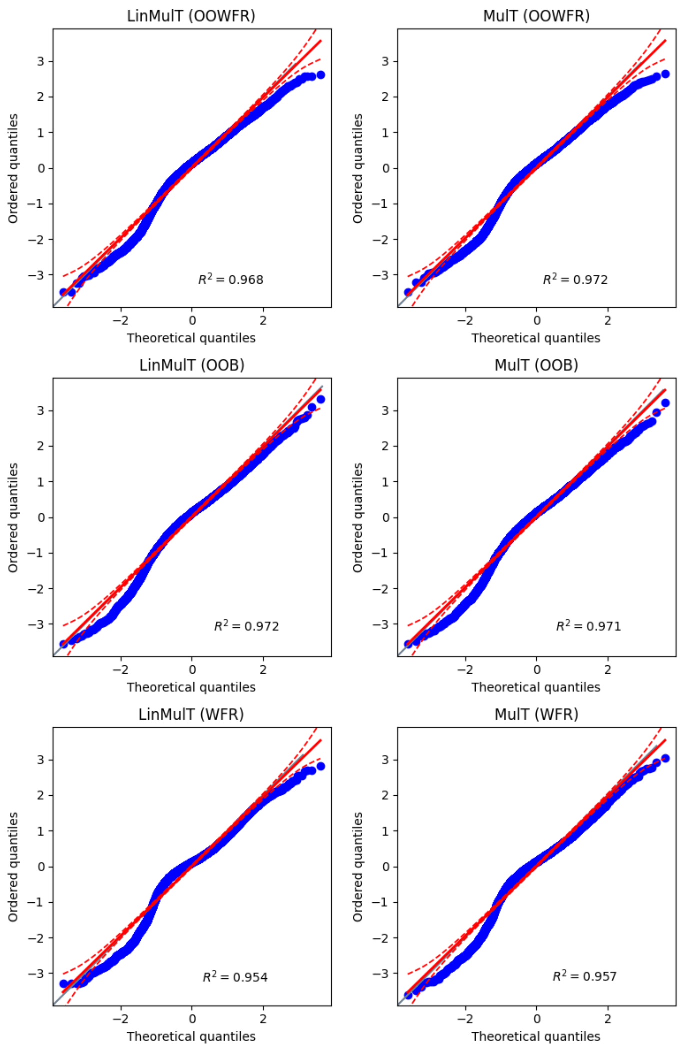

Figure A.2: Q–Q plots for the MOSEI dataset. Columns represent LinMulT and MulT models, and rows correspond to three feature sets: OOWFR, OOB, and WFR. Each plot compares predicted sentiment quantiles to theoretical quantiles to evaluate residual normality.

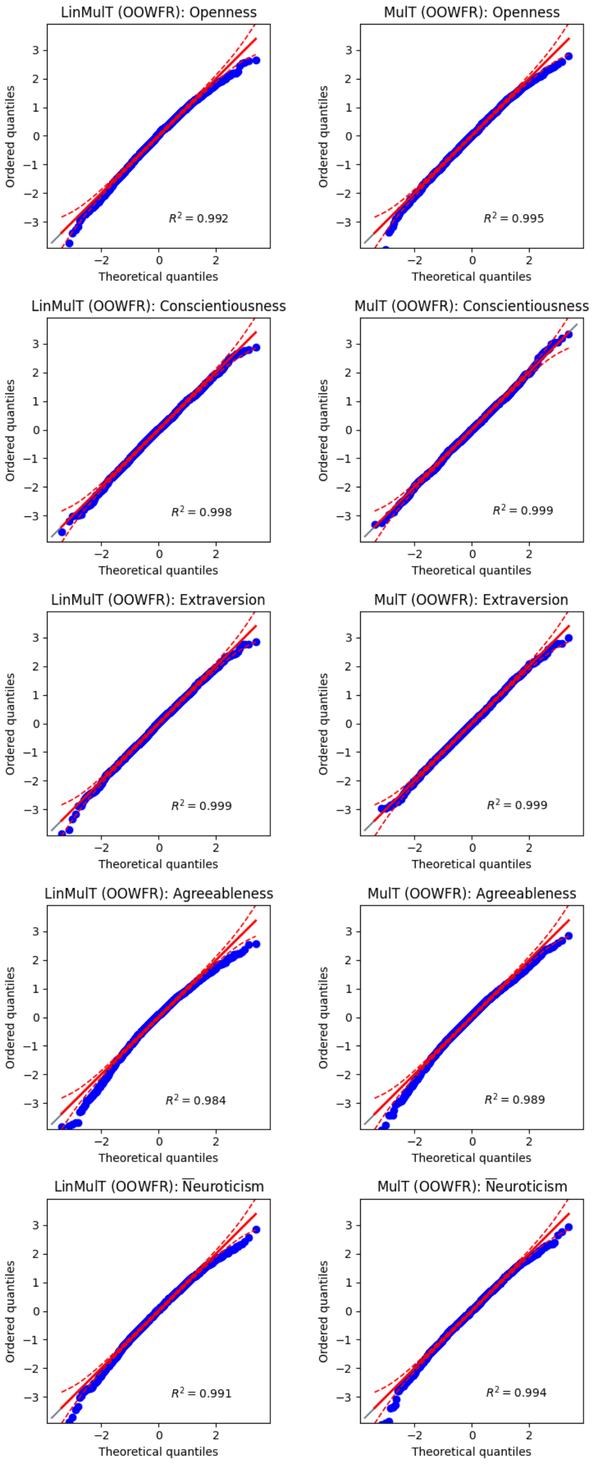

Figure A.3: Q–Q plots for the FI dataset using the OOWFR feature set with LinMulT and MulT models. Rows represent Big Five traits, comparing predicted trait quantiles to theoretical quantiles to evaluate residual normality.

ID	Meaning	Software
AU1	Inner Brow Raiser	OpenFace & OpenGraphAU
AU2	Outer Brow Raiser	OpenFace & OpenGraphAU
AU4	Brow Lowerer	OpenFace & OpenGraphAU
AU5	Upper Lid Raiser	OpenFace & OpenGraphAU
AU6	Cheek Raiser	OpenFace & OpenGraphAU
AU7	Lid Tightener	OpenFace & OpenGraphAU
AU9	Nose Wrinkler	OpenFace & OpenGraphAU
AU10	Upper Lip Raiser	OpenFace & OpenGraphAU
AU11	Nasolabial Deepener	OpenGraphAU
AU12	Lip Corner Puller	OpenFace & OpenGraphAU
AU13	Sharp Lip Puller	OpenGraphAU
AU14	Dimpler	OpenFace & OpenGraphAU
AU15	Lip Corner Depressor	OpenFace & OpenGraphAU
AU16	Lower Lip Depressor	OpenGraphAU
AU17	Chin Raiser	OpenFace & OpenGraphAU
AU18	Lip Pucker	OpenGraphAU
AU19	Tongue Show	OpenGraphAU
AU20	Lip Stretcher	OpenFace & OpenGraphAU
AU22	Lip Funneler	OpenGraphAU
AU23	Lip Tightener	OpenFace & OpenGraphAU
AU24	Lip Pressor	OpenGraphAU
AU25	Lips Part	OpenFace & OpenGraphAU
AU26	Jaw Drop	OpenFace & OpenGraphAU
AU27	Mouth Stretch	OpenGraphAU
AU28	Lip Suck	OpenFace
AU32	Lip Bite	OpenGraphAU
AU38	Nostril Dilator	OpenGraphAU
AU39	Nostril Compressor	OpenGraphAU
AU45	Blink	OpenFace
AU1 (L)	Inner Brow Raiser (Left)	OpenGraphAU
AU1 (R)	Inner Brow Raiser (Right)	OpenGraphAU
AU2 (L)	Outer Brow Raiser (Left)	OpenGraphAU
AU2 (R)	Outer Brow Raiser (Right)	OpenGraphAU
AU4 (L)	Brow Lowerer (Left)	OpenGraphAU
AU4 (R)	Brow Lowerer (Right)	OpenGraphAU
AU6 (L)	Cheek Raiser (Left)	OpenGraphAU
AU6 (R)	Cheek Raiser (Right)	OpenGraphAU
AU10 (L)	Upper Lip Raiser (Left)	OpenGraphAU
AU10 (R)	Upper Lip Raiser (Right)	OpenGraphAU
AU12 (L)	Nasolabial Deepener (Left)	OpenGraphAU
AU12 (R)	Nasolabial Deepener (Right)	OpenGraphAU
AU14 (L)	Dimpler (Left)	OpenGraphAU
AU14 (R)	Dimpler (Right)	OpenGraphAU

Table A.1: List of AUs detected by OpenFace and OpenGraphAU.

Model 1	Model 2	T	DoF	Alternative	p-val	CI95%	Cohen-d	BF10	Power	Bonferroni p-val	Decision
LinMulT (OOWFR)	MulT (OOWFR)	-18.354	685	two-sided	1.688e-61	[-0.1 -0.08]	0.070	8.333e+57	0.451	8.441e-61	H_a
LinMulT (OOWFR)	LinMulT (OOB)	5.511	685	two-sided	5.053e-08	[0.11 0.24]	0.147	1.146e+05	0.971	2.527e-07	H_a
LinMulT (OOWFR)	MulT (OOB)	5.232	685	two-sided	2.237e-07	[0.1 0.23]	0.140	2.730e+04	0.955	1.118e-06	H_a
LinMulT (OOWFR)	LinMulT (WFR)	-12.935	685	two-sided	2.134e-34	[-0.31 -0.22]	0.211	1.035e+31	1.000	1.067e-33	H_a
LinMulT (OOWFR)	MulT (WFR)	-11.926	685	two-sided	6.318e-30	[-0.28 -0.2]	0.189	3.867e+26	0.999	3.159e-29	H_a

Table A.2: Two-sided paired t-test results comparing the predictions of the proposed LinMulT model using the OOWFR feature set to other variations of LinMulT and MulT on the MOSI dataset. The table includes t-statistics (T), degrees of freedom (DoF), p-values, 95% confidence intervals (CI95%), effect sizes (Cohen-d), Bayes Factors (BF10), statistical power, Bonferroni-corrected p-values, and the decision to reject or fail to reject the null hypothesis (H_a or H_0).

Model 1	Model 2	T	DoF	Alternative	p-val	CI95%	Cohen-d	BF10	Power	Bonferroni p-val	Decision
LinMulT (OOWFR)	MulT (OOWFR)	8.687	4582	two-sided	5.088e-18	[0.01 0.02]	0.021	2.922e+14	0.285	2.544e-17	H_a
LinMulT (OOWFR)	LinMulT (OOB)	-1.273	4582	two-sided	0.203	[-0.02 0.]	0.010	0.037	0.102	1.000	H_0
LinMulT (OOWFR)	MulT (OOB)	-1.346	4582	two-sided	0.178	[-0.02 0.]	0.010	0.041	0.106	0.891	H_0
LinMulT (OOWFR)	LinMulT (WFR)	-20.167	4582	two-sided	1.011e-86	[-0.11 -0.09]	0.108	6.015e+82	1.000	5.056e-86	H_a
LinMulT (OOWFR)	MulT (WFR)	-1.712	4582	two-sided	0.087	[-0.02 0.]	0.009	0.072	0.091	0.435	H_0

Table A.3: Two-sided paired t-test results comparing the predictions of the proposed LinMulT model using the OOWFR feature set to other variations of LinMulT and MulT on the MOSEI dataset. The table includes t-statistics (T), degrees of freedom (DoF), p-values, 95% confidence intervals (CI95%), effect sizes (Cohen-d), Bayes Factors (BF10), statistical power, Bonferroni-corrected p-values, and the decision to reject or fail to reject the null hypothesis (H_a or H_0).

Trait	Model 1	Model 2	T	DoF	Alternative	p-val	CI95%	Cohen-d	BF10	Power	Bonferroni p-val	Decision
Openness	LinMulT (OOWFR)	MulT (OOB)	11.678	1999	two-sided	1.585e-30	[0.01 0.02]	0.148	1.015e+27	1.000	7.925e-30	H_a
Openness	LinMulT (OOWFR)	LinMulT (OOB)	5.083	1999	two-sided	4.067e-07	[0. 0.01]	0.064	9.272e+03	0.819	2.034e-06	H_a
Openness	LinMulT (OOWFR)	MulT (WFR)	-3.842	1999	two-sided	1.261e-04	[-0.01 -0.]	0.068	38.875	0.863	6.303e-04	H_a
Openness	LinMulT (OOWFR)	LinMulT (WFR)	4.053	1999	two-sided	5.248e-05	[0. 0.01]	0.076	88.900	0.923	2.624e-04	H_a
Openness	LinMulT (OOWFR)	MulT (OOWFR)	-13.450	1999	two-sided	1.537e-39	[-0.02 -0.01]	0.117	8.941e+35	0.999	7.685e-39	H_a
Conscientiousness	LinMulT (OOWFR)	MulT (OOB)	16.909	1999	two-sided	4.682e-60	[0.02 0.03]	0.227	2.252e+56	1.000	2.341e-59	H_a
Conscientiousness	LinMulT (OOWFR)	LinMulT (OOB)	8.493	1999	two-sided	3.875e-17	[0.01 0.02]	0.113	5.829e+13	0.999	1.938e-16	H_a
Conscientiousness	LinMulT (OOWFR)	MulT (WFR)	28.686	1999	two-sided	7.394e-152	[0.05 0.05]	0.429	7.343e+147	1.000	3.697e-151	H_a
Conscientiousness	LinMulT (OOWFR)	LinMulT (WFR)	15.288	1999	two-sided	5.543e-50	[0.02 0.03]	0.237	2.141e+46	1.000	2.771e-49	H_a
Conscientiousness	LinMulT (OOWFR)	MulT (OOWFR)	22.690	1999	two-sided	1.310e-101	[0.02 0.03]	0.200	5.611e+97	1.000	6.551e-101	H_a
Extraversion	LinMulT (OOWFR)	MulT (OOB)	-9.652	1999	two-sided	1.407e-21	[-0.02 -0.01]	0.112	1.404e+18	0.999	7.034e-21	H_a
Extraversion	LinMulT (OOWFR)	LinMulT (OOB)	-14.313	1999	two-sided	2.604e-44	[-0.02 -0.02]	0.163	4.918e+40	1.000	1.302e-43	H_a
Extraversion	LinMulT (OOWFR)	MulT (WFR)	1.802	1999	two-sided	0.072	[-0. 0.01]	0.034	0.127	0.326	0.358	H_0
Extraversion	LinMulT (OOWFR)	LinMulT (WFR)	-0.100	1999	two-sided	0.920	[-0. 0.]	0.002	0.025	0.051	1.000	H_0
Extraversion	LinMulT (OOWFR)	MulT (OOWFR)	3.585	1999	two-sided	3.447e-04	[0. 0.01]	0.031	15.131	0.280	0.002	H_a
Agreeableness	LinMulT (OOWFR)	MulT (OOB)	-1.058	1999	two-sided	0.290	[-0. 0.]	0.017	0.044	0.115	1.000	H_0
Agreeableness	LinMulT (OOWFR)	LinMulT (OOB)	-7.379	1999	two-sided	2.319e-13	[-0.01 -0.01]	0.115	1.125e+10	0.999	1.159e-12	H_a
Agreeableness	LinMulT (OOWFR)	MulT (WFR)	11.883	1999	two-sided	1.620e-31	[0.02 0.02]	0.226	9.740e+27	1.000	8.098e-31	H_a
Agreeableness	LinMulT (OOWFR)	LinMulT (WFR)	10.362	1999	two-sided	1.525e-24	[0.01 0.02]	0.206	1.201e+21	1.000	7.624e-24	H_a
Agreeableness	LinMulT (OOWFR)	MulT (OOWFR)	6.754	1999	two-sided	1.882e-11	[0. 0.01]	0.078	1.516e+08	0.935	9.412e-11	H_a
$\overline{\text{N}}$euroticism	LinMulT (OOWFR)	MulT (OOB)	-11.177	1999	two-sided	3.556e-28	[-0.02 -0.01]	0.144	4.744e+24	1.000	1.778e-27	H_a
$\overline{\text{N}}$euroticism	LinMulT (OOWFR)	LinMulT (OOB)	-13.631	1999	two-sided	1.599e-40	[-0.02 -0.02]	0.175	8.463e+36	1.000	7.997e-40	H_a
$\overline{\text{N}}$euroticism	LinMulT (OOWFR)	MulT (WFR)	4.347	1999	two-sided	1.450e-05	[0. 0.01]	0.076	301.504	0.925	7.252e-05	H_a
$\overline{\text{N}}$euroticism	LinMulT (OOWFR)	LinMulT (WFR)	-7.022	1999	two-sided	2.979e-12	[-0.02 -0.01]	0.124	9.211e+08	1.000	1.489e-11	H_a
$\overline{\text{N}}$euroticism	LinMulT (OOWFR)	MulT (OOWFR)	-22.949	1999	two-sided	1.200e-103	[-0.02 -0.02]	0.185	6.038e+99	1.000	6.001e-103	H_a

Table A.4: Two-sided paired t-test results comparing the trait-wise OCEAN predictions of the proposed LinMulT (OOWFR) to MulT (OOWFR) on the FI dataset. The table includes t-statistics (T), degrees of freedom (DoF), p-values, 95% confidence intervals (CI95%), effect sizes (Cohen-d), Bayes Factors (BF10), statistical power, Bonferroni-corrected p-values, and the decision to reject or fail to reject the null hypothesis (H_a or H_0).

Dataset		Traits	F	df1	df2	p-val	η_p^2
ELEA	Perceived	Meta	4.59	16	47	<.001	0.61
		OCEAN	4.31	16	47	<.001	0.59
	Self-Report	Meta	0.55	16	47	<.001	0.16
		OCEAN	0.92	16	47	<.001	0.24
AMI	A	Meta	3.26	29	90	<.001	0.51
		OCEAN	4.41	29	90	<.001	0.59
	B	Meta	5.20	33	102	<.001	0.63
		OCEAN	3.97	33	102	<.001	0.59
	C	Meta	6.94	33	102	<.001	0.69
		OCEAN	7.92	33	102	<.001	0.72
	D	Meta	5.84	30	93	<.001	0.65
		OCEAN	7.10	30	93	<.001	0.69
UDIVA	Animals	Meta	6.50	13	14	<.001	0.86
		OCEAN	5.69	13	14	<.001	0.84
	Ghost	Meta	2.28	13	14	<.001	0.68
		OCEAN	2.23	13	14	<.001	0.67
	Talk	Meta	1.98	13	14	<.001	0.65
		OCEAN	2.07	13	14	<.001	0.66
	Lego	Meta	1.95	13	14	<.001	0.64
		OCEAN	1.94	13	14	<.001	0.65

Table A.5: Results of the PERMANOVA analysis across the ELEA, AMI and UDIVA datasets.

Database	Backbone	P	R	F1	AP
CEW	EyeNet	0.992 ± 0.007	0.987 ± 0.009	0.990 ± 0.006	0.999 ± 0.001
	ResNet-50	0.994 ± 0.008	0.993 ± 0.005	0.994 ± 0.003	0.999 ± 0.001
	DenseNet-121	0.998 ± 0.003	0.993 ± 0.005	0.995 ± 0.002	0.999 ± 0.001
	<u>CLIP ResNet-50</u>	0.996 ± 0.006	0.996 ± 0.004	<u>0.996 ± 0.005</u>	0.999 ± 0.001
	CLIP ViT-B/16	0.996 ± 0.006	0.997 ± 0.004	**0.996 ± 0.003**	0.999 ± 0.001
ZJU Eye	EyeNet	0.908	0.939	0.923	0.943
	ResNet-50	0.929	0.929	**0.929**	0.965
	<u>DenseNet-121</u>	0.938	0.917	<u>0.927</u>	0.959
	CLIP ResNet-50	0.901	0.883	0.892	0.941
	CLIP ViT-B/16	0.950	0.883	0.915	0.957
RT-BENE^{img}	EyeNet	0.878	0.858	0.868	0.930
	ResNet-50	0.932	0.893	**0.912**	0.946
	ResNet-50-224px	0.929	0.837	0.881	0.918
	<u>DenseNet-121</u>	0.891	0.876	<u>0.883</u>	0.926
	CLIP ResNet-50	0.908	0.835	0.870	0.932
	CLIP ViT-B/16	0.927	0.821	0.871	0.933

Table A.6: Performance of the backbone models on the CEW, ZJU Eye, and RT-BENE^{img} datasets. 10-fold cross-validation is performed on the CEW dataset, while, in the case of the ZJU Eye and RT-BENE^{img}, I evaluated on the test set. The highest F1 score is in bold, and the second highest score is underlined.

Model	Train DB	Test DB	P	R	F1	AP
EyeNet	CEW	ZJU Eye	0.749	0.766	0.758	0.847
ResNet-50	CEW	ZJU Eye	0.687	0.749	0.717	0.813
DenseNet-121	CEW	ZJU Eye	0.636	0.759	0.692	0.791
CLIP-ResNet-50	CEW	ZJU Eye	0.806	0.742	0.773	0.868
CLIP-ViT-B/16	CEW	ZJU Eye	0.819	0.705	0.758	0.857
EyeNet	RT-BENE^{img}	ZJU Eye	0.683	0.766	0.722	0.822
<u>ResNet-50</u>	RT-BENE^{img}	ZJU Eye	0.817	0.795	<u>0.806</u>	0.852
DenseNet-121	RT-BENE^{img}	ZJU Eye	0.880	0.824	**0.851**	0.936
CLIP-ResNet-50	RT-BENE^{img}	ZJU Eye	0.807	0.654	0.722	0.820
CLIP-ViT-B/16	RT-BENE^{img}	ZJU Eye	0.676	0.693	0.684	0.775
EyeNet	ZJU Eye	CEW	0.433	1.000	0.604	0.499
ResNet-50	ZJU Eye	CEW	0.540	0.822	0.652	0.668
DenseNet-121	ZJU Eye	CEW	0.694	0.866	0.771	0.825
CLIP-ResNet-50	ZJU Eye	CEW	0.474	0.968	0.636	0.565
CLIP-ViT-B/16	ZJU Eye	CEW	0.452	0.930	0.608	0.549
EyeNet	RT-BENE^{img}	CEW	0.811	0.784	0.797	0.859
ResNet-50	RT-BENE^{img}	CEW	0.881	0.907	**0.894**	0.950
<u>DenseNet-121</u>	RT-BENE^{img}	CEW	0.792	0.885	<u>0.836</u>	0.903
CLIP-ResNet-50	RT-BENE^{img}	CEW	0.693	0.880	0.775	0.793
CLIP-ViT-B/16	RT-BENE^{img}	CEW	0.662	0.830	0.733	0.740
EyeNet	ZJU Eye	RT-BENE^{img}	0.177	0.217	0.195	0.116
ResNet-50	ZJU Eye	RT-BENE^{img}	0.459	0.554	0.502	0.497
DenseNet-121	ZJU Eye	RT-BENE^{img}	0.636	0.680	0.657	0.666
CLIP-ResNet-50	ZJU Eye	RT-BENE^{img}	0.098	0.452	0.161	0.098
CLIP-ViT-B/16	ZJU Eye	RT-BENE^{img}	0.132	0.619	0.217	0.124
EyeNet	CEW	RT-BENE^{img}	0.529	0.845	0.651	0.497
ResNet-50	CEW	RT-BENE^{img}	0.555	0.852	0.672	0.514
<u>DenseNet-121</u>	CEW	RT-BENE^{img}	0.624	0.801	<u>0.701</u>	0.740
CLIP-ResNet-50	CEW	RT-BENE^{img}	0.737	0.834	**0.782**	0.781
CLIP-ViT-B/16	CEW	RT-BENE^{img}	0.559	0.782	0.652	0.558

Table A.7: Performance of the backbone models using cross-dataset evaluation. The highest F1 ratings and corresponding models are in bold. The second-highest ratings and corresponding models are underlined.

Train DB	Test DB	P_{state}	R_{state}	$F1_{state}$	$P_{presence}$	$R_{presence}$	$F1_{presence}$
EyeBlink8	RT-BENE^{seq}	0.139	0.462	0.213	0.313	0.746	0.441
RN15	RT-BENE^{seq}	0.337	0.592	0.430	0.676	0.731	0.702
RN30	RT-BENE^{seq}	0.295	0.480	0.365	0.811	0.721	0.763
RT-BENE^{seq}	EyeBlink8	0.958	0.679	0.795	0.950	0.983	0.966
RN15	EyeBlink8	0.531	0.954	0.682	0.983	0.974	0.978
RN30	EyeBlink8	0.704	0.967	0.815	0.983	0.974	0.978
RT-BENE^{seq}	RN15	0.797	0.464	0.586	0.858	0.868	0.863
EyeBlink8	RN15	0.838	0.787	0.811	0.919	0.838	0.877
RN30	RN15	0.834	0.807	0.820	0.941	0.930	0.935
RT-BENE^{seq}	RN30	0.851	0.571	0.683	0.908	0.860	0.883
EyeBlink8	RN30	0.655	0.768	0.707	0.873	0.804	0.837
RN15	RN30	0.404	0.890	0.555	0.819	0.905	0.860
RT-BENE^{seq}	TalkingFace	0.834	0.490	0.617	0.968	1.000	0.984
EyeBlink8	TalkingFace	0.961	0.805	0.876	1.000	0.984	0.992
RN15	TalkingFace	0.947	0.614	0.745	1.000	0.967	0.983
RN30	TalkingFace	0.994	0.714	0.831	1.000	1.000	1.000

Table A.8: Eye state recognition and blink presence detection performance of the proposed BlinkLinMulT during cross-dataset evaluation.

ACKNOWLEDGMENTS

I am profoundly grateful to my supervisor, András Lőrincz, for his invaluable teachings and wise counsel throughout my PhD journey. His support was pivotal in making this work possible.

My heartfelt thanks go to Dávid Sztahó and Gábor Gosztolya for their meticulous feedback and insightful suggestions, which significantly improved the quality of my dissertation. Their expertise was instrumental in refining my work and enhancing its clarity.

I would like to thank Zsolt János Viharos at MTA SZTAKI, where I took my first steps in research during my early university years. That experience introduced me to the field and sparked the interest that set me on this path.

To all members of the NIPG group, I am grateful for their support and camaraderie. Special thanks to Kristian Arian Fenech, Ellák Somfai, László Kopácsi, Krisztián Kis, Bruno Melício, and Kaan Karaköse for their constant encouragement and collaborative research efforts. My thanks also go to Gyöngyvér Ferencz for her dedicated administrative assistance. For the lovely discussions on a range of topics throughout our studies, I thank Zoltán Ádám Milacski, Gábor Baranyi, Kinga Bettina Faragó, Viktor Varga, Áron Fóthi, Imre Molnár, Zsolt Csibi, and Xiang Linyun. My sincere thanks extend to everyone who contributed through professional discussions or casual chats, enriching my research experience.

On a personal note, my deepest gratitude goes to the pillars of my life: my parents, my brother, and my godparents—the most important people in my world, to whom no amount of thanks could ever feel enough. Their endless love and steadfast encouragement have been my constant source of strength and inspiration throughout this journey. To my close relatives and friends, who provided much-needed breaks filled with laughter and joy, I owe a tremendous debt of gratitude.

I am forever grateful for each of you, as you have all contributed immensely not just to the success of this work, but to the person I have become.

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